

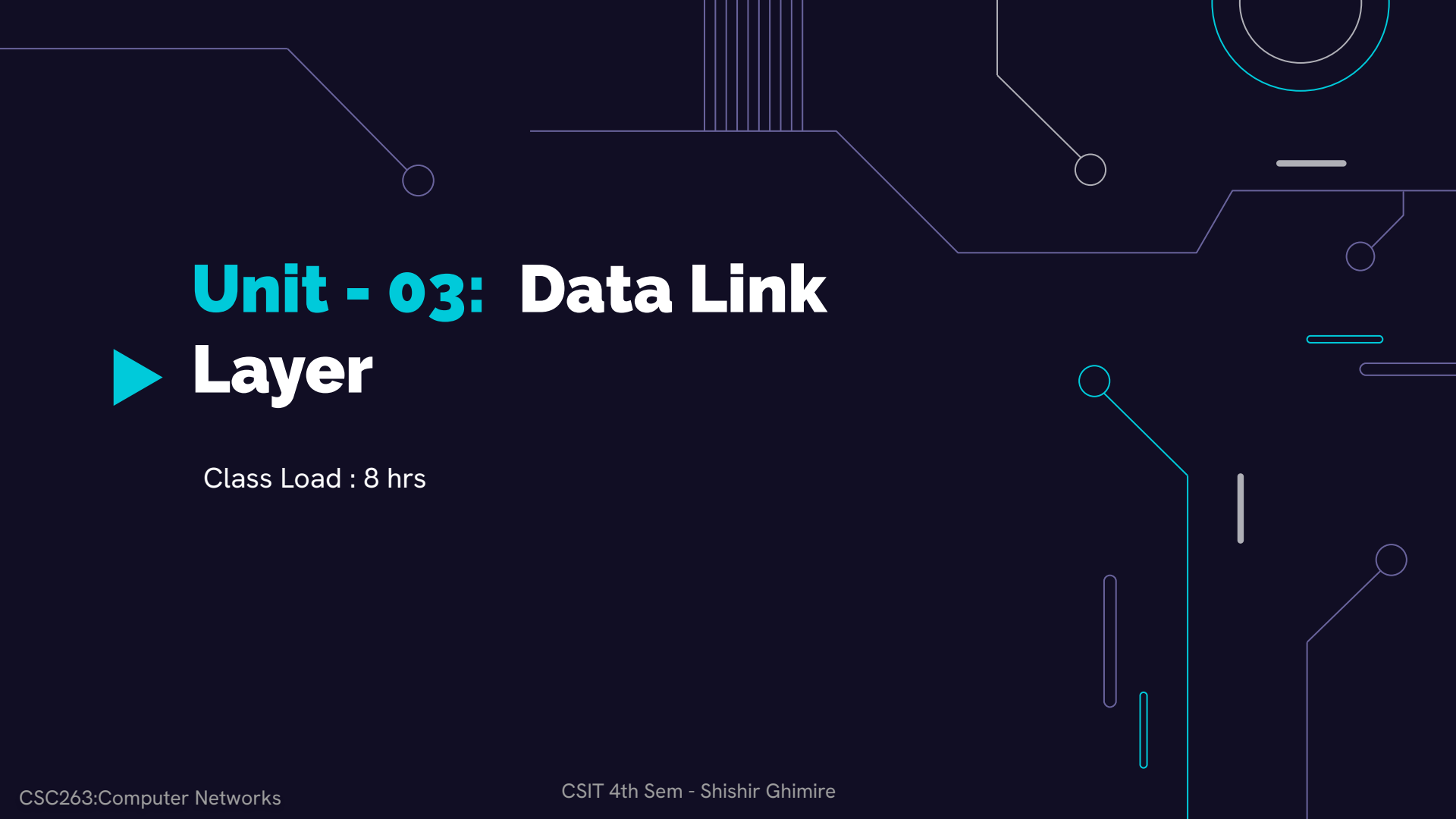
Computer Networks



Course Code: CSC263
Year/ Semester: II/IV

Compiled by Shishir Ghimire

Credit Hours: 3hrs



Unit - 03: Data Link ▶ Layer

Class Load : 8 hrs

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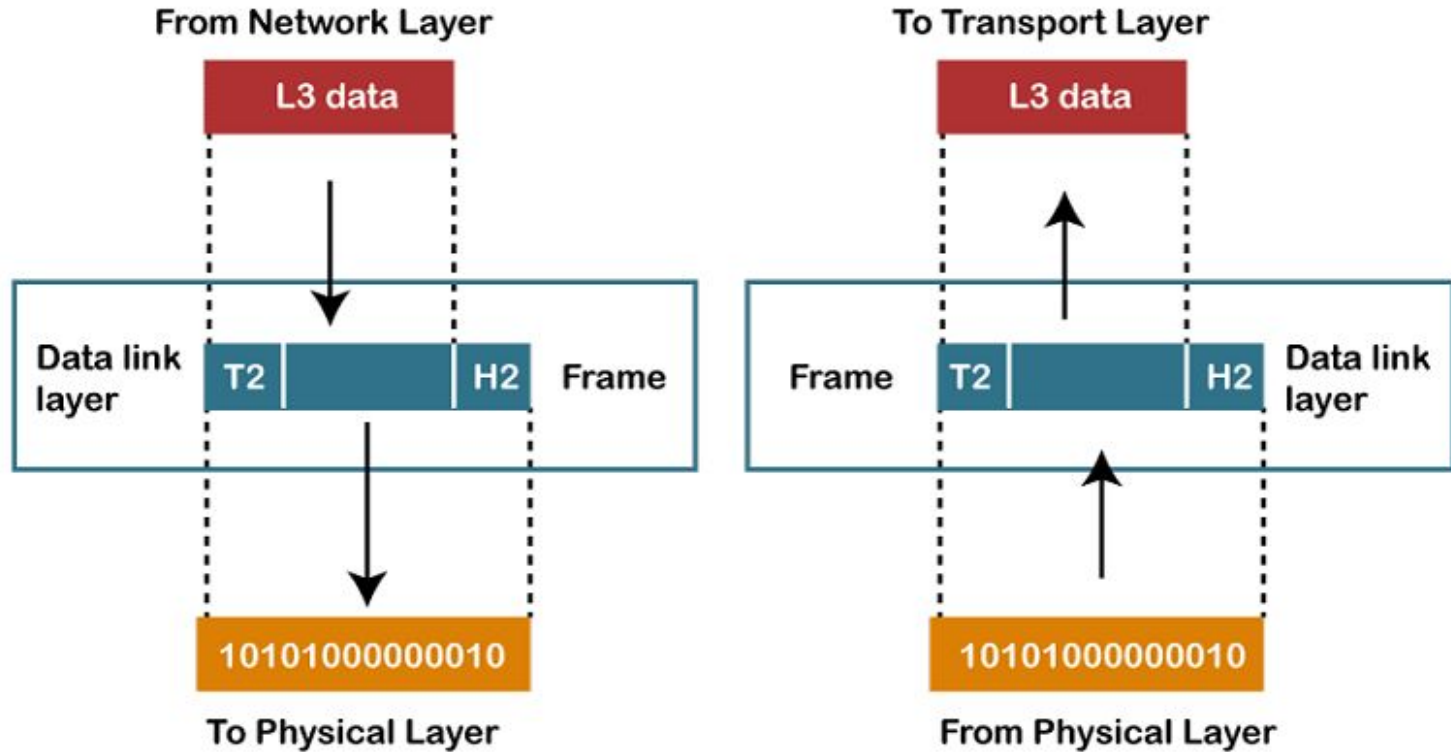
Unit 3: Data Link Layer (8Hrs.)

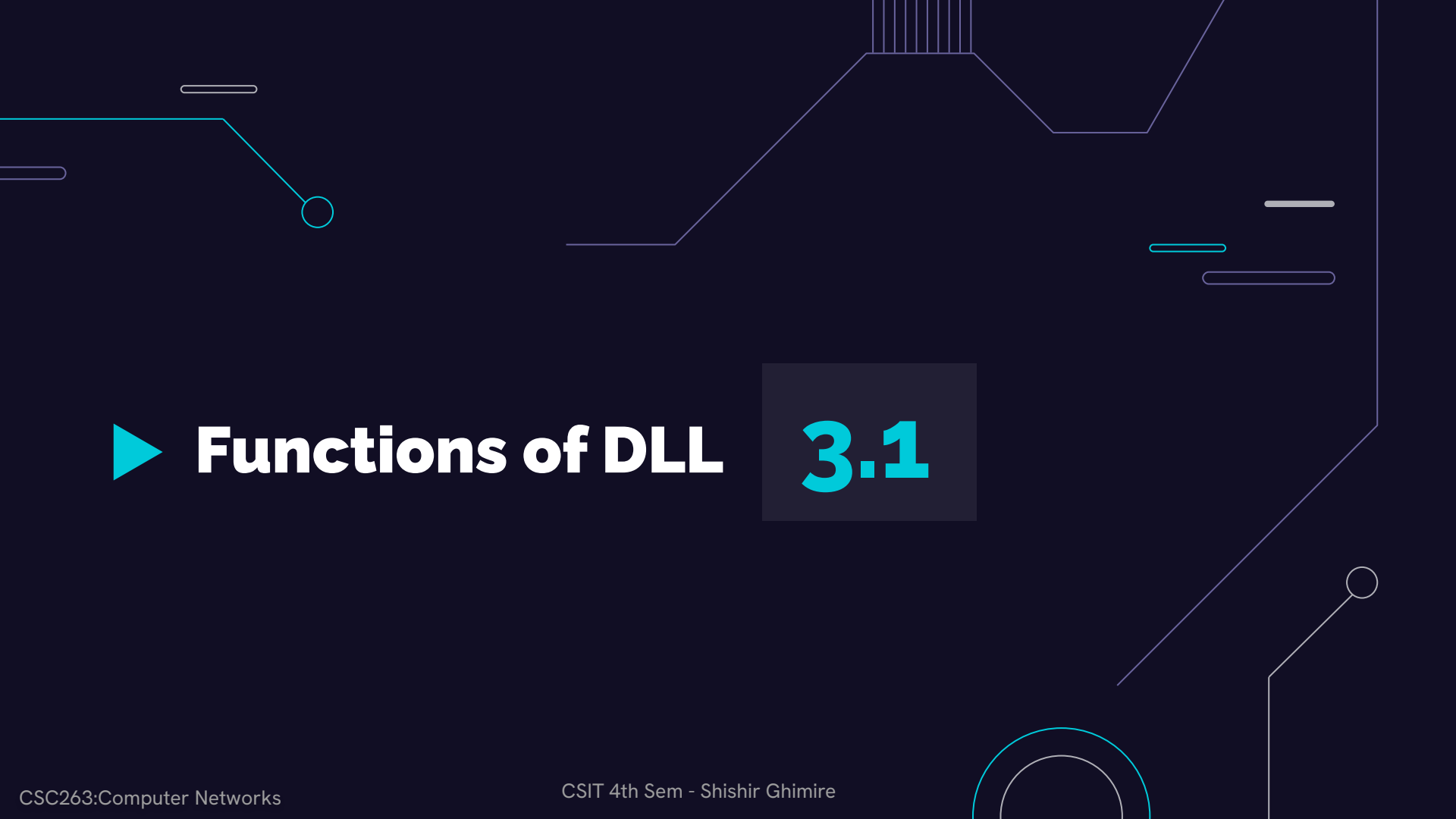
- 3.1. Function of Data Link Layer (DLL)
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- 3.3. Framing and Flow Control Mechanisms
- 3.4. Error Detection and Correction techniques
- 3.5. Channel Allocation Techniques (ALOHA, Slotted ALOHA)
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- 3.7. Wireless LAN: Spread Spectrum, Bluetooth, Wi-Fi
- 3.8. Overview Virtual Circuit Switching, Frame Relay& ATM
- 3.9. DLL Protocol: HDLC, PPP

► Data Link Layer (Revisit):

- The data link layer is responsible for the **node-to-node delivery** of the message.
- The **main function** of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.
- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its **MAC address**.
- The Data Link Layer is divided into **two sublayers**:
 - Logical Link Control (LLC)
 - Media Access Control (MAC)
- The MAC sub-layer is responsible for data encapsulation and accessing media.
- **PDU : Frame (Ethernet Frame)**

► Data Link Layer (Revisit):



The background features a dark blue field with white and light blue geometric lines resembling a circuit board. These lines include straight paths, right-angle turns, and small circles at various points. A prominent light blue line starts from the left edge, goes right, then diagonals down to a circle. Other lines are scattered across the top and right sides, some ending in small horizontal bars or circles.

► **Functions of DLL**

3.1

► Functions of Data Link Layer:

→ Framing:

- ◆ The Data Link Layer breaks down the **stream of bits** received from the Network Layer (Layer 3) into **frames**. This can be accomplished by attaching **special bit patterns** to the beginning and end of the frame.
- ◆ **Purpose:** Framing helps in defining the **start and end** of a frame, allowing the receiver to identify and extract the data.

→ Physical Addressing (Media Access Control - MAC):

- ◆ The Data Link Layer adds a **header to the frame**, which includes the source and destination MAC addresses.
- ◆ **Purpose:** MAC addresses are used to **uniquely identify devices** on a local network. This addressing helps in the proper delivery of frames to the intended recipient.

→ Error Detection:

- ◆ The Data Link Layer is responsible for **detecting errors** that may occur during the transmission of frames.
- ◆ **Purpose:** By using **error detection mechanisms** such as checksums or cyclic redundancy checks (CRC), the layer ensures the integrity of the transmitted data.

► Functions of Data Link Layer:

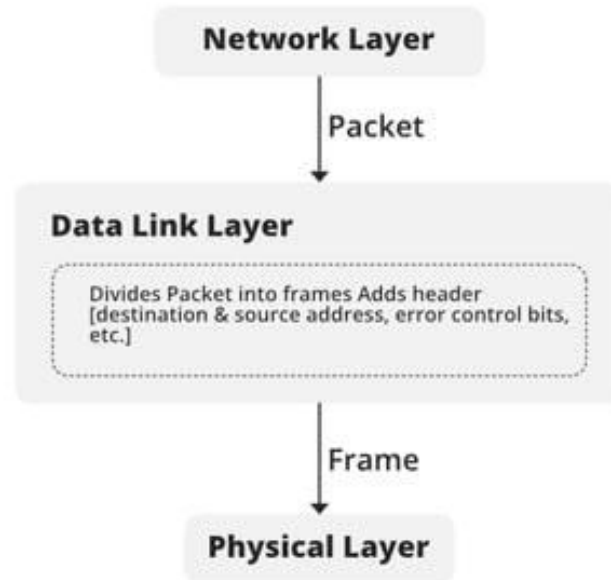
→ Flow Control:

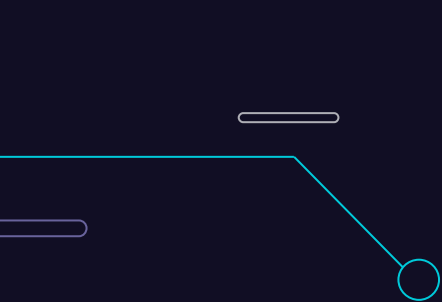
- ◆ The Data Link Layer manages the **flow of data** between devices to prevent congestion and ensure that a fast sender **does not overwhelm** a slow receiver.
- ◆ **Purpose:** Flow control mechanisms prevent data loss and ensure efficient communication by regulating the pace of data transmission.

→ Access Control:

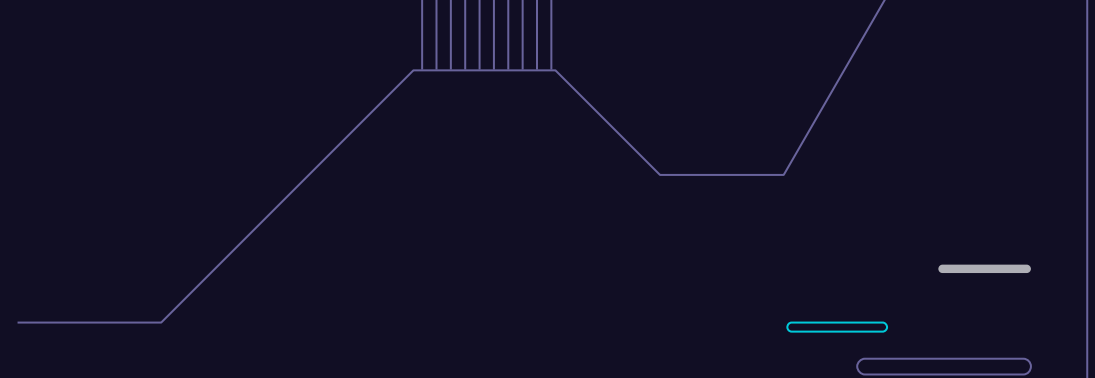
- ◆ The Data Link Layer determines how **devices gain access** to the physical medium (communication channel).
- ◆ **Purpose:** Access control methods, such as **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) in Ethernet networks, manage the **order and timing** of data transmissions to avoid collisions.

► Data Link Layer:



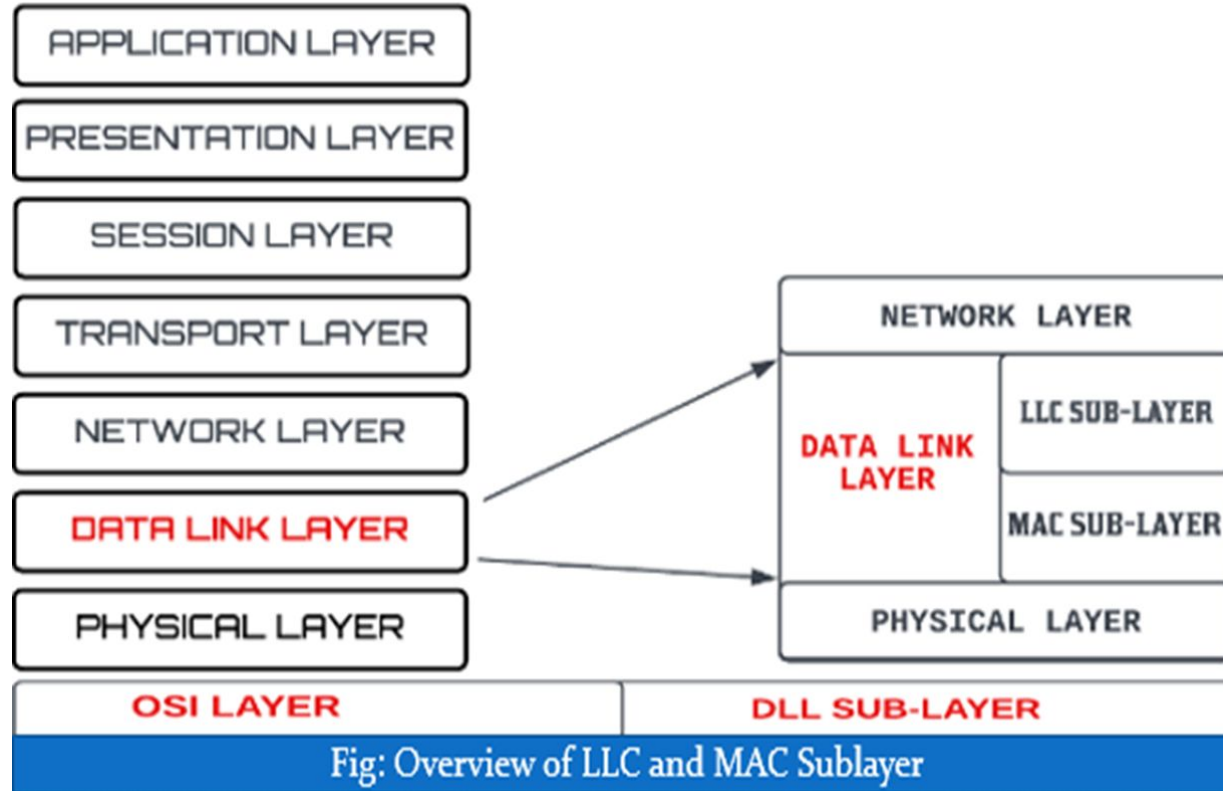


► Overview of LLC and MAC



3.2

► Overview:



► LLC (Logical Link Control) Sublayer:

- ❖ It is uppermost layer of DLL.
- ❖ The primary function of LLC is to **multiplex protocols** over the MAC layer and **de-multiplex** while receiving.
- ❖ LLC provides **hop-hop** flow control and error control .i.e its main function is flow control.
- ❖ It allows **multipoint communication** over computer network.
- ❖ It may also be used to **retransmit error frame** detected by MAC sublayer.

► MAC (Media Access Control) Sublayer:

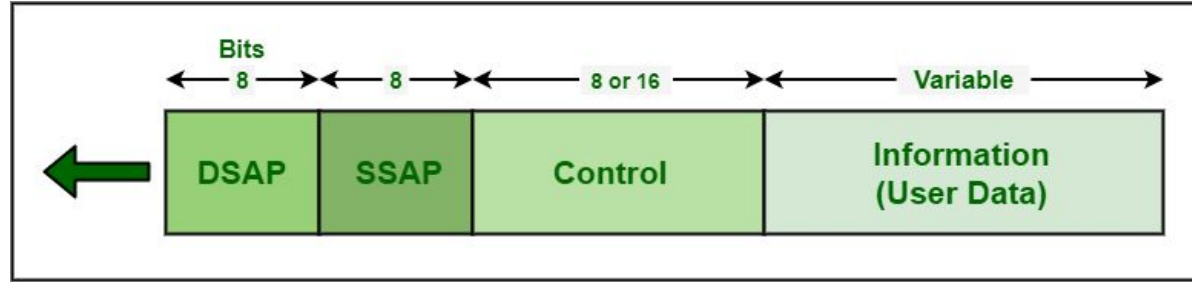
- It is lower layer of DLL.
- It is responsible for **encapsulating frames** for transmission via physical medium.
- Its main **purpose is framing** and to direct those data frames across a network using some kind of **addressing**(many protocols use MAC address).
- Implemented by hardware, typically in the computer **NIC**.
- On a network, the **network interface card (NIC)** has a unique hardware address(MAC address-programmed in NIC ROM) which **identifies** a computer or device.



► Problems in Framing:

- **Detecting start of the frame:** When a frame is transmitted every station must be able to detect it looking out for **special sequence** of bits called **SFD** (Starting Frame Delimiter).
- **How do station detect a frame:** Every station listens to link for SFD pattern through a sequential circuit. If SFD is detected, **sequential circuit alerts station**. Station checks destination address to **accept or reject** frame.
- **Framing incompatibility:** Different networking devices and protocols may use **different framing methods**, which can lead to framing incompatibility issues..
- **Handling errors:** Framing errors may occur due to **noise or other transmission errors**, which can cause a station to **misinterpret** the frame. Therefore, error detection and correction mechanisms, such as cyclic redundancy check (CRC), are used to ensure the integrity of the frame.
- **Framing overhead:** Every frame has a header and a trailer that contains control information such as source and destination address, error detection code, and other protocol-related information. This overhead **reduces the available bandwidth** for data transmission, especially for small-sized frames.
-

► LLC Frame Format:



- **Destination Service Access Point (DSAP)**
 - IEEE 802.2 header begins with a 1 byte field, which identifies the **receiving upper-layer process**. DSAP is generally an **8-bit** long field that is used to represent the **logical addresses** of the network layer **entity** meant to **receive** the message.
- **Source Service Access Point (SSAP)**
 - Following the DSAP address is the 1-byte address, which identifies the **sending upper-layer process**. SSAP is also an **8-bit** long field that is used to represent the **logical addresses** of the network layer **entity** meant to **create** a message

► LLC Frame Structure:

Control :

- The Control field employs three different formats, depending on the type of LLC frame used:
 - **Information (I) frame:** Carries upper-layer information and **some** control information.
 - **Supervisory (S) frame:** Provides control information. An S frame can **request and suspend** transmission, **reports** on status, and **acknowledge** receipt of I frames. S frames do **not** have an Information field.
 - **Unnumbered (U) frame:** Used for **control purposes** and is not sequenced. Some U frames have **an** Information field.

Data:

- Variable-length field bounded by the **MAC format** implemented. Usually contains IEEE 802.2 Subnetwork Access Protocol (SNAP) **header** information, as well as **application-specific** data.

The background features a dark blue field with white and light blue geometric lines resembling a circuit board. These lines include straight paths, right-angle turns, and small circles at various points, creating a technical, network-like aesthetic.

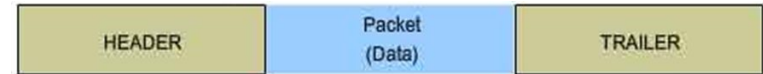
▶ Framing & Flow Control Mechanisms

3.3

► Framing:

- A frame is a unit of data that includes both the data to be transmitted and control information, such as frame start and end delimiters, addressing, error checking, and possibly flow control information.
- Frames are manageable group of bits to be transmitted.
- The data link layer, needs to pack bits into frames, so that each frame is distinguishable from another.
- The Data Link layer prepares a packet for transport across the local media by encapsulating it with a header and a trailer to create a frame.

- **Parts of a Frame:**



- **Data :** The packet from the Network layer
- **Header:** Contains control information, such as addressing, and is located at the beginning of the PDU.
- **Trailer :** It contains error detection and error correction bits.

► Problems in Framing:

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- **How do station detect a frame:** Every station listens to link for SFD pattern through a sequential circuit. If SFD is detected, **sequential circuit alerts station**. Station checks destination address to **accept or reject** frame.
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- **Framing overhead:** Every frame has a header and a trailer that contains control information such as source and destination address, error detection code, and other protocol-related information. This overhead **reduces the available bandwidth** for data transmission, especially for small-sized frames.

► Types of Framing:

1. Fixed Size Framing:

- The frame is of fixed size and there is **no need to provide** boundaries to the frame, length of the frame itself acts as delimiter.
- It does not require any additional boundary bits to identify the start and end of the frame.
- **Drawback:** It suffers from internal fragmentation if data size is less than frame size
- **Solution:** Padding

2. Variable Size Framing:

- The size of each frame to be transmitted **may be different.**

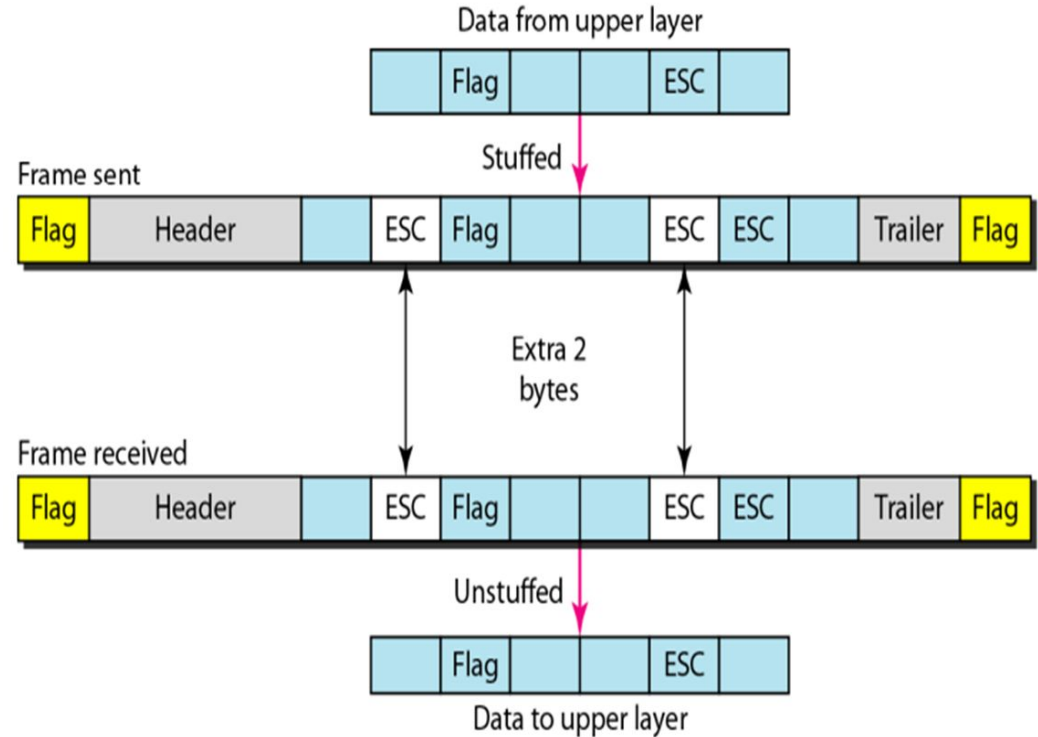
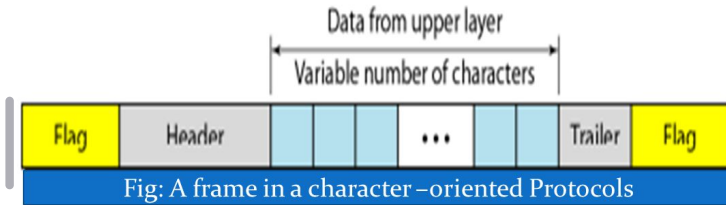
► Types of Framing:

- So additional mechanism are used to mark the end of one frame and the beginning of the next frame.
- Two ways to define frame delimiters:
 - a. **Length Field:** Length field is used to determine the size of frame.
 - b. **End Delimiters:** Pattern is used as a delimiter to define the size of frame. If the pattern occurs in the message then 2 approach is used:
 - **Character-Oriented Protocols/Byte Oriented Stuffing:** A **byte** is stuffed in the message to differentiate from the delimiter. This is also called character-oriented framing.
 - **Bit-Oriented Protocols/Bit Stuffing:** A **pattern of bits** of arbitrary length is stuffed in the message to differentiate from the delimiter. This is also called bit-oriented framing.

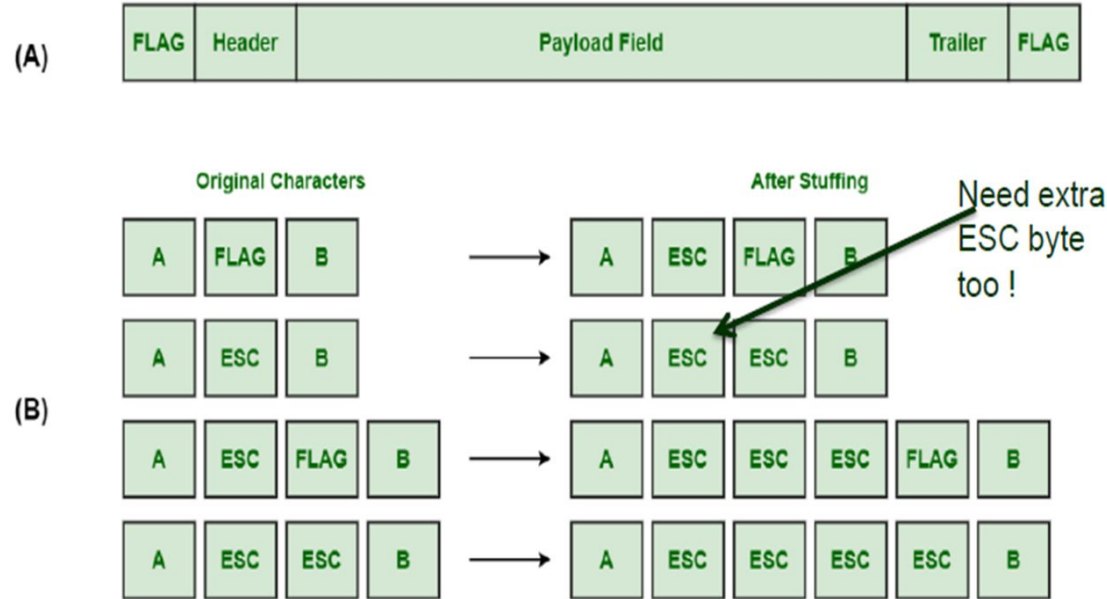
► Character-Oriented Approach:

- In byte stuffing, a special byte called the **escape character(ESC)** is stuffed before every byte in the message with the same pattern as the flag byte.
- The escape character, which has a predefined bit pattern. Whenever the receiver encounters **ESC character**, it remove it from the data section and treats next character as data, not a delimiting flag.
- Byte stuffing is the process of **adding 1 extra byte** whenever there is a flag or escape character in a text.
- If the ESC sequence is found **in the message byte**, then **another ESC byte** is stuffed before it.

► Character-Oriented Approach:



► Character-Oriented Approach:



A Character Stuffing

(A) A frame delimited by flag bytes

(B) Four examples of byte sequences before and after byte stuffing

► Bit Stuffing Approach:

- Bit stuffing is the process of **adding one extra 0 whenever five consecutive 1s follow a 0** in the data , so that the receiver does not mistake the pattern **01111110** for a flag.

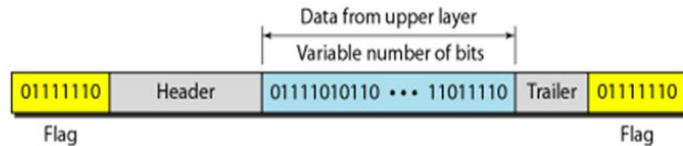


Fig: A frame in a bit-oriented protocol

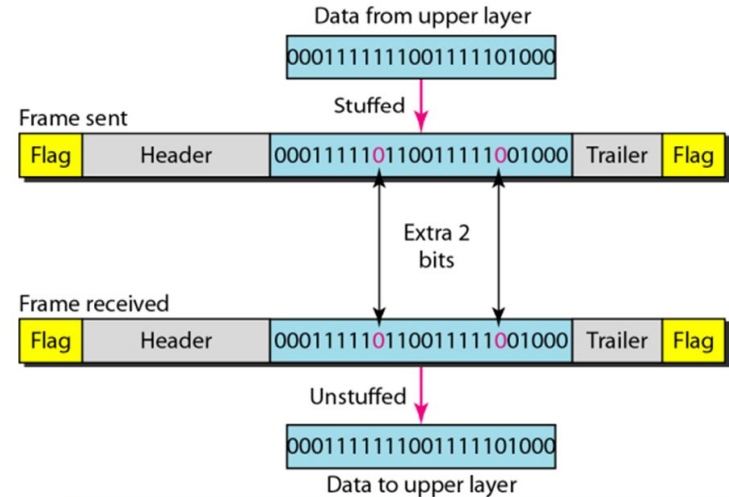
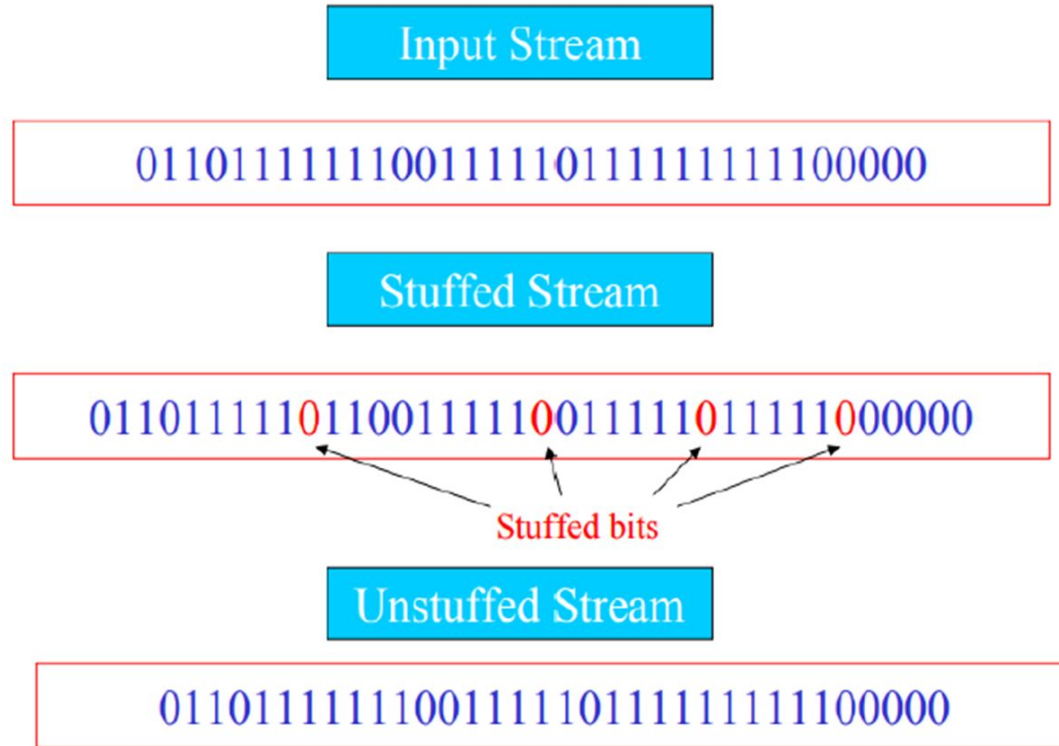


Fig: Bit stuffing and unstuffing

► Bit Stuffing Approach:

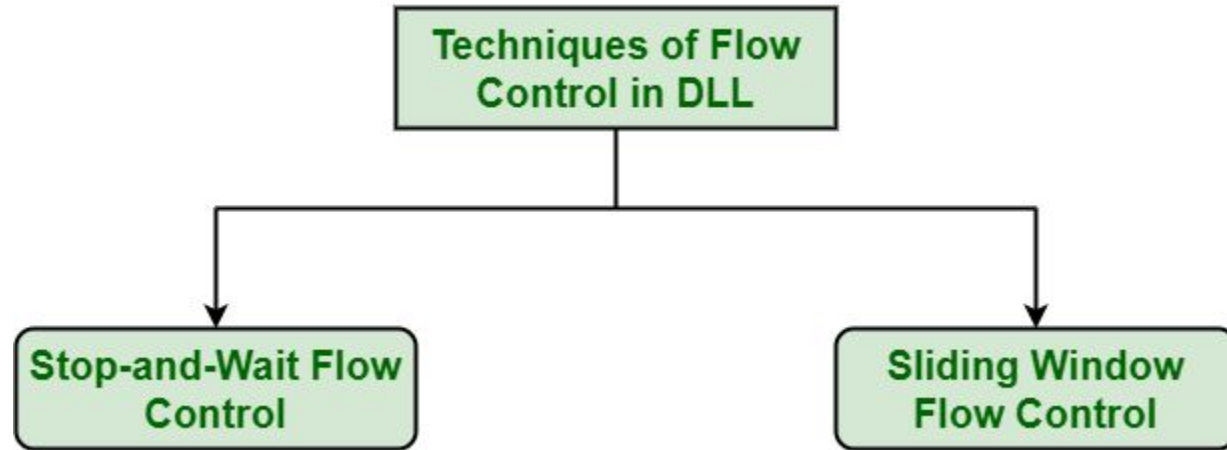


► Flow Control:

❖ Data link control = flow control + error control.

- ❖ Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgement.
- ❖ It is also called Speed Matching Mechanism.
- ❖ If receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- ❖ Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.

► Flow Control Techniques:

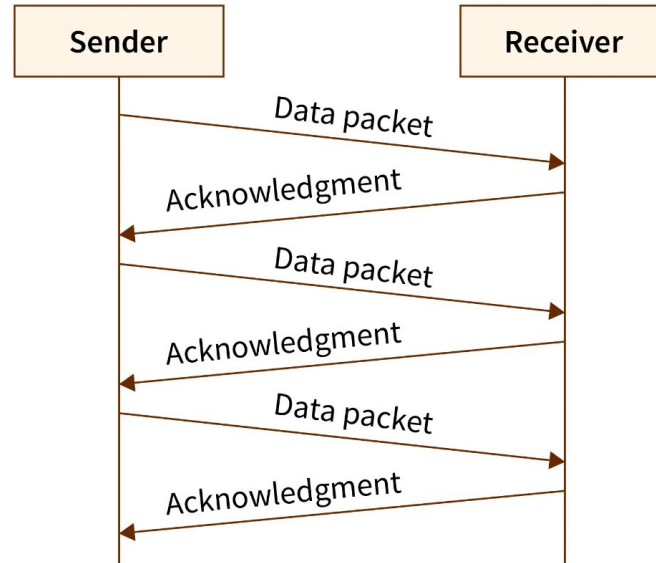


► Stop & Wait Flow Control:

- This method is the **easiest and simplest** form of flow control.
- In this method, basically message or data is broken down into various multiple frames, and then receiver **indicates its readiness** to receive frame of data.
- When **acknowledgement** is received, then only sender will **send or transfer** the next frame.
- This process is continued until sender transmits **EOT (End of Transmission)** frame. In this method, **only one of frames** can be in transmission at a time.
- **Advantages:**
 - This method is very easiest and simple and each of the frames is **checked** and **acknowledged** well.
 - This method is also very **accurate**.
- **Disadvantages:**
 - This method is fairly **slow**. In this, only one packet or frame can be sent **at a time**.
 - It is very inefficient and makes the transmission process very slow.

► Stop & Wait Flow Control:

STOPN-AND-WAIT PROTOCOL

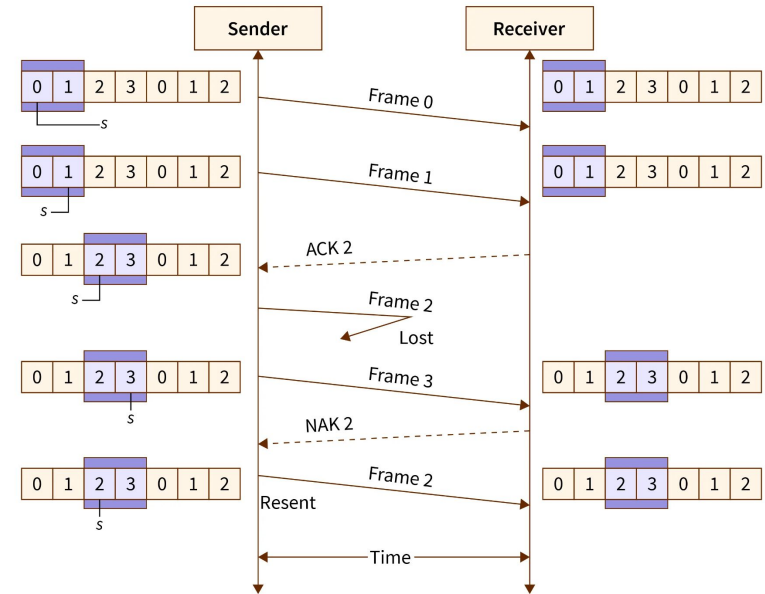


► Sliding Window Flow Control:

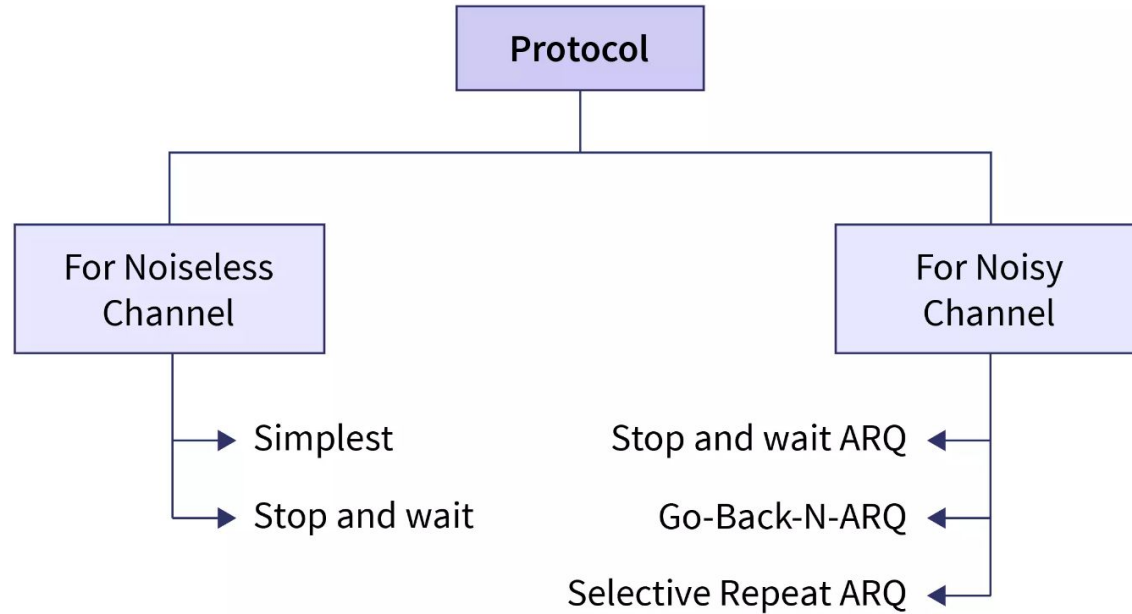
- This method **improves the efficiency** of stop and wait protocol by allowing **multiple frames** to be transmitted **before** receiving an **acknowledgement**.
- **Working Principle:**
 - Both the sender and receiver has **finite size buffer** called as **windows**. In this method, **both** the sender and receiver **agree** upon **total number of data frames** after which **acknowledgement** is needed to be transmitted.
 - The sender sends multiple frames in a sequence, **without waiting** for acknowledgement. When its sending window is **filled**, it **waits** for acknowledgement. Upon **receiving** the acknowledgement, it **advances** the window and transmits the next frame.
- **Advantages:**
 - **Multiples frames** can be sent one after another.
 - This method **increases** efficiency.

► Sliding Window Flow Control:

- The sender sends the frames 0 and 1 from the first window (because the window size is 2).
- The receiver after receiving the sent frames, sends an acknowledgment for frame 2 (as frame 2 is the next expected frame).
- The sender then sends frames 2 and 3. Since frame 2 is lost on the way, the receiver sends back a "NAK" signal (a non-acknowledgment) to inform the sender that frame 2 has been lost. So, the sender retransmits frame 2.

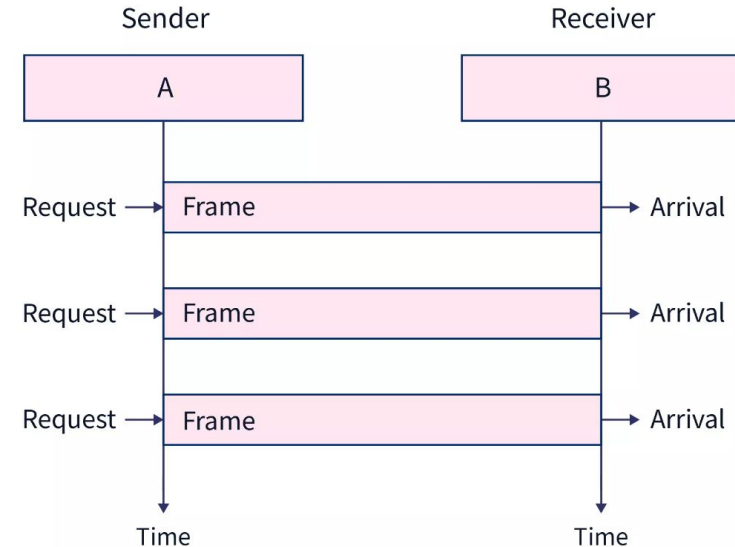


► Flow Control Protocols:



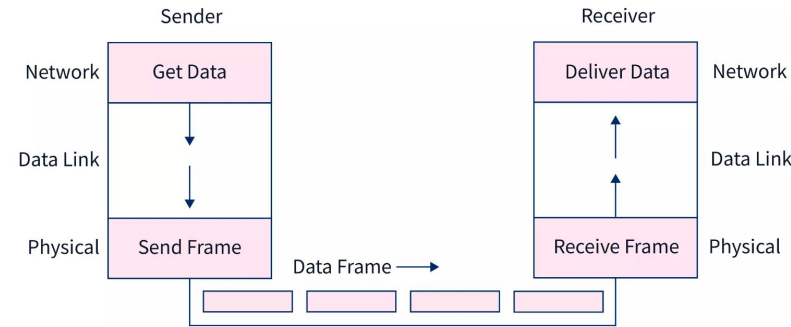
► Noiseless Channel ► Simplest Protocol:

- Simplest protocol with **no** flow and error control .
- It is a **unidirectional** protocol in which data frames are traveling in only one direction-from the **sender to receiver**.



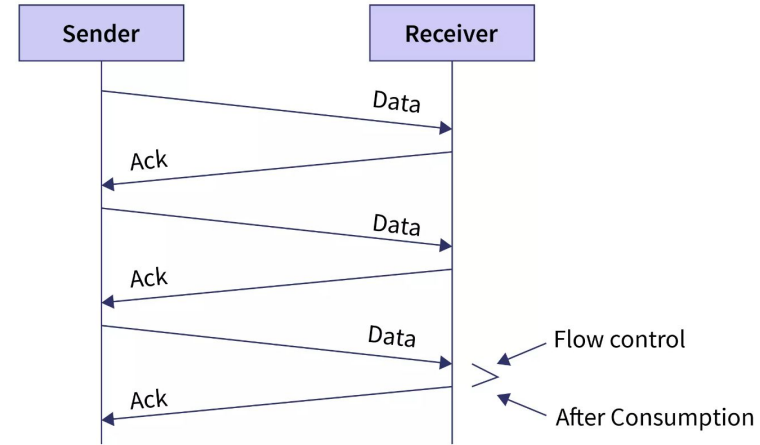
► Noiseless Channel ► Simplest Protocol:

- The following assumptions have been made for developing the simplest protocol:
 - The transmission channel is **completely noiseless** (a channel in which no frames are lost, corrupted, or duplicated).
 - The transmission channel is assumed to be **ideal** in which there is no data loss, no data duplication, and no data corruption.
 - The **buffer space** for storing the frames at the sender's end and the receiver's end **is infinite**.
 - There is no **error and flow control** mechanism.



► Stop and Wait Protocol:

- Stop-and-Wait Protocol is **data link layer protocol** for transmission of frames over **noiseless** channels.
- It provides **unidirectional** data transmission **with** flow control facilities but **without** error control facilities.
- The idea behind Stop-and-Wait Protocol is straight forward. i.e. after transmitting one frame, the sender **waits** for an **acknowledgement** from the receiver before transmitting the **next frame**.



► Advantages of Stop and Wait Protocol:

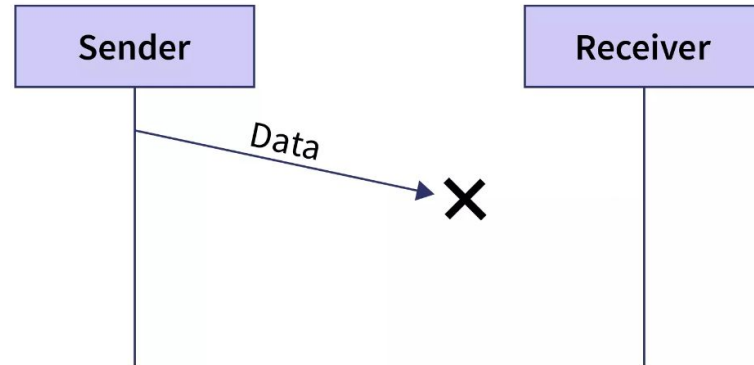
- This protocol provides **flow control**.
- One of the main advantages of the stop-and-wait protocol is its simplicity.
- Stop and Wait protocol is useful in LAN as its **efficiency decreases** with the increasing distance between sender and receiver.
- If we increase the data packet size, the efficiency is going to **increase**. Hence, it is **suitable** for transmitting **big data packets**.
- Stop & wait protocol is accurate as the sender sends the next frame to the receiver **only when the acknowledgment** of the previous packet is received. So there is **less** chance of the **frame being lost**.

► Problems of Stop and Wait Protocol (Disad):

- Problems Occur Due to Lost Data

In this case, two problems occur in data transmission:

- For an acknowledgment, the sender has to wait for an **infinite amount of time**.
- The receiver will also have to wait for an infinite amount of time **to receive** the data.

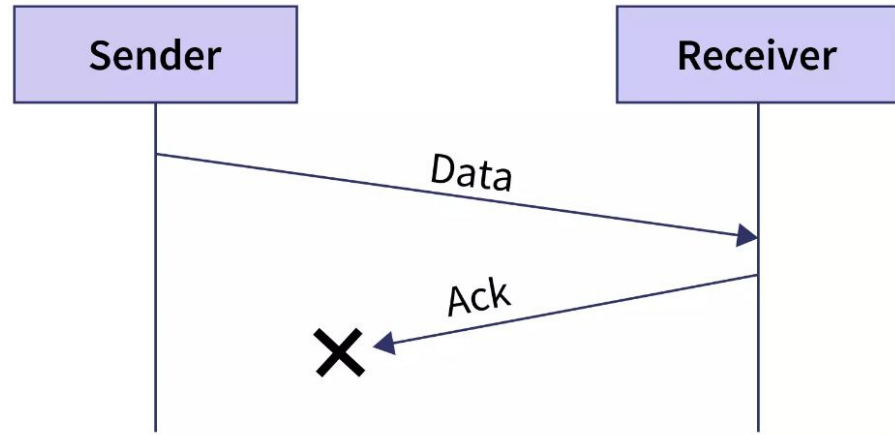


► Problems of Stop and Wait Protocol (Disad):

- Problems Occur Due to Lost Acknowledgment

In this case, one problem occur in data transmission:

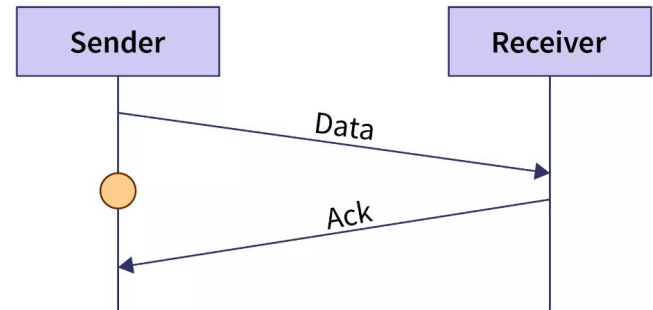
- The sender has to wait for an infinite amount of time due to acknowledgment loss in the network.



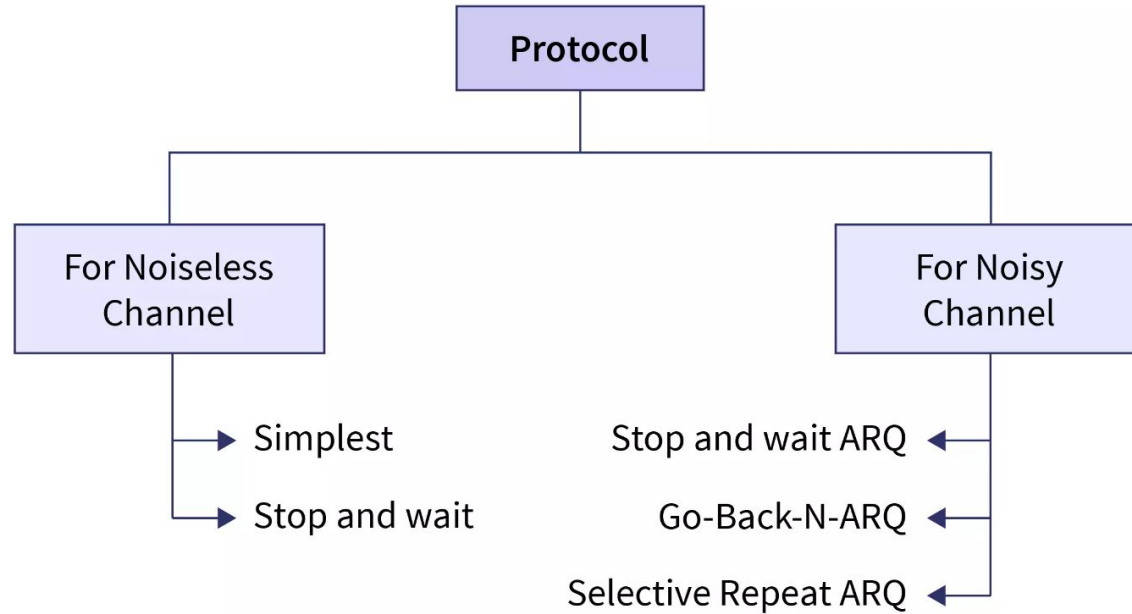
► Problems of Stop and Wait Protocol (Disad):

- Problem Due to the Delay in Data or Acknowledgment

- The sender sends the data packet, and the receiver will also send the acknowledgment, but the acknowledgment is received by the sender after the timeout period on sender side.
- As the acknowledgment is received late, so acknowledgment can be wrongly considered as the acknowledgment of some other data packet.



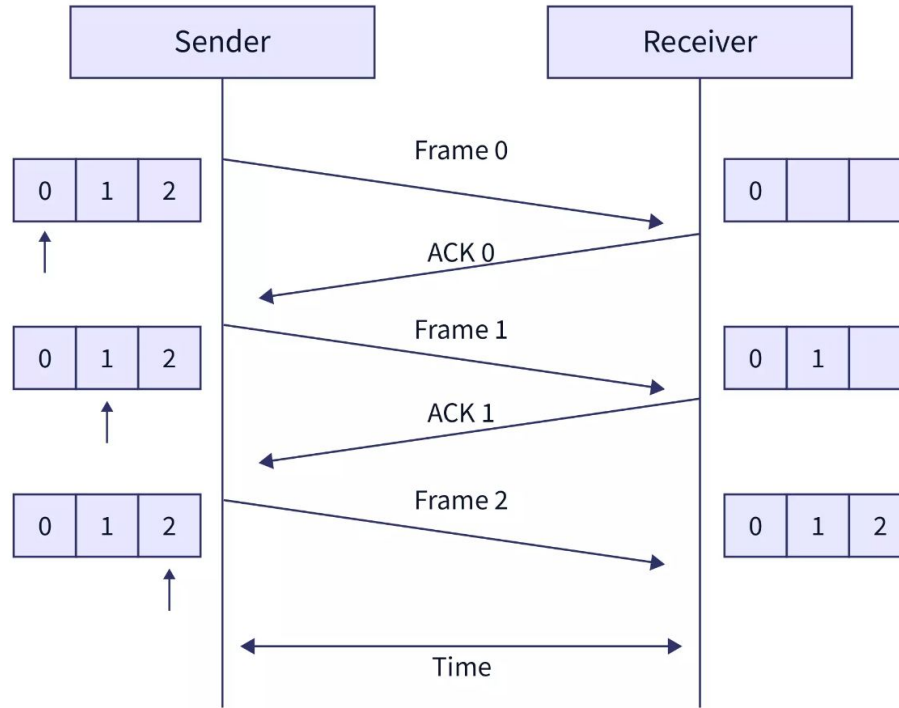
► Flow Control Protocols:



► Stop and Wait ARQ (Automatic Repeat Request) :

- Stop and wait ARQ is a type of **sliding window protocol**.
- **Error correction** in Stop-and-Wait ARQ is done by **keeping a copy** of the sent frame and **retransmitting** of the frame when the **timer expires**.
- Initially, the sender sends one frame as the **window size is 1**. The receiver on the other end receives the frame and **sends the ACK** for the correctly received frame. The sender waits for the ACK until the **timer expires**. If the sender **does not receive** the ACK within the timer limit, it **re-transmits** the frame for which the ACK has **not been** received.
- In Stop-and-Wait ARQ, we use **sequence numbers** to number the frames. The sequence numbers are based on **modulo-2 arithmetic**.
- Acknowledgment **number** always **announces** in modulo-2 arithmetic the sequence number of the **next frame expected**.

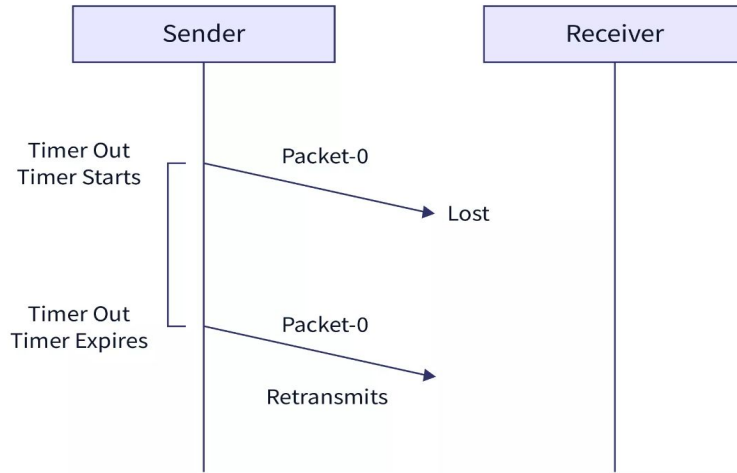
► Stop and Wait ARQ (Automatic Repeat Request) :



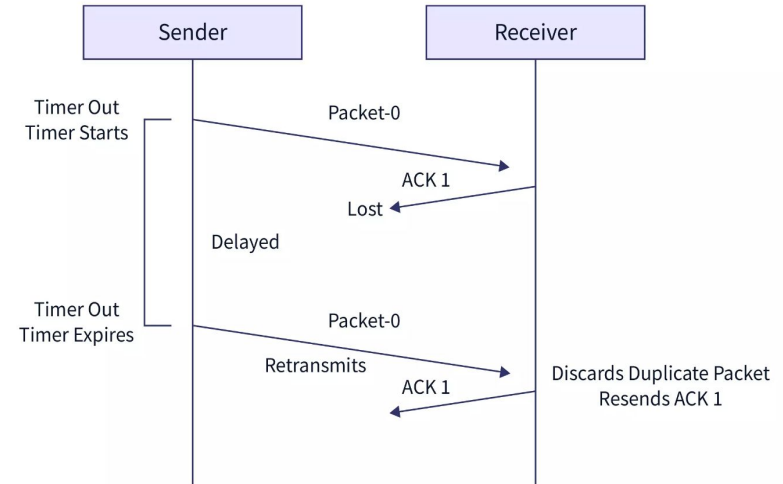
► Advantages of Stop and Wait ARQ :

- The stop and wait ARQ can be used in both the **data link** layer and the **transport layer**.
- The stop and wait ARQ provides both **error** management and **flow** control management.
- The stop and wait ARQ works in **half-duplex mode** as at a time only the sender can **send** the message **or** the **receiver** can send the ACK.
- The stop and wait ARQ ensures that the information is received by the receiver **in the order** it was sent.

► Problems in Stop and Wait ARQ :

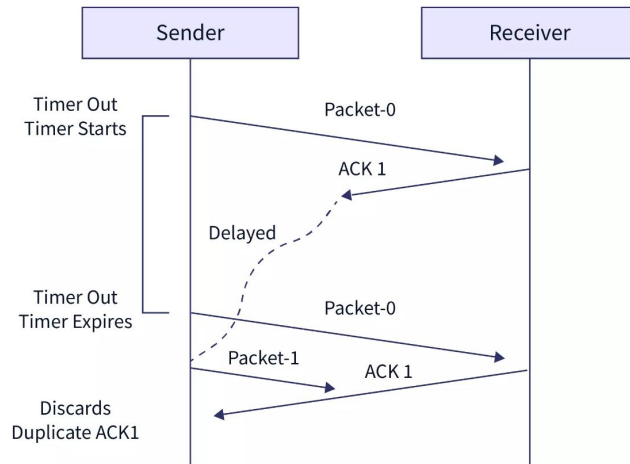


Problem of Lost Data Packet

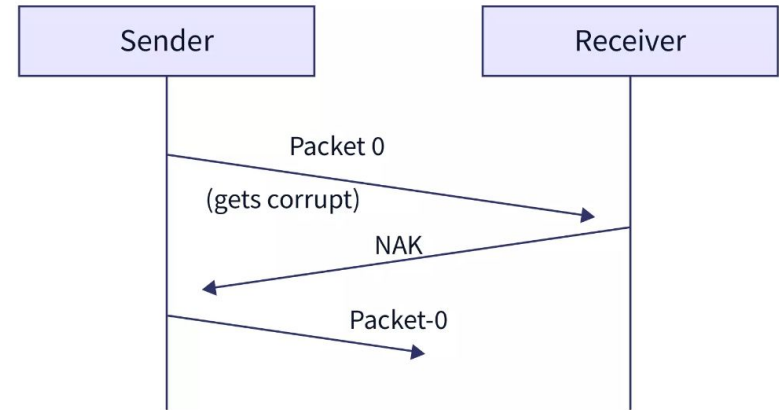


Problem of Lost Acknowledgement

► Problems in Stop and Wait ARQ :



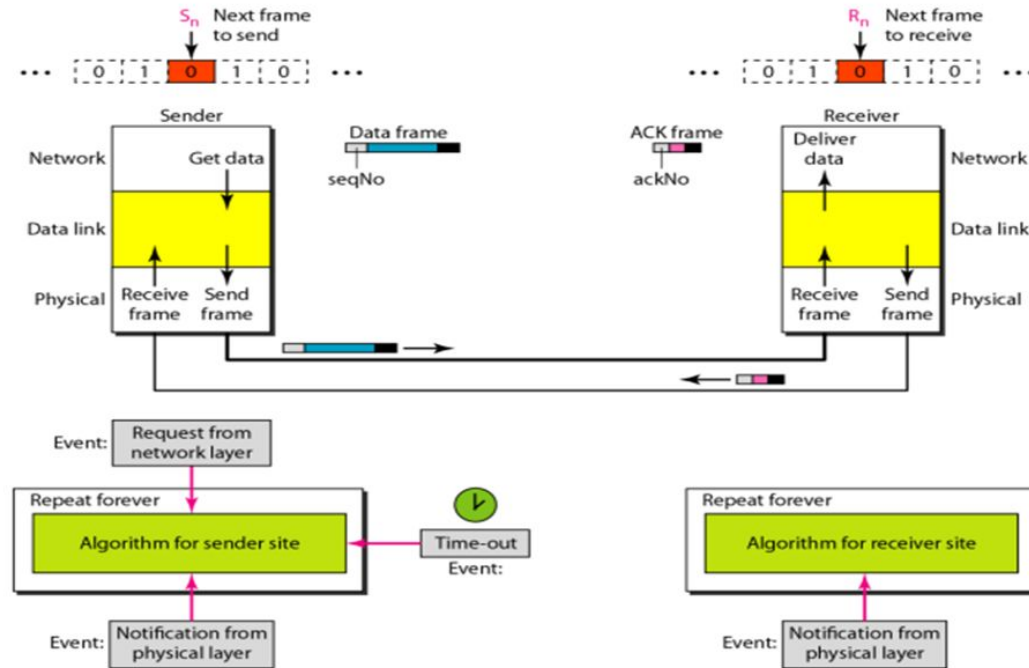
Problem of Delayed Acknowledgement



Problem of Damaged Packet

► Design of Stop and Wait ARQ:

Stop-and-Wait ARQ



► Go-Back-N ARQ:

- Go-Back-N ARQ protocol is a type of **sliding window protocol** where Sender window **size is N** and receiver window **size is always 1**.
- This protocol is based on the method of using the Sliding Window Protocol as **the basis of** data exchange, where the '**N**' in the protocol represents the window size.
- The term ARQ represents the **Automatic Repeat Request**, which refers to sending **multiple frames simultaneously** from the sender to the receiver end.
- The frames shared in the protocol are **sequenced** for better efficiency, to **avoid any retransmission** of shared data, and differentiate between the frames.

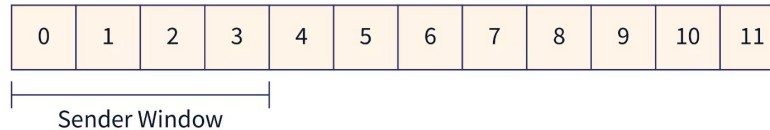
► Go-Back-N ARQ:

- The protocol is designed to share multiple frames at a time, with the receiver end, before expecting any acknowledgment from it.
- **Pipelining** is allowed in the Go Back N ARQ protocol. Pipelining means sending multiple frames **before receiving the acknowledgment** for the previously/first sent frame.
- The **acknowledgement number** is cumulative and it defines the **sequence number** of the next frame expected to arrive.
- The sequence number are in **modulo 2^m** . For sending window size 4 (2^2) the sequence number will be 0, 1, 2, 3, 0, 1, 2, 3 and so on.

► Go-Back-N ARQ:

Sender Window:

- The sender window is a **fixed-sized window** that defines the number of frames that are transmitted from sender to receiver at once. The **integer 'N'** in the Go Back 'N' is the **frame size**.
- For example in **Go Back 4 ARQ**, the size of the sender **window is 4**.



Receiver Window:

- The Receiver window in the Go Back N ARQ protocol is **always of size 1**. This means that the receiver takes at most 1 frame at a single time.

► Working of Go-Back-N ARQ:

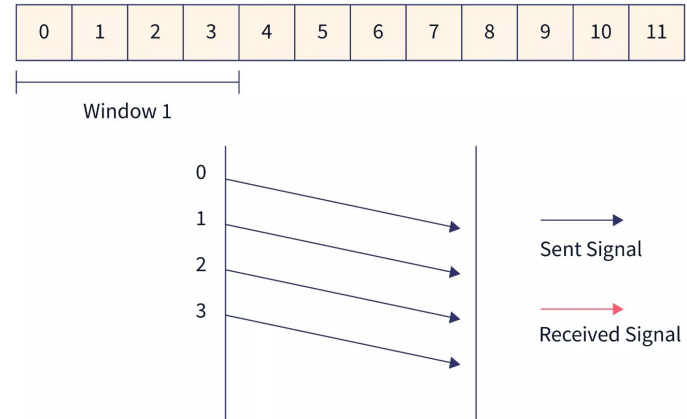
- Data packets are divided into **multiple frames**. Each frame contains information about the destination address, the error control mechanism it follows, etc. These multiple frames are numbered so that they can be **distinguished** from each other.



Frames

► Working of Go-Back-N ARQ:

- The integer 'N' in Go Back 'N' ARQ tells us the size of the window i.e. the number of frames that are sent at once from sender to receiver. Suppose the window size 'N' is equal to 4. Then, 4 frames (frame 0, frame 1, frame 2, and frame 3) will be sent first from sender to receiver.

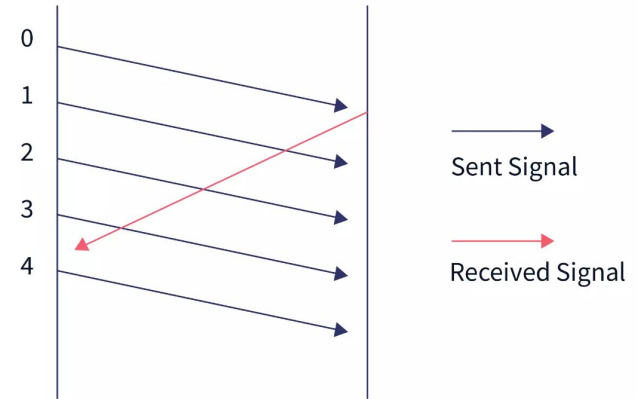


► Working of Go-Back-N ARQ:

- The receiver sends the acknowledgment of **frame 0**. Then the sliding window moves by one and **frame 4** is sent.



Window 2

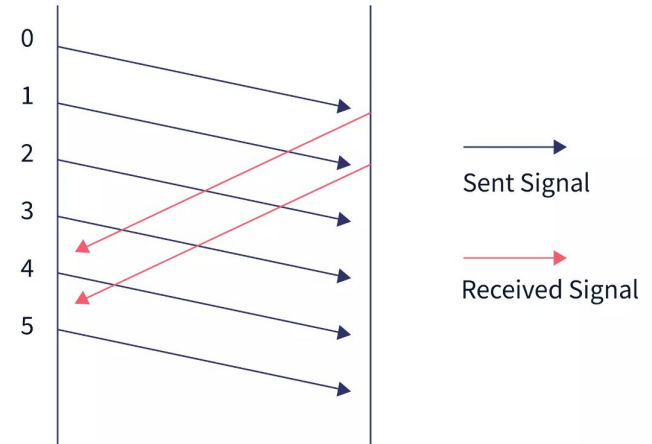


► Working of Go-Back-N ARQ:

- The receiver sends the acknowledgment of **frame 1**. Then the sliding window moves by one and **frame 5** is sent.

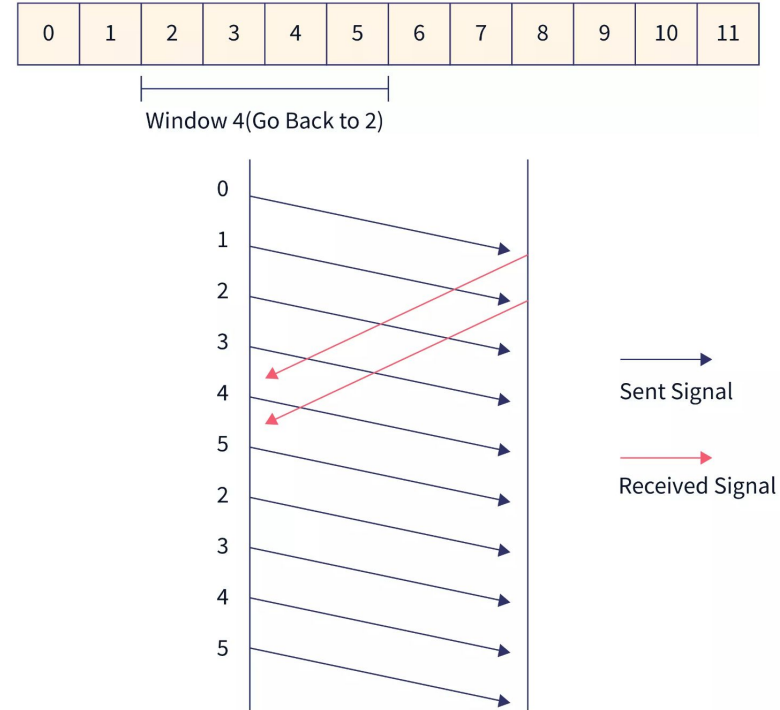


Window 3

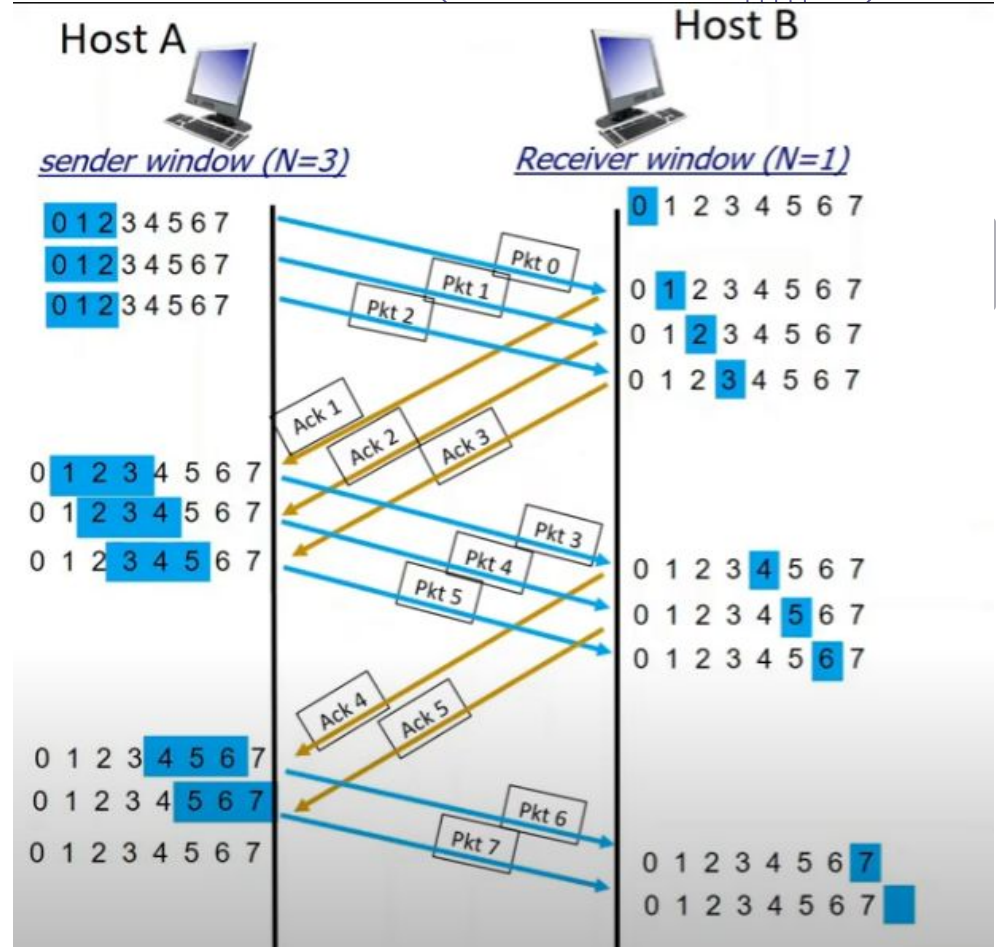


► Working of Go-Back-N ARQ:

- The sender waits for the acknowledgment for a **fixed amount of time**. If the sender does not get the acknowledgment for a frame in time, it considers the frame to be **corrupted**. Then the sliding window moves to the start of the corrupted frame and all the frames in the window are **retransmitted**.
- For example, if the sender does not receive the acknowledgment for **frame 2**, it retransmits all the frames in the windows i.e. **frames [2, 3, 4, 5]**.

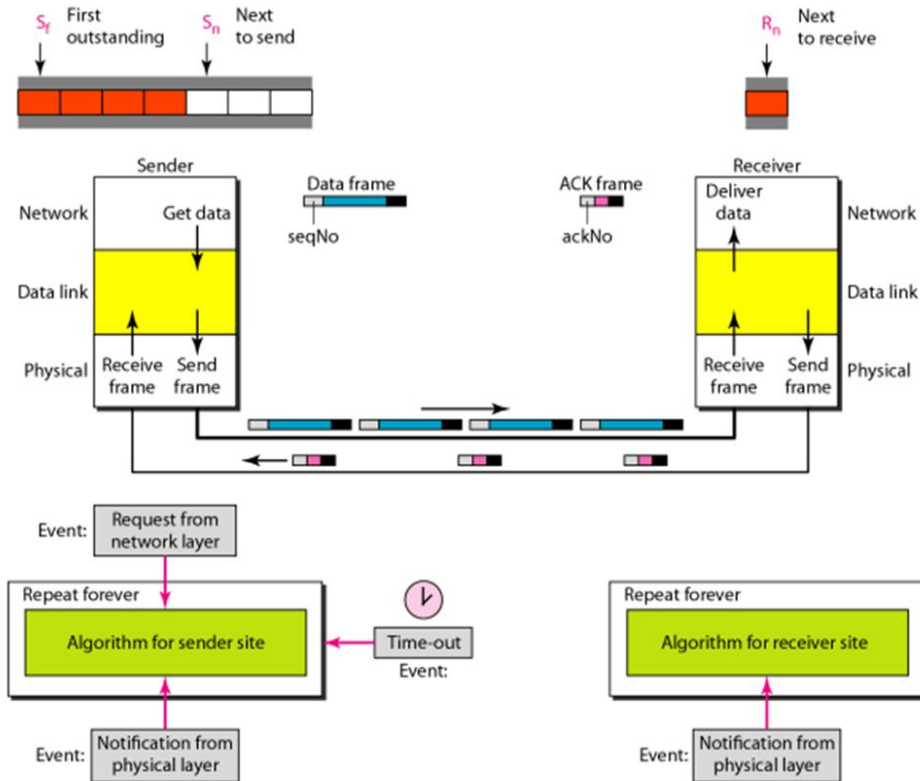


► Go-Back-3 ARQ:



► Design of Go-Back-N ARQ:

Design of Go-Back-N ARQ



► **QUESTION:** Station A needs to send a message consisting of **9 packets** to station B using a sliding window (**window size 3**) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If **every 5th packet** that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

- 1. 12
- 2. 14
- 3. 16
- 4. 18

► Selective Repeat ARQ:

- The selective repeat ARQ is one of the **Sliding Window Protocol** strategies that is used where **reliable in-order delivery** of the data packets is required.
- In the selective repeat Protocol, the sequence numbers are **modulo 2^m** , where m is the size of the **sequence number field in bits**.
- In this protocol, remains the same for the sender and the receiver side, i.e., the Sender window's N = Receiver window's N.
- If the sender does not receive the acknowledgment for a frame within a certain time, that frame is **retransmitted without** influencing other frames.
- This protocol also shares a **NACK** (Negative Acknowledgement) for either lost or damaged frames for that frame to be retransmitted.

► Working of Selective Repeat ARQ:

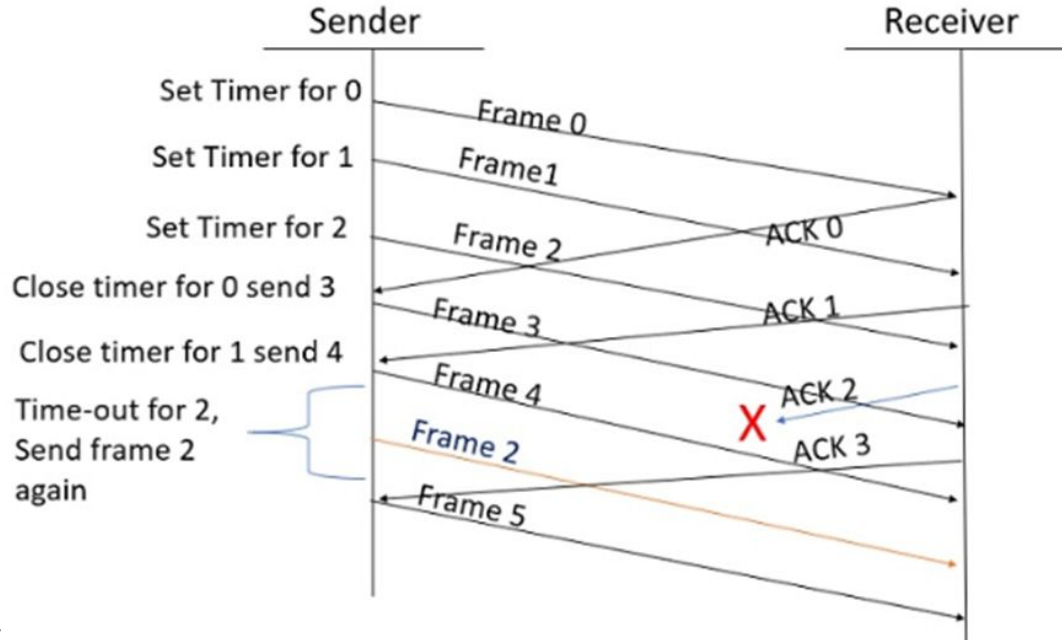
- Initially, the sender sends **several frames** according to the **window size**.
- The receiver on the other end receives the frames and sends the **ACK** for the correct frames and **NACK** for lost or damaged frames.
- The sender **re-transmits** the frames for which the NACK was sent by the receiver.
- After **receiving all** the frames, the receiver **sorts** the frame according to the **sequence number** for further usage.

► Working of Selective Repeat ARQ:

The image below shows the transmission of frames. Let us suppose that the window size of the sender and the receiver is 3.

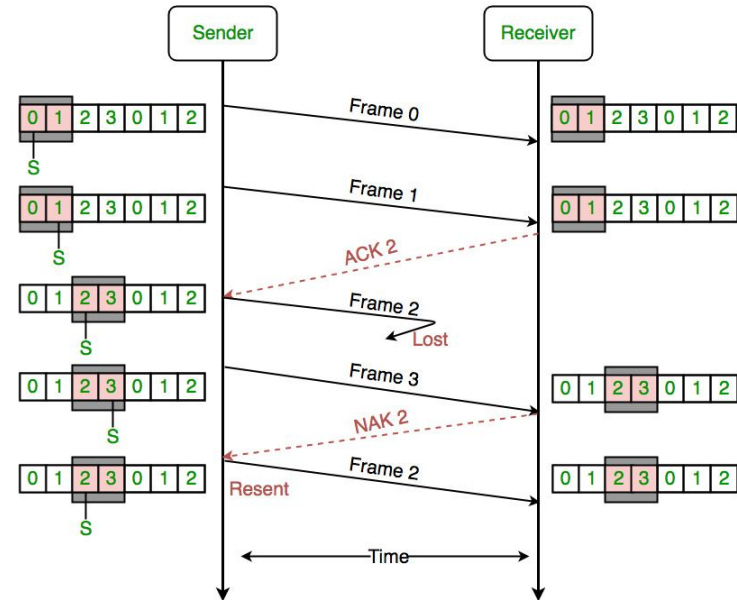


Window Size 3



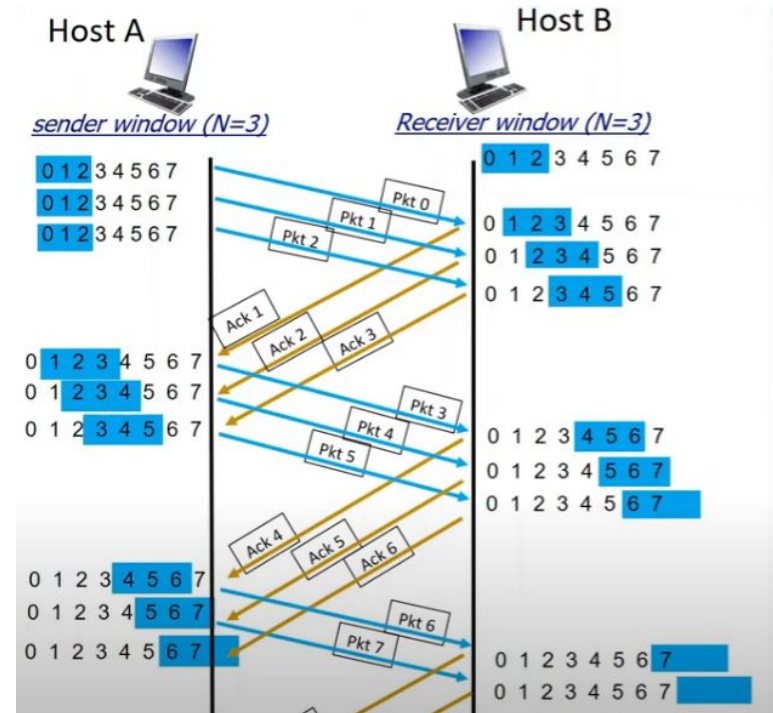
► Working of Selective Repeat ARQ:

The image below shows the transmission of frames. Let us suppose that the window size of the sender and the receiver is 2.



► Working of Selective Repeat ARQ:

The image below shows the transmission of frames. Let us suppose that the window size of the sender and the receiver is 3.

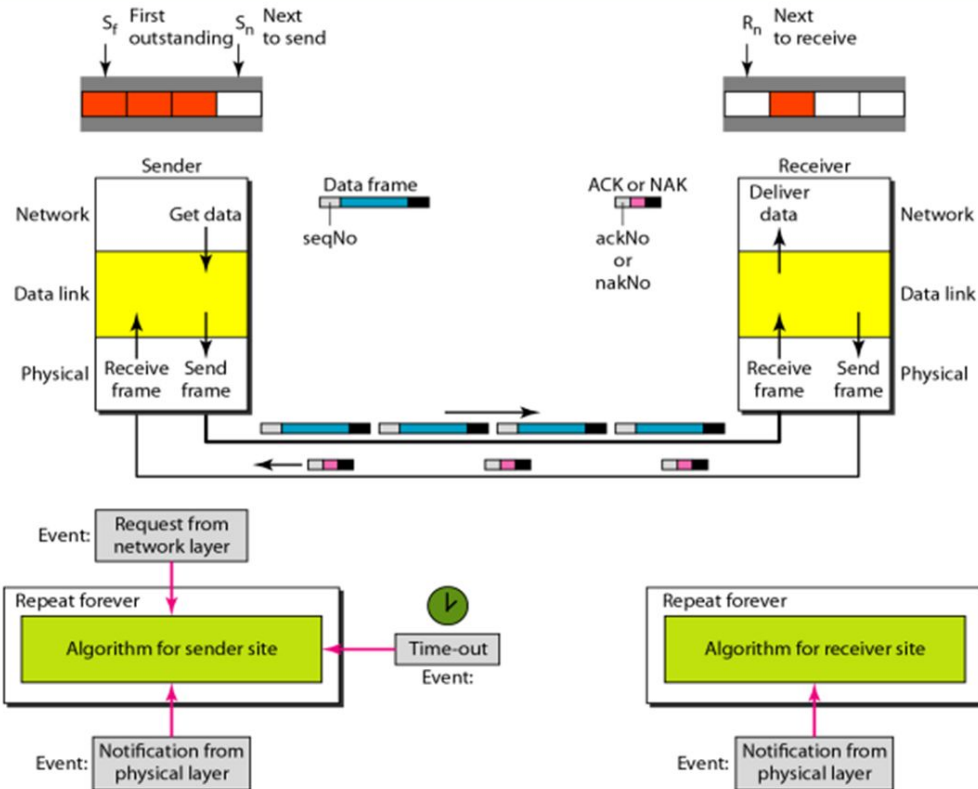


► Working of Selective Repeat ARQ:

- **Step 1** – Frame 0 sends from sender to receiver and set timer.
- **Step 2** – Without waiting for acknowledgement from the receiver another frame, Frame1 is sent by sender by setting the timer for it.
- **Step 3** – In the same way frame 2 is also sent to the receiver by setting the timer without waiting for previous acknowledgement.
- **Step 4** – Whenever sender receives the ACK0 from receiver, within the frame 0 timer then it is closed and sent to the next frame, frame 3.
- **Step 5** – Whenever the sender receives the ACK1 from the receiver, within the frame 1 timer then it is closed and sent to the next frame, frame 4.
- **Step 6** – If the sender doesn't receive the ACK2 from the receiver within the time slot, it declares timeout for frame 2 and resends the frame 2 again, because it thought the frame 2 may be lost or damaged.

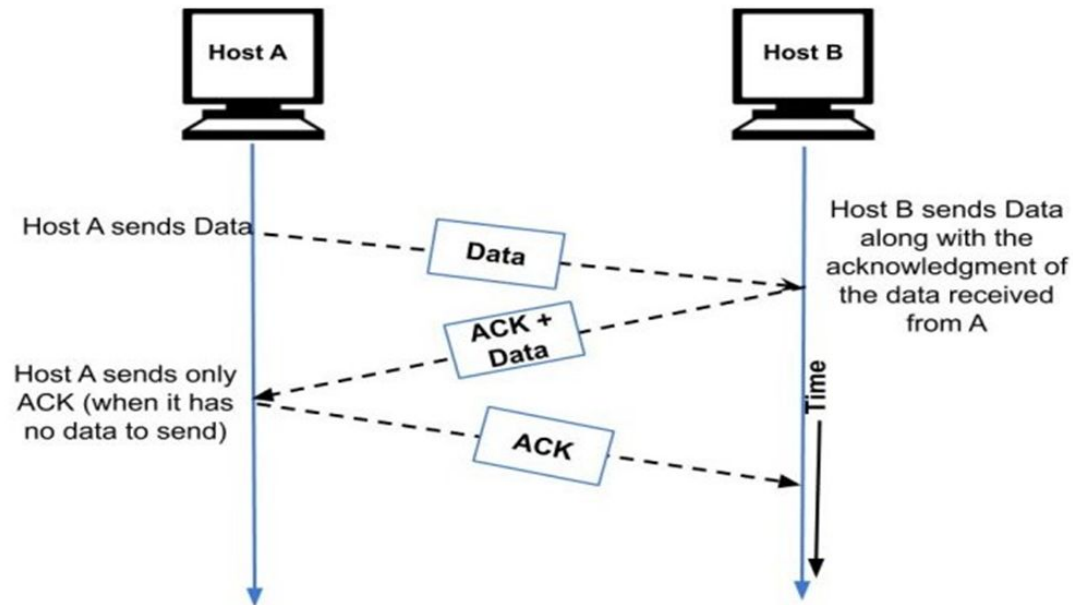
► Design of Selective Repeat ARQ:

Design of Selective Repeat ARQ



► Piggybacking:

- Piggybacking is a method of **attaching acknowledgment** to the outgoing data frame. i.e bidirectional transmission.



► Piggybacking:

Working Mechanism:

- When a data frame arrives, the receiver waits and does not send the control frame (acknowledgment) back immediately.
- The receiver waits until its network layer moves to the next data packet. Acknowledgment is associated with this outgoing data frame. Thus the acknowledgment travels along with the next data frame. This technique in which the outgoing acknowledgment is delayed temporarily is called Piggybacking.

Advantages of Piggybacking

- The major advantage of piggybacking is the better use of available channel bandwidth. This happens because an acknowledgment frame needs does not to be sent separately.
- To avoid the delay and rebroadcast of frame transmission, piggybacking uses a very short-duration timer.

Disadvantages of Piggybacking

- The disadvantage of piggybacking is the additional complexity.
- If the data link layer waits long before transmitting the acknowledgment (blocks the ACK for some time), the frame will rebroadcast.

The background features a dark blue gradient with white and light blue geometric lines resembling a circuit board. These lines include straight paths, right-angle turns, and small circles at junctions. A prominent light blue line starts from the left edge, goes down, and then right to a small circle. Other lines are scattered across the top and right sides, some ending in small horizontal bars or circles.

▶ Error Detection and Correction Techniques:

3.4

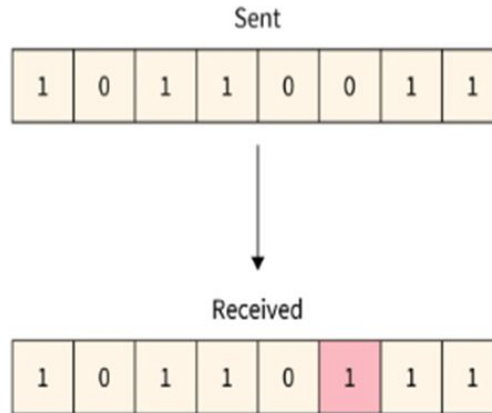
► Overview:

- When bits are transmitted over the computer network, they are **subject to** get **corrupted** due to interference and network problems.
- The corrupted bits leads to **counterfeit data** being received by the receiver and are called **errors**.
- Networks must be able to transfer data from one device to another with **complete accuracy**.
- Data can be corrupted during transmission.
- For reliable communication, errors must be **detected and corrected**.

► Types of Errors:

Single Bit Error

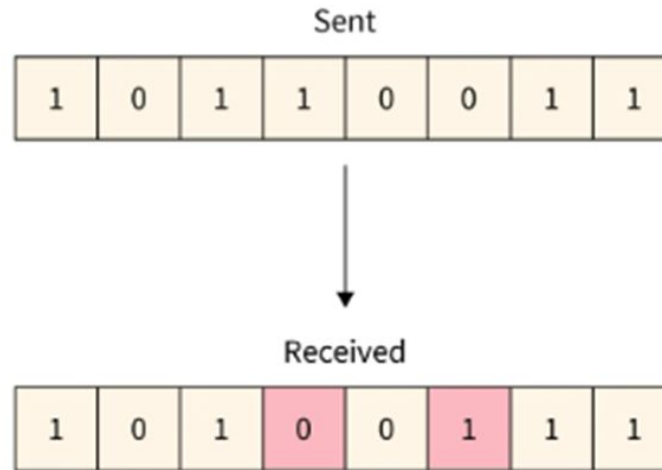
In single bit error, **only one bit** of the frame received is **corrupt**, and the corrupted bit can be located anywhere in the frame.



► Types of Errors:

Multiple Bit Error

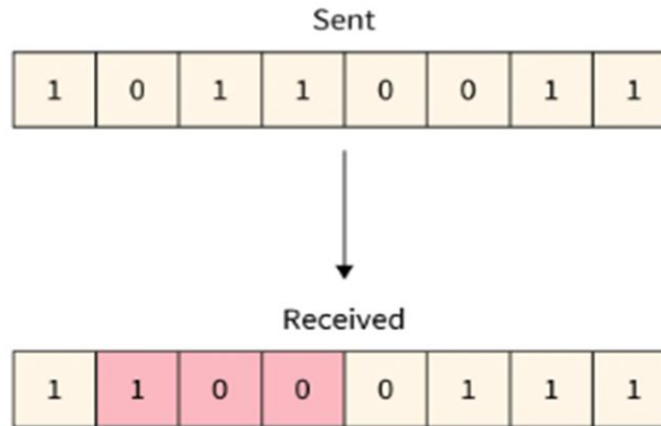
In multiple bit error, **More than one bit** received in the frame is found to be corrupted.



► Types of Errors:

Burst Error:

In burst error, more than one **consecutive bit** is corrupted in the received frame.



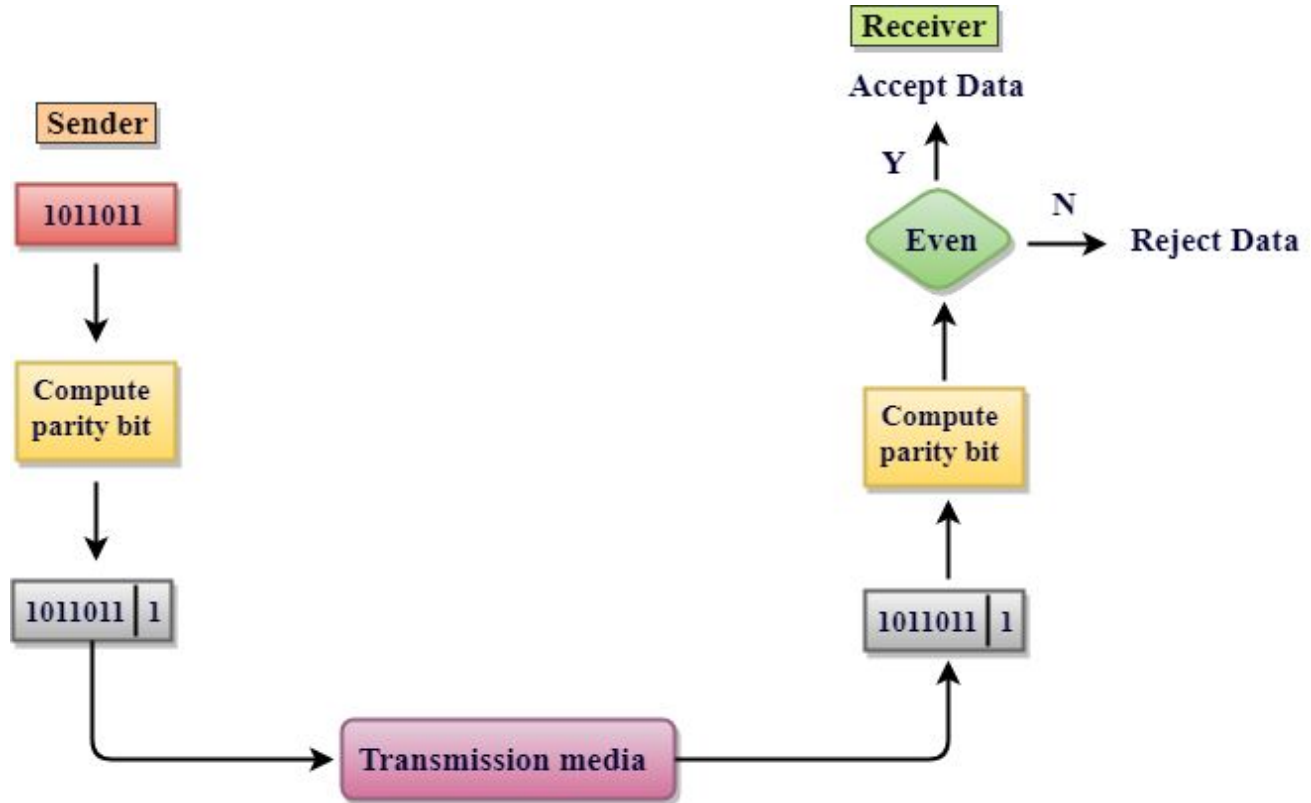
► Error Detection Mechanism:

- Process of detecting error is called **error detection**.
- **Types** of error detection mechanisms:
 - a. Simple Parity Check
 - b. Two-dimensional Parity check
 - c. Checksum
 - d. CRC(Cyclic Redundancy Check)

► Simple Parity Check:

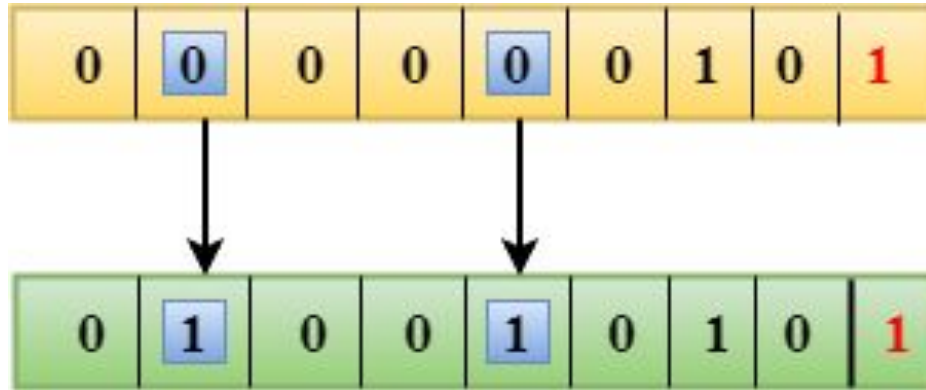
- ❖ Data sent from the sender **undergoes** parity check :
 - **1 is added** as a **parity bit** to the data block if the data block has an **odd number of 1's**.
 - **0 is added** as a parity bit to the data block if the data block has an **even number of 1's**.
- ❖ This procedure is used for making the **number of 1's even**. This is commonly known as **even parity checking**.

► Simple Parity Check:



► Disadvantage of Simple Parity Check:

- It can only detect **single-bit errors** which are very rare.
- If **two bits** are interchanged, then it **cannot** detect the errors.



► Two-Dimensional Parity Check :

- Performance can be **improved** by using **Two-Dimensional** Parity Check which **organizes** the data in the form of a table.
- **Parity check bits** are calculated for **each row**, which is equivalent to a simple parity check bit.
- Parity check bits are **also** calculated for **all columns** then both are sent along with the data.
- At the side of receiving end these data are **compared with** the parity bits calculated on the received data.

Original data 10110011 :: 10101011 :: 01011010 :: 11010101

1 0 1 1 0 0 1 1	1	Row parities
1 0 1 0 1 0 1 1	1	
0 1 0 1 1 0 1 0	0	
1 1 0 1 0 1 0 1	1	
Column parities		
1 0 0 1 0 1 1 1	1	

101100111 :: 101010111 :: 010110100 :: 110101011 :: 100101111

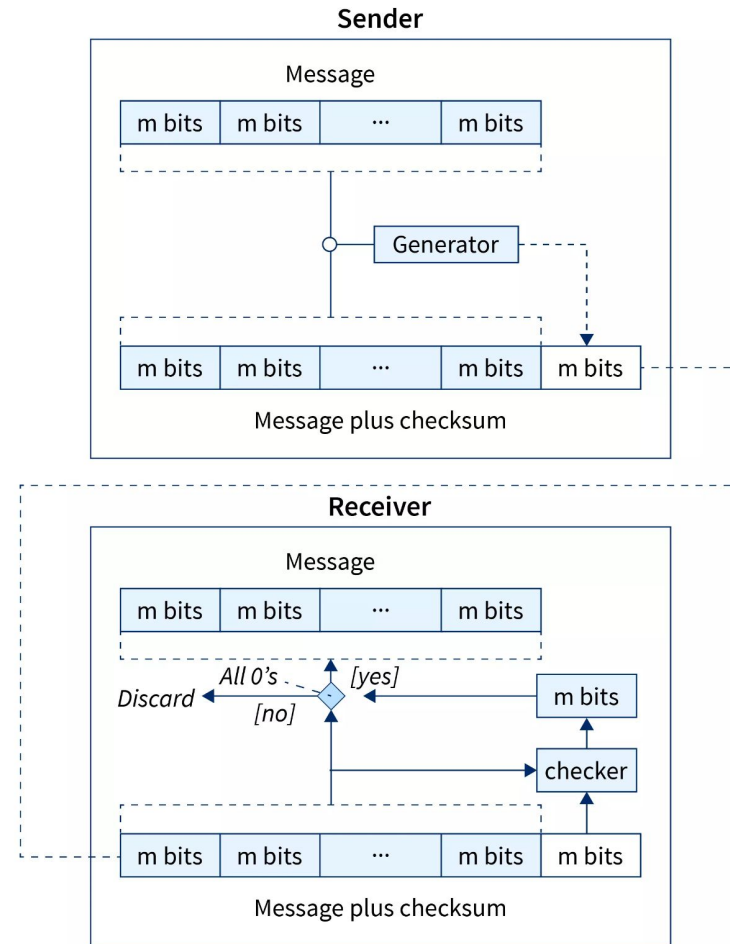
Data to be sent

► Checksum:

- Checksum is an error detection which detects the error by **dividing** the data into **segments of equal size** and then use **1's complement** to find the **sum of the segments** and then the sum is **transmitted** with the data to the receiver.
- The **same process** is done by the receiver and at the receiver side.
- All **zeros in the sum** indicates the **correctness** of the data.
- **Checksum** is the **redundant bits** that are **attached** with actual data.
- Checksum Method is an error detection algorithm that can be applied to **any length message**.

► Working Mechanism

■



► Working Mechanism (At Sender's Side):

- Step 1 : First of all, **break** the given data into "k" an **equal number** of blocks, i.e., "N" bits in all the "K" blocks of the message.
- Step 2 : Perform **addition** of all the "k" divisions
- Step 3 : If there is a **carry bit**, add it.
- Step 4 : Now find the **1's complement** of the sum. For finding 1's complement of any binary number, just replace every zero with one and replace everyone with zero(e.g., 10001 - 1's Complement 01110) This complemented sum is known as **Checksum**.
- Step 5 : Checksum is **appended** to the message to be sent to the receiver.

► Working Mechanism (At Receiver's Side):

- Step 1 : Perform **addition** on all "K" data blocks
- Step 2 : If there is a carry bit, add it.
- Step 3 : Find the **1's complement** of the sum.
- Step 4 : If we got a result that contains **only 0**, then ACCEPT the data, otherwise, REJECT the data.

► Working Mechanism:

Original Data

10011001	11100010	00100100	10000100
1	2	3	4

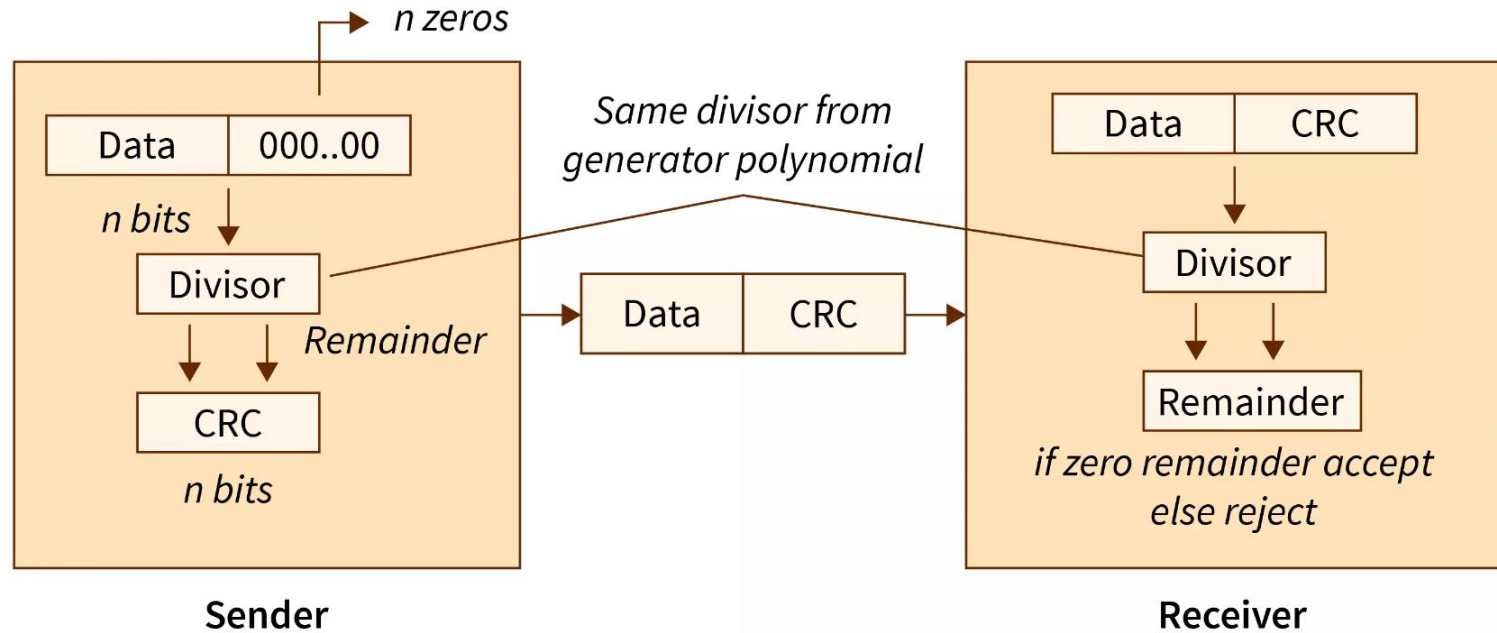
k=4, m=8

Sender		Receiver	
1	10011001	1	10011001
2	11100010	2	11100010
	<u>101111011</u>		<u>101111011</u>
	└─→ 1		└─→ 1
	01111100		01111100
3	00100100	3	00100100
	<u>10100000</u>		<u>10100000</u>
4	10000100	4	10000100
	<u>100100100</u>		<u>100100100</u>
	└─→ 1		└─→ 1
Sum:	00100101		00100101
Checksum:	11011010		11011010
		Sum:	11111111
		Complement:	00000000
		Conclusion : Accept Data	

► Cyclic Redundancy Check (CRC):

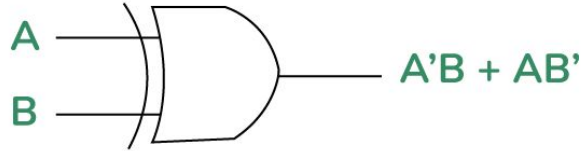
- The checksum scheme uses the addition method but CRC uses **binary division**.
- A **bit sequence** commonly known as cyclic redundancy check is **added** to the **end of the bits** in CRC. This is done so that the **resulting data unit** will be **divisible** by the **second binary number** that is predetermined.
- The **receiving data units** on the receiver's side need to be divided by the **same number**.
- These data units are **accepted** and found to be correct only on the condition that the remainder of this division is **zero**. The remainder **other than zero** shows that the data is **not correct**. So, they need to be discarded.

► CRC Working Mechanism:



► CRC Working Mechanism:

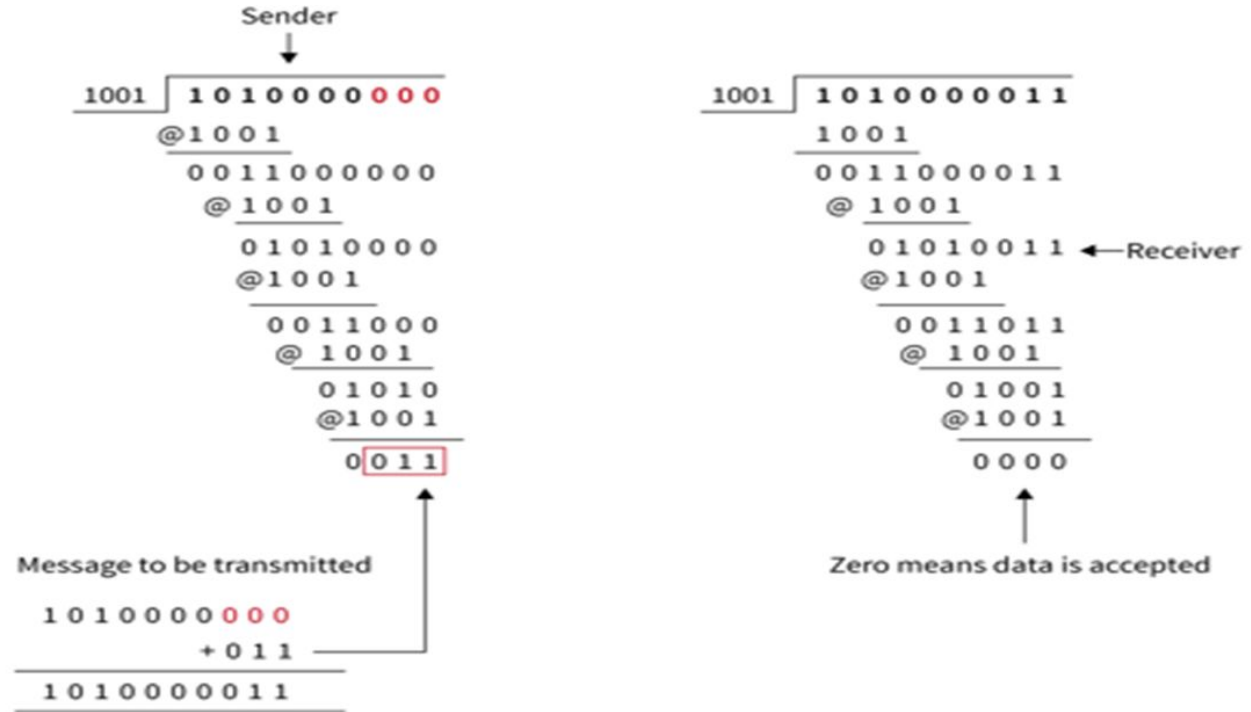
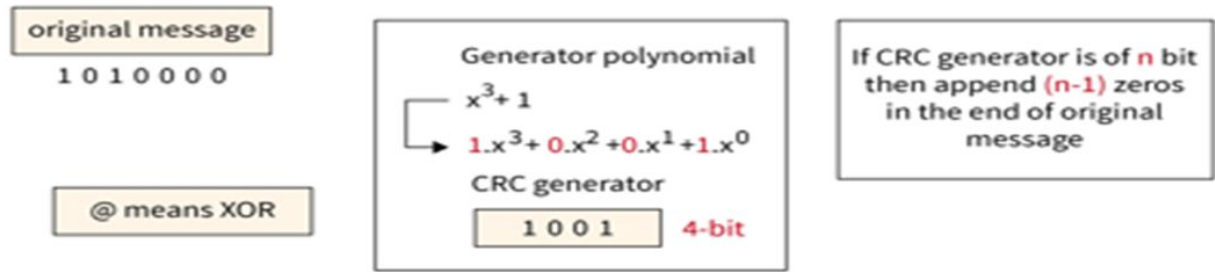
XOR Gate



Truth Table

A (Input 1)	B (Input 2)	$X = A'B + AB'$
0	0	0
0	1	1
1	0	1
1	1	0

CRC Working Mechanism:



► Q.A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

Q.A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

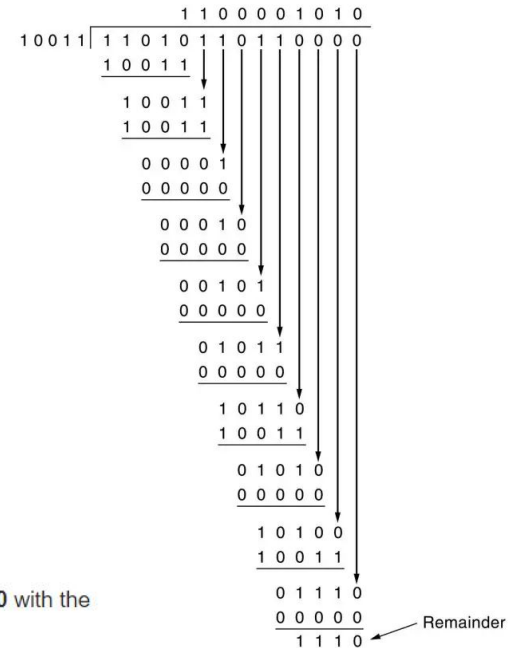
Solution-

- The generator polynomial $G(x) = x^4 + x + 1$ is encoded as 10011.
- Clearly, the generator polynomial consists of 5 bits.
- So, a string of 4 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 11010110110000.

From here, CRC = 1110.

Now,

- The code word to be transmitted is obtained by replacing the last 4 zeroes of 11010110110000 with the CRC.
- Thus, the code word transmitted to the receiver = 11010110111110.



► Q.A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1 .

1. What is the actual bit string transmitted?
2. Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?

Solution:

Part-01:

- The generator polynomial $G(x) = x^3 + 1$ is encoded as 1001.
- Clearly, the generator polynomial consists of 4 bits.
- So, a string of 3 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 10011101**000**.

From here, CRC = 100.

Now,

- The code word to be transmitted is obtained by replacing the last 3 zeroes of 10011101**000** with the CRC.
- Thus, the code word transmitted to the receiver = 10011101**100**.

$$\begin{array}{r} 100011100 \\ 1001 \overline{) 10011101000} \\ \underline{1001} \\ 00001 \\ \underline{0000} \\ 00011 \\ \underline{0000} \\ 00110 \\ \underline{0000} \\ 01101 \\ \underline{1001} \\ 01000 \\ \underline{1001} \\ 00010 \\ \underline{0000} \\ 00100 \\ \underline{0000} \\ 0100 \end{array} \leftarrow \text{CRC}$$

Solution:

Part-02:

According to the question,

- Third bit from the left gets inverted during transmission.
- So, the bit stream received by the receiver = 10111101100.

Now,

- Receiver receives the bit stream = 10111101100.
- Receiver performs the binary division with the same generator polynomial as-

From here,

- The remainder obtained on division is a non-zero value.
- This indicates to the receiver that an error occurred in the data during the transmission.
- Therefore, receiver rejects the data and asks the sender for retransmission.

$$\begin{array}{r} 10101000 \\ 1001 \overline{) 10111101100} \\ \underline{1001} \\ 00101 \\ \underline{0000} \\ 01011 \\ \underline{1001} \\ 00100 \\ \underline{0000} \\ 01001 \\ \underline{1001} \\ 00001 \\ \underline{0000} \\ 00010 \\ \underline{0000} \\ 00100 \\ \underline{0000} \\ 0100 \end{array} \leftarrow \text{Remainder}$$

► Q. Consider a polynomial x^3+x^2+1 for the divisor (CRC polynomial) and a polynomial $x^6+x^4+x^2+1$ for the data to be divided for parity checking. Perform CRC on both ends.

► Error Correction:

- Error correction can be done in two ways:
 - **Backward Error Correction:** When the receiver detects an error in the data received, it **requests back** the sender to retransmit the data unit.
 - **Forward Error Correction:** When the receiver detects some error in the data received, it **executes error correcting code**, which helps it to auto-recover and to correct some kinds of errors.

► Error Correction Technique:

- We can **detect** the error using a **single additional bit** but we cannot use this bit for the **correction** purpose.
- It is important to know the exact **location of the error** if we want to correct that error. For example, for finding out the single-bit error, the error detection code checks out that the error is actually in **one of the seven bits**.
- Let **d** represents the number of **data bits** and **r** represents the number of **redundant bits**. The formula below is used for **finding** the **r** number of redundant bits :

$$2^r \geq D + r + 1$$

- The above formula is used to find out the value of **r**. **For example**, suppose 4 will be the value of **d**, then 3 will be the only and smallest value that satisfies this particular relation.

► Hamming Code:

- A hamming code is a technique developed by **R.W Hamming** for finding out the position of the error bit.
- This Hamming code is based on the **relationship between the redundant bits and data units** and its main advantage is that it can be applied to data units of any length.
- It is used to **detect and correct** single bit error.
- All bits position that are power of 2 are marked as **parity bits** and remaining bits are **data bits**.
- Commonly use hamming code is 7 bit hamming code.
 - **Parity bits:** The parity bits are the special type of bits that are added to the original data of binary bits to make the total 1s either **even or odd**.
 - **Even parity:** For checking the even parity, the following concept is used: The value of the even parity bit will be 0 if the total **occurrence of 1s is even** and the value of the **parity bit can be 1** if the total occurrence of 1s is odd.
 - **Odd Parity:** For checking the even parity, the following concept is used: The value of the parity bit will be 1 if the total **occurrence of 1s is even** and the value of the **parity bit can be 0** if the total occurrence of 1s is odd.

► Hamming Code:

Algorithm of Hamming Code:

1. Add the information in 'D' bits to the redundant bits 'R' to make data as D+R.
2. A decimal value will be assigned by the position of each (D+R) digit.
3. In positions $1, 2 \dots 2^k$, the 'R' bits will be placed.
4. The parity bits are again calculated on the receiver's end. The position of an error defines the parity bit's decimal value.

► Hamming Code Example:

Example:

1010 is supposed to be the original data that needs to be transferred.

'D' total number of data bits = 4

Redundant bits number:

$$2^r \geq D + r + 1$$

$$2^r \geq 4 + r + 1$$

So, the above-given relation is completely satisfied when 3 will be the value of r.

► Hamming Code:

Determining the position of the redundant bits:

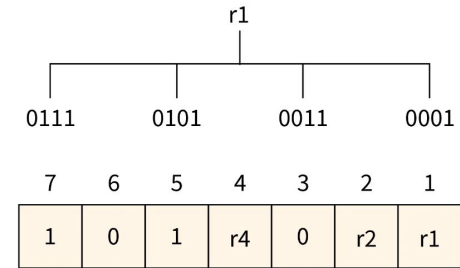
- 3 is the number of redundant bits. r1, r2, and r4 are used to show these three bits. We can find out the position of these bits for the raised power of 2. So according to this, their positions will be $1, 2^1, 2^0$.
 - r1 position is 1
 - r2 position is 2
 - r4 position is 4
- Refer to the below image for the data with redundant bits

7	6	5	4	3	2	1
1	0	1	r4	0	r2	r1

► Hamming Code Example:

Determining the parity bits > Determining the r1 bit:

- The r1 bit value is calculated based on a parity check performed on the bits available at the position whose binary conversion contains 1 at the first position.
- Refer to the below image for the finding r1 bit value:

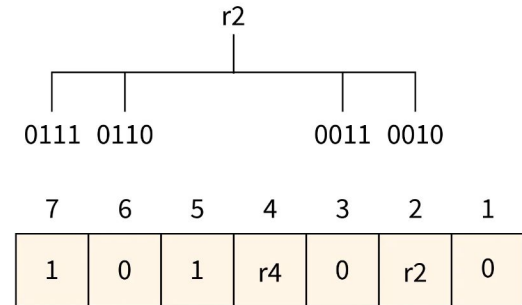


- From the above figure, we can find the position having 1 as its first bit in binary representation is 1,3,5,6. Now the even-parity check method is applied to these bit positions. The number of 1's at these bit positions 2 which is an even number so the 0 is the value of the r1 bit.

► Hamming Code Example:

Determining the parity bits > Determining the r2 bit:

- The r2 bit value is calculated based on a parity check performed on the bits available at the position whose binary conversion contains 1 at the second position.
- Refer to the below image for the finding r2 bit value

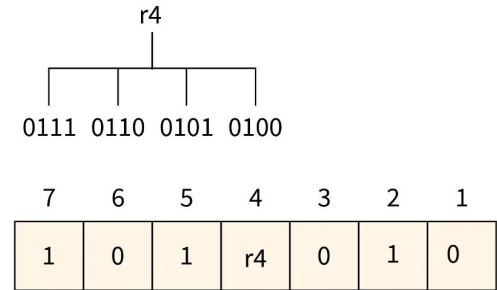


- From the above figure, we can find the position having 1 as its second bit in binary representation is 2, 3, 6, 7. Now the even-parity check method is applied to these bit positions. The number of 1's at these bit positions is 1 which is an odd number so 1 is the value of the r2 bit.

► Hamming Code Example:

Determining the parity bits > Determining the r4 bit:

- The r4 bit value is calculated based on a parity check performed on the bits available at the position whose binary conversion contains 1 at the third position.
- Refer to the below image for the finding r4 bit value



- From the above figure, we can find the position having 1 as its third bit in binary representation is 4, 5, 6, 7. Now the even-parity check method is applied to these bit positions. The number of 1's at these bit positions 2 which is an even number so the 0 is the value of the r4 bit.

► Hamming Code Example:

Data to be transmitted:

7	6	5	4	3	2	1
1	0	1	0	0	1	0

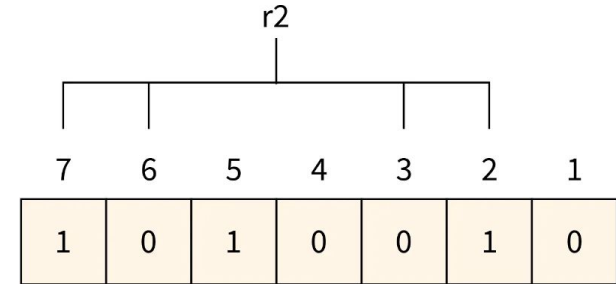
- Suppose when the data is received at the receiver end the value of the 4th bit is changed to 1 from 0. Now parity bits are calculated again to find the position of the error.
- R1 bit 1, 3, 5 and 7 are the positions of the bits for the r1 bit.

r1						
7	6	5	4	3	2	1
1	0	1	1	0	1	0

► Hamming Code Example:

Parity Bit Recalculation:

- We can find from the above figure that the r1 binary representation is 1100. After applying the even-parity check method on the bits appearing in the r4 bits we get an even number of totals 1's. So 0 is the value for the r1.
- R2 bit 2, 3, 6, and 7 are the positions of the bits for the r2

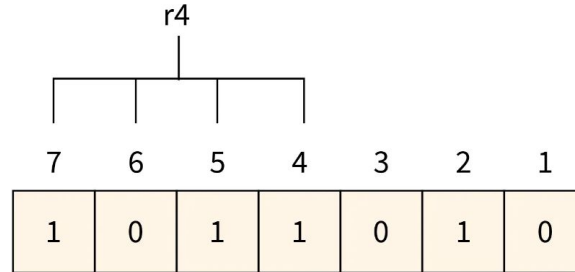


- We can find from the above figure that the r2 binary representation is 1001. After applying the even-parity check method on the bits appearing in the r4 bits we get an even number of totals 1's. So 0 is the value for the r2.

► Hamming Code Example:

Parity Bit Recalculation:

- R4 bit 4, 5, 6 and 7 are the positions of the bits for the r4 bit.



- We can find from the above figure that the r4 binary representation is 1011. After applying the even-parity check method on the bits appearing in the r4 bits we get an odd number of totals 1's. So 1 is the value for the r4.
- Redundant bits r4 r2 r1 binary representation is 100 which is represented as 4 in decimal value. So there is an error in the data at the 4th position and the value of the bit corresponding to the 4th position must be changed from 1 to 0 for error correction.

► Hamming Code:

Hamming Code:



None of
the bits are
corrupted.

Any one of the
(p+m) bits can
be corrupted.

$$p\text{-bits} \rightarrow 2^p \geq (p + m) + 1$$

Q: If data size is 4 bits, how many parity bits are needed?

Sol. $m = 4$

$p = 3$

► Hamming Code:

Hamming Code:

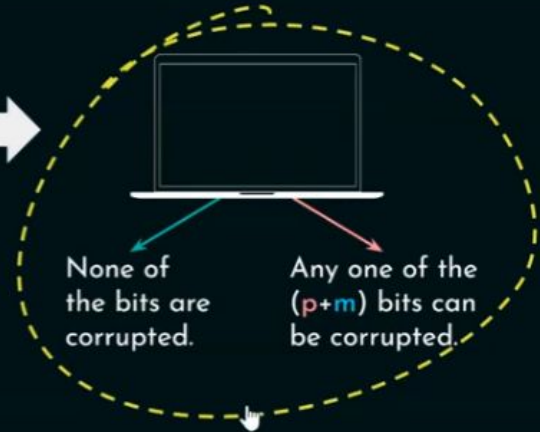
$m_1 m_2 m_3 m_4$

$m = 0 \ 1 \ 1 \ 0$

$p = p_1 p_2 p_3$



$p_1 \ p_2 \ m_1 \ p_3 \ m_2 \ m_3 \ m_4$
1 2 3 4 5 6 7

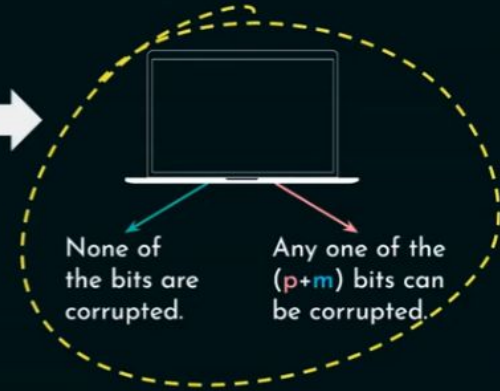
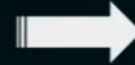


► Hamming Code:

Hamming Code:

p_1 p_2 m_1 p_3 m_2 m_3 m_4
1 2 3 4 5 6 7

c_1	c_2	c_3	
0	0	0	---
0	0	1	---
0	1	0	---
0	1	1	---
1	0	0	---
1	0	1	---
1	1	0	---
1	1	1	---



$p_1 \rightarrow c_3$ (1, 3, 5, 7)

$p_2 \rightarrow c_2$ (2, 3, 6, 7)

$p_3 \rightarrow c_1$ (4, 5, 6, 7)

► Hamming Code:

Hamming Code:

$m_1 m_2 m_3 m_4$

$m = 0 \ 1 \ 1 \ 0$

$p_1 (1, 3, 5, 7)$

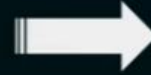
$p_2 (2, 3, 6, 7)$

$p_3 (4, 5, 6, 7)$

$p_1 \ p_2 \ m_1 \ p_3 \ m_2 \ m_3 \ m_4$

1 2 3 4 5 6 7

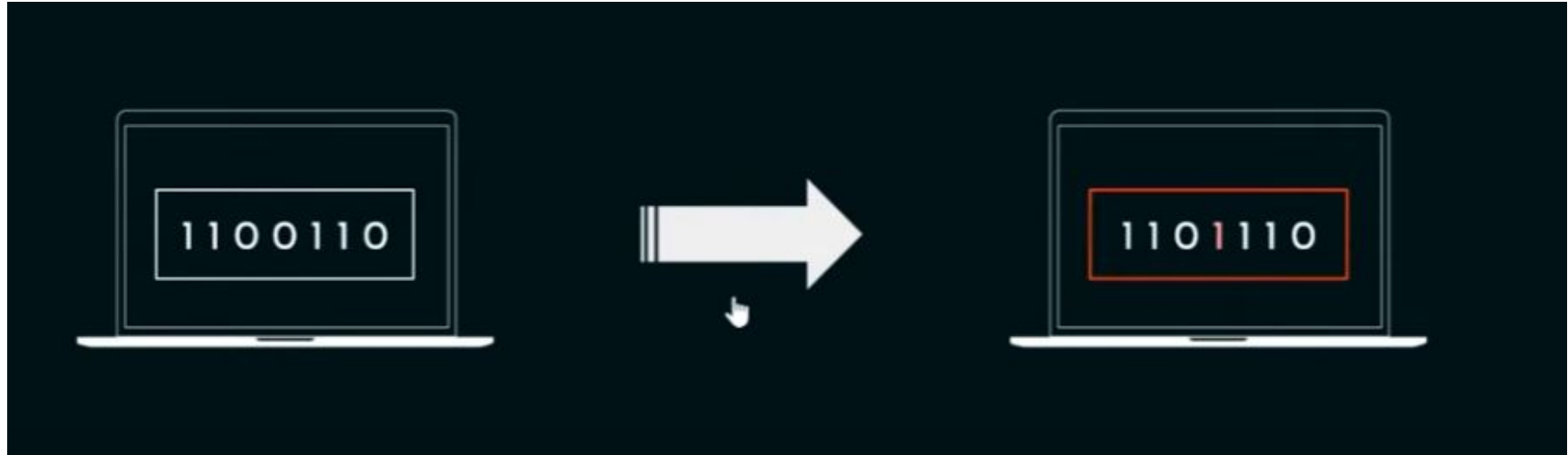
1 1 0 0 1 1 0



None of
the bits are
corrupted.

Any one of the
($p+m$) bits can
be corrupted.

► Hamming Code:



► Hamming Code:

p_1	p_2	m_1	p_3	m_2	m_3	m_4
1	2	3	4	5	6	7
1	1	0	0	1	1	0

c_1	(4, 5, 6, 7)
c_2	(2, 3, 6, 7)
c_3	(1, 3, 5, 7)

c_1	→	1
c_2	→	0
c_3	→	0

p_1	p_2	m_1	p_3	m_2	m_3	m_4
1	2	3	4	5	6	7
1	1	0	1	1	1	0

► Hamming Code:

c_1	c_2	c_3	
0	0	0	---> ✓
0	0	1	---> 1
0	1	0	---> 2
0	1	1	---> 3
1	0	0	---> 4
1	0	1	---> 5
1	1	0	---> 6
1	1	1	---> 7

$c_1 \longrightarrow 1$
 $c_2 \longrightarrow 0$
 $c_3 \longrightarrow 0$

p_1	p_2	m_1	p_3	m_2	m_3	m_4
1	2	3	4	5	6	7
1	1	0	1	1	1	0

c_1 (4, 5, 6, 7)
 c_2 (2, 3, 6, 7)
 c_3 (1, 3, 5, 7)

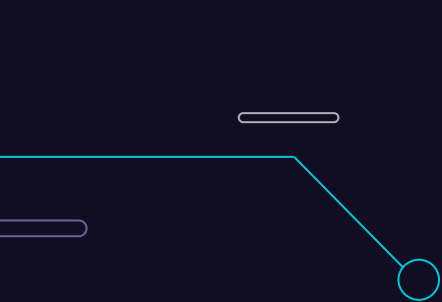
► Hamming Code:

Advantages of Hamming code:

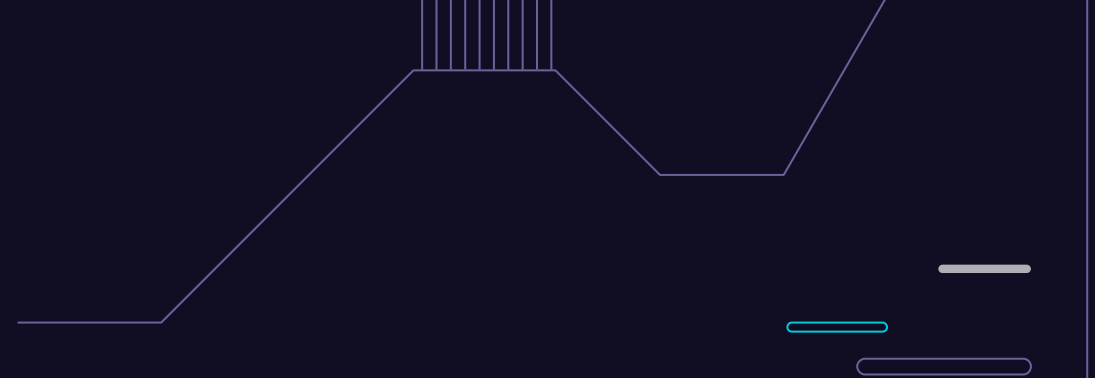
- Hamming code method is effective on networks where the data streams are given for the **single-bit errors**.
- Hamming code not only provides the detection of a bit error but also helps you to indent bit containing error so that **it can be corrected**.
- The **ease of use** of hamming codes makes it best them suitable for use in computer memory and single-error correction.

Disadvantages of Hamming code:

- Single-bit error detection and correction code. However, if **multiple bits** are founded error, then the outcome may result in another bit which should be correct to be changed. This can cause the data to be **further errored**.
- Hamming code algorithm can solve only **single bits issues**.



Channel Allocation Techniques (ALOHA, Slotted ALOHA)



3.5

► Overview:

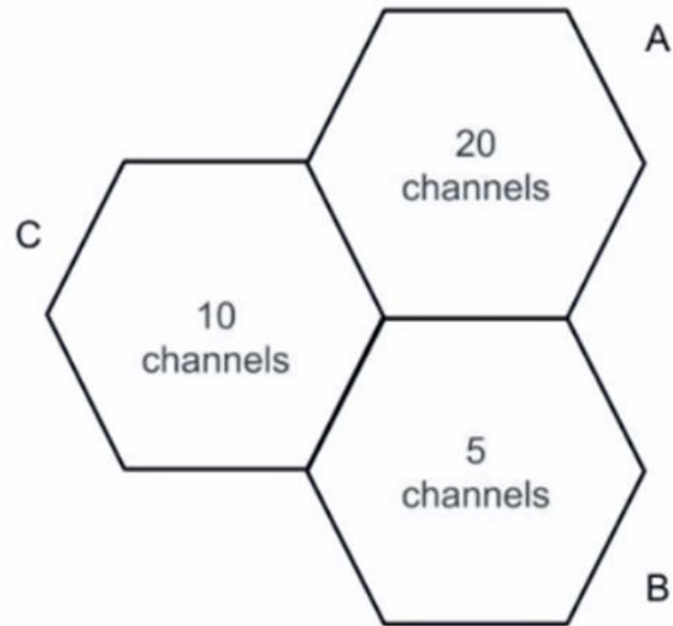
- Channel allocation is the process in which a **single channel** is divided and **allotted to multiple user** in order to perform user specific tasks.
- If there are **N number** of user and channel is divided into **N equal sized** sub channel.
- Each user is **assigned** one channel.
- There are different channel allocation techniques:
 - Fixed Channel allocation
 - Dynamic channel allocation
 - Hybrid channel allocation
 - Borrowing Channel allocation

► Static Channel Allocation (SCA):

- ❖ In SCA, each user or device is assigned a **fixed or pre-determined** frequency channel.
- ❖ Once channels are assigned, they remain **unchanged** until **manually** reconfigured i.e. no real time adjustments.
- ❖ If there are **N users**, the bandwidth is divided into **N equal-sized** portions, with each user being assigned **one portion**.
- ❖ Since each user has a **private frequency band**, there is no interference between users.
- ❖ This scheme is also referred to as **fixed channel allocation** or fixed channel assignment.
- ❖ The **FDM and TDM** are examples of static channel allocation.

► Static Channel Allocation (SCA):

- In figure below,
 - Cell A: assigned 20 channel
 - Cell B: assigned 10 channels
 - Cell C: assigned 5 channel



► Dynamic Channel Allocation (DCA):

- No permanent channel allocation is required.
- Assign channel based on traffic, if traffic is increased ,more channel is assigned else less channel is assigned.
- Channel are assigned to a cell only when it is required.
- This allocation scheme optimises bandwidth usage and results is faster transmissions.
- Dynamic channel allocation is further divided into centralized and distributed allocation.

► Dynamic Channel Allocation (DCA):

- **Initial state:** A,B,C channel are available and 5 cell are available.
- User **call or requests** for channel allocation
- Dynamic allocation process **examine the demand** and allocates channels accordingly.
- Let's say the algorithm assigns the following channels to users or cells based on demand:
 - User 1: Channel A
 - User 2: Channel B
 - User 3: Channel C
 - User 4: Channel A
 - User 5: Channel B

► Dynamic Channel Allocation: Possible

Assumptions:

- Station Model:

- This assumption considers a network scenario where there are **multiple stations**, and each station **operates independently**, producing its own frames of data.
- N stations in the context means there are 'N' independent network stations.

- Single Channel Assumption:

- This assumption implies that there is **only one communication channel** available for all stations. All stations are considered **equivalent** and can **both** send and receive on this **common channel**.
- In other words, there is a **shared medium (channel)** through which all communication takes place.

► Dynamic Channel Allocation: Possible Assumptions:

- **Collision Assumption:**

- Collision occurs when **two or more** stations attempt to transmit data at the **same time**, resulting in their signals overlapping.
- Collisions are treated as **errors**, and when a collision is detected, the frames involved in the collision need to be **retransmitted**.
- This assumption implies that the network protocol must have **mechanisms** in place to **detect collisions** and handle them appropriately.

- **Time Division:**

- Time in the network can be **divided into slots** or can be continuous.
- **Slotted time division** involves dividing time into **discrete slots**, and each station gets a turn to transmit during its **assigned slot**.
- **Continuous time division** may involve stations accessing the channel whenever they have data to transmit, **without predefined slots**.

► Dynamic Channel Allocation: Possible Assumptions:

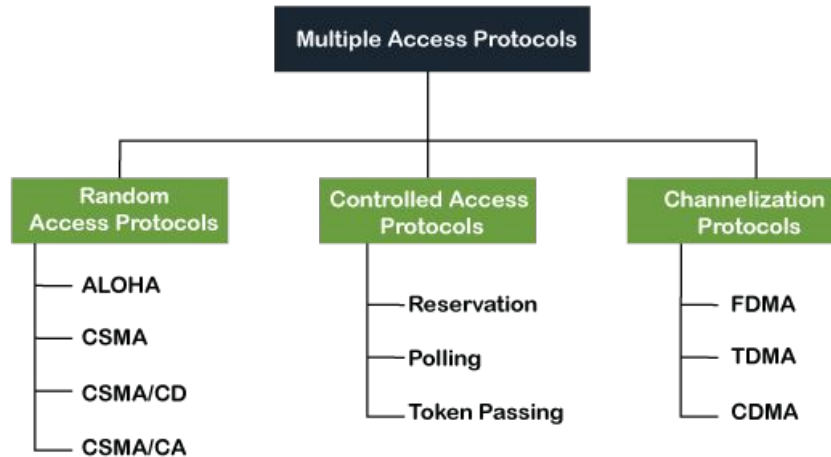
- **Carrier Sense or No Carrier Sense:**
 - Carrier sensing is the **ability of a station to detect** whether the channel is busy (someone else is transmitting) before attempting to send data.
 - If **carrier sense** is **employed**, stations will check if the **channel is clear** before initiating a transmission. If the **channel is busy**, they may delay their transmission.
 - **No carrier sense** implies that stations **do not check** the channel before transmitting, and **collisions may occur** more frequently.

► SCA vs DCA:

Parameter	Static Channel Allocation	Dynamic Channel Allocation
Nature of Allocation	Fixed assignment of channels to devices.	Channels are dynamically assigned based on real-time conditions.
Efficiency	May lead to underutilization of channels.	Generally more efficient as channels can be dynamically reassigned.
Adaptability	Limited adaptability to changing traffic patterns.	Adapts well to dynamic changes in network conditions and traffic.
Interference Management	Susceptible to interference, with no immediate remedy.	Can dynamically avoid or mitigate interference by changing channels.
Complexity	Simple to implement and manage.	More complex due to the need for dynamic allocation algorithms.
Resource Utilization	May result in inefficient use of available spectrum.	Optimizes the use of available channels, reducing waste of resources.
Flexibility	Limited flexibility to respond to real-time changes.	Offers high flexibility to adapt to changing communication patterns.
Implementation Overhead	Low overhead, as assignments remain static.	Higher implementation overhead due to dynamic allocation algorithms.
Scalability	May face scalability issues with a growing number of devices.	More scalable, as channels can be dynamically allocated based on demand.
Suitability	Suitable for relatively stable and predictable environments.	Suitable for dynamic and unpredictable communication environments.

► Multiple Access Protocol:

- DL layer sufficient for **dedicated link** between sender and receiver, In dedicated channel DLL ensures direct and exclusive communication path **without interference**.
- Multiple Access Control Protocols are used to manage the **access of multiple users** or devices to a **shared communication channel**.
- **Purpose** is to **decrease collisions and prevent crosstalk** among multiple stations.



► Random Access Protocol:

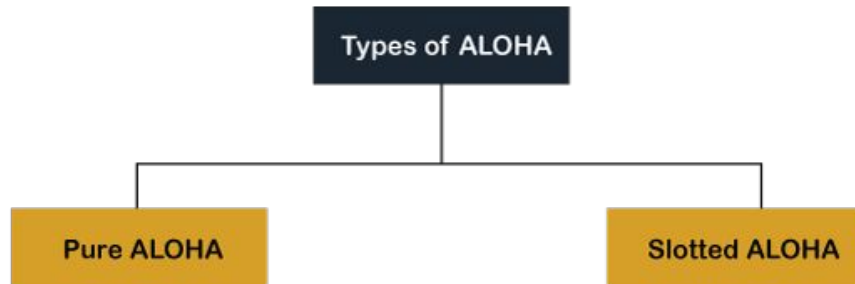
- This protocol ensures that all stations have **equal superiority**, meaning no station has a higher priority than another.
- In **random access** or contention methods, no station is superior to another station and none is assigned the **control over another**. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the **protocol** to make a **decision** on whether or not to send.
- Depending on the status of the medium, any station can transfer data (idle or busy). It has **two characteristics**:
 - There is no set timing for data transmission.
 - There is no set order in which stations deliver data.
- Random access protocols are further classified into the following categories:
 - ALOHA
 - CSMA
 - CSMA/CD
 - CSMA/CA

► ALOHA:

- ALOHA is a **multiple-access protocol** that allows data to be transmitted via a shared network channel. It functions in the open systems interconnection (OSI) model's medium access control sublayer (MAC sublayer).
- ALOHA is designed for **wireless LAN (Local Area Network)** but can also be used in a shared medium to transmit data. In aloha, any station can transmit data to a channel at any time. It **does not require** any carrier sensing.
- ALOHA is an **early** random access protocol used in computer networks. It was first used in the **ALOHAnet network** at the University of Hawaii in the 1970s.
- Each node or station in ALOHA transmits a frame **without attempting to determine** whether the transmission channel is idle or busy. If the channel is idle, the frames will be successfully transmitted.
- If two frames try to occupy the channel at the **same time**, they will collide and be discarded. These stations may opt to retransmit the corrupted frames until successful transmission occurs.

► ALOHA Rules:

- At any time, **any station** can transmit data to a channel.
- Collisions and data frames **may be lost** during data transmission over multiple stations.
- There is **no need** for carrier sensing.
- Aloha has no collision detection because the **acknowledgment of the frames** exists.
- It necessitates data retransmission after a **random** amount of time.

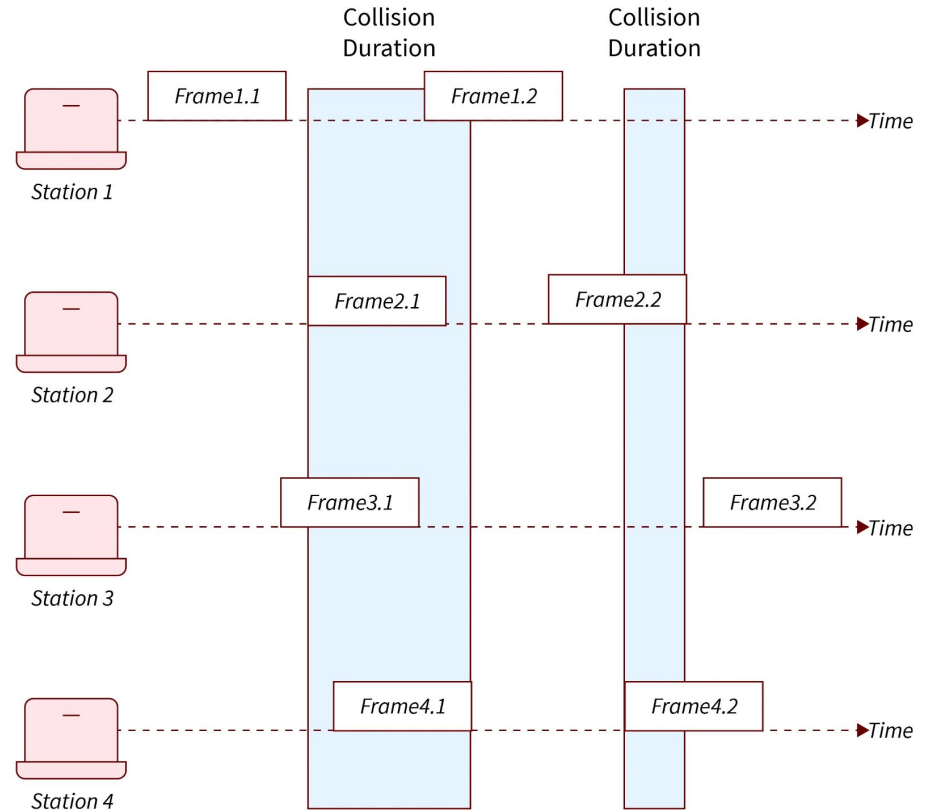


► Pure ALOHA:

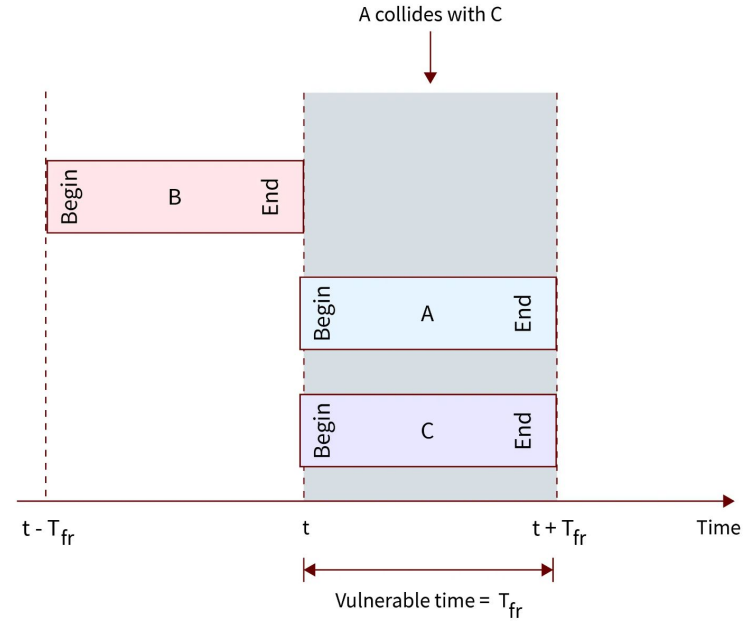
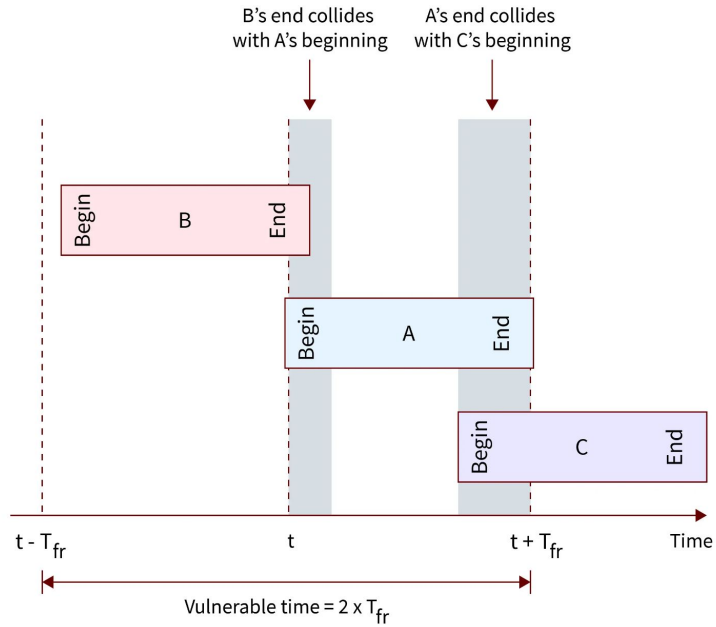
- We employ Pure Aloha **whenever** data is available for transmission over a channel at stations.
- Pure ALOHA is a **simple and early** random access protocol.
- In this protocol, any device can transmit data **anytime without** checking whether the channel is in use or idle.
- This can lead to **collisions**, where two or more devices transmit **simultaneously**, resulting in data corruption and lost packets.
- Pure ALOHA employs a “**send and hope**” mechanism to **recover** from these collisions. Devices send their data and **hope it reaches** its destination without colliding with other transmissions.
- If a collision occurs, the devices involved wait for a **random amount of time** called as **backoff time (T_b)** before resending the data.
- New arrivals are transmitted immediately (no slots)
 - No need for synchronization
 - No need for fixed length packets

► Pure ALOHA:

- Four stations **compete** with one another for access to the **shared channel**.
- The diagram demonstrates that each station sends **two frames** for a total of eight frames on the shared medium.
- Due to several frames competing for the same shared channel, some of these frames **collide**.
- The picture above reveals that just two frames survive: **frame 1.1** from station 1 and **frame 3.2** from station 3.
- Even if **one bit** from **one frame** coexists on the channel with one bit from **another frame**, a collision occurs, and both are destroyed.



▶ ALOHA:

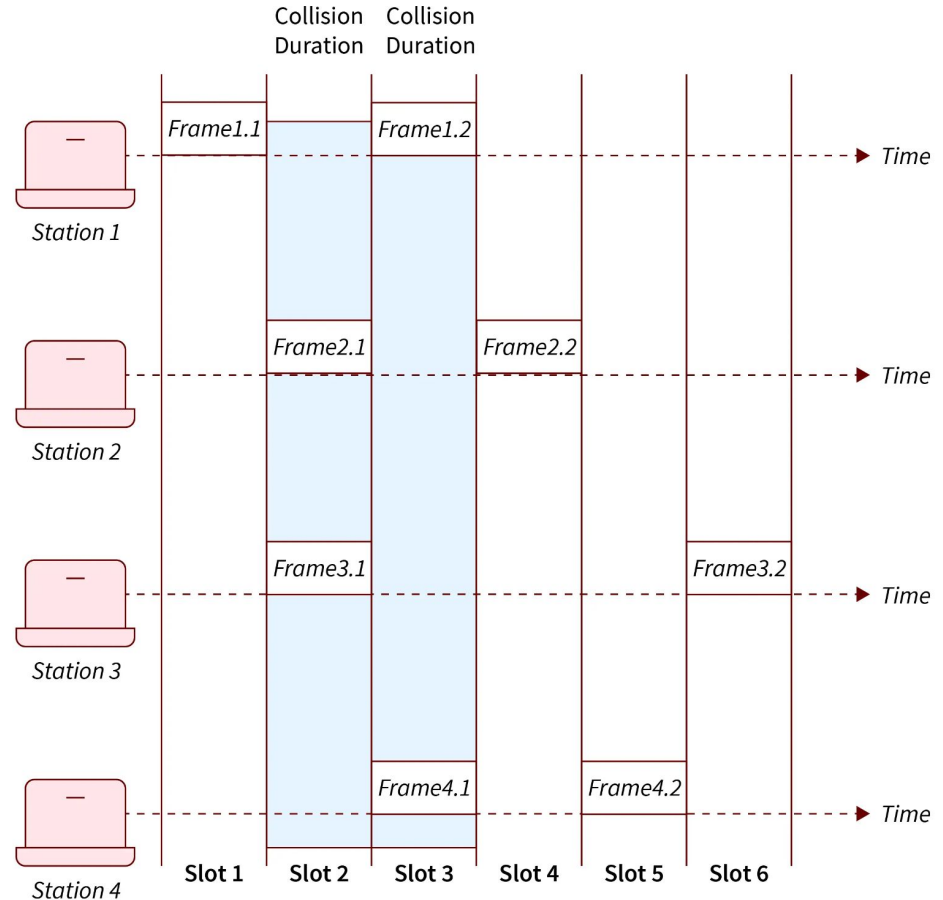


► Slotted ALOHA:

- Pure Aloha has a very **high chance** of hitting a frame, hence the slotted Aloha is intended to **outperform** its efficiency.
- In slotted Aloha, the shared channel is **split** into **fixed time intervals** called slots.
- As a result, if a station wants to send a frame to a shared channel, it can **only** do so at the **start of the slot**, and **only one frame** can be sent to each slot.
- Additionally, the station **must wait** until the beginning of the slot for the subsequent transmission **if it is unable** to transfer data at the beginning of the slot.
- However, sending a frame at the **start** of two or more station time slots still carries the **risk of a collision**.
- If no other nodes attempt transmission during that slot, the transmission is successful.
 - Otherwise “collision”
 - Collided packet are retransmitted after a random delay

► Slotted ALOHA:

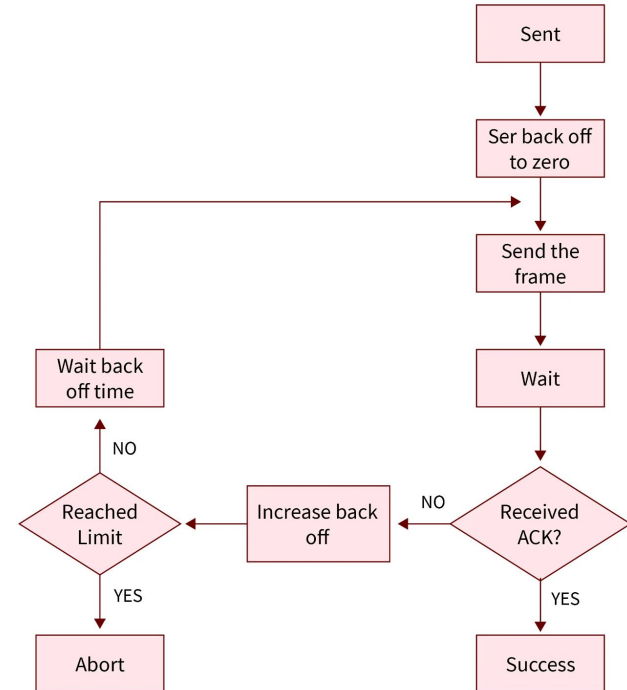
- The figure shows how the channel is divided into **slots**.
- A station can start its transmission only **at the start** of the slot.
- So the only possible condition for collision is if two or more stations start transmission in the same slot.
- This condition is shown in **Frame 1.2** of Station 1 and **Frame 4.1** of Station 4.



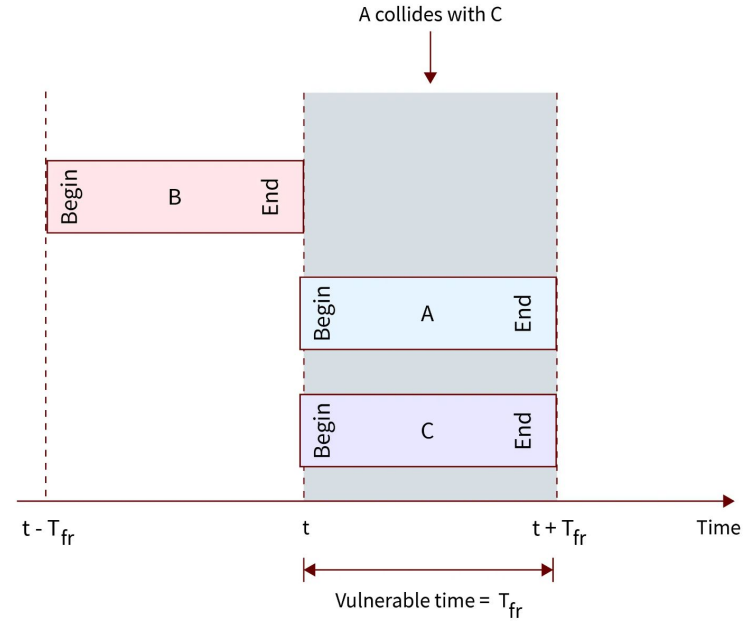
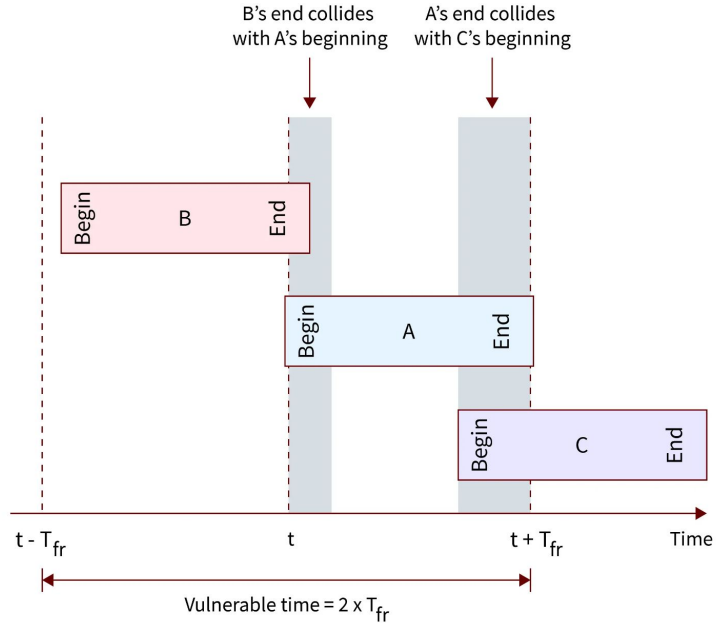
► Protocol Flow Chart for ALOHA:

- A station having data to transmit **will start** its transmission.
- It **waits** for some **random time** (depending on the type of Aloha Protocol). For example, it is $2 \times \text{Transmission Time}$ in the case of Pure Aloha.
- If the **acknowledgment** is received, then the transmission is successful.
- If **no acknowledgment** is received, it means a collision has occurred. Then the station uses a backoff strategy to **retransmit** the packet again.
- After various iterations, the **station aborts** the transmission if there is no acknowledgment.

Protocol flow chart for ALOHA



▶ ALOHA:



▶ ALOHA:

- In Pure Aloha in computer network:
 - Vulnerable Time is equal to twice the transmission time.
 - Efficiency is 18.4%.
 - The average number of successful ALOHA transmissions is given by $S = G * e^{-2G}$.
- In Slotted Aloha:
 - Vulnerable Time is equal to the transmission time.
 - Efficiency is 36.8%.
 - The average number of successful ALOHA transmissions is given by $S = G * e^{-G}$.
- The number of collisions is relatively less in Slotted Aloha than in Pure Aloha.

Suppose $G = 0.4$ (traffic load is 0.4 or 40%).

$$PS = 0.4 \cdot e^{-0.4}$$

$$PS \approx 0.4 \cdot 0.6703$$

$$PS \approx 0.2681$$

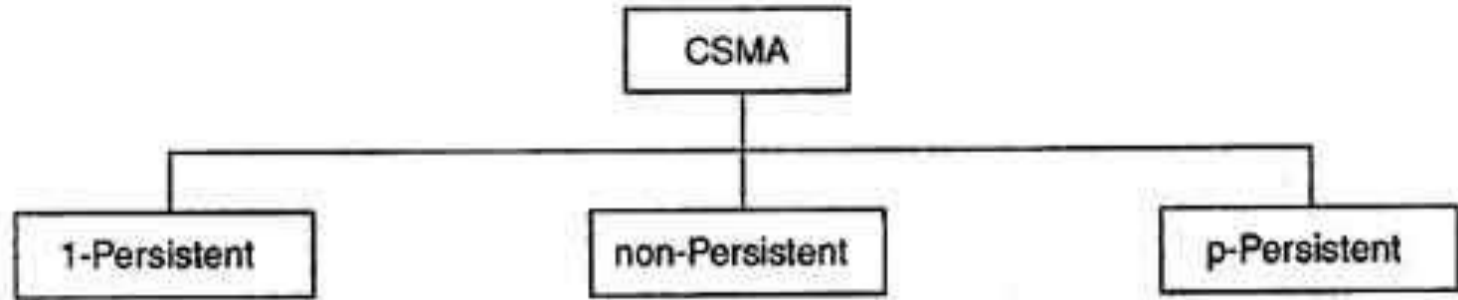
Therefore, with a traffic load (G) of 0.4, the calculated probability of successful transmission (PS) is approximately 0.2681 or 26.81%.

S.no.	On the basis of	Pure Aloha	Slotted Aloha
1.	Basic	In pure aloha, data can be transmitted at any time by any station.	In slotted aloha, data can be transmitted at the beginning of the time slot.
2.	Introduced by	It was introduced under the leadership of Norman Abramson in 1970 at the University of Hawaii.	It was introduced by Robert in 1972 to improve pure aloha's capacity.
3.	Time	Time is not synchronized in pure aloha. Time is continuous in it.	Time is globally synchronized in slotted aloha. Time is discrete in it.
4.	Number of collisions	It does not decrease the number of collisions to half.	On the other hand, slotted aloha enhances the efficiency of pure aloha. It decreases the number of collisions to half.
5.	Vulnerable time	In pure aloha, the vulnerable time is $= 2 \times T_t$	Whereas, in slotted aloha, the vulnerable time is $= T_t$
6.	Successful transmission	In pure aloha, the probability of the successful transmission of the frame is - $S = G * e^{-2G}$	In slotted aloha, the probability of the successful transmission of the frame is - $S = G * e^{-G}$
7.	Throughput	The maximum throughput in pure aloha is about 18%.	The maximum throughput in slotted aloha is about 37%.

► CSMA (Carrier Sense Multiple Access):

- *Principle:* "Sense before transmit or Listen before talking"
- CSMA stands for Carrier Sense Multiple Access (CSMA). CSMA is one of the **network protocols** which works on the principle of 'carrier sense'. CSMA is a protocol developed to **increase** the performance of the network and **reduce** the chance of collision in the network.
- If any device wants to send data then the device **first senses or listens** to the network medium to check whether the shared network is free or not. If the channel is found **idle** then the device will transmit its data.
- This sense **reduces** the chance of collision in the network but this method is **not able** to eliminate the collision.
- Carrier Sense Multiple Access (CSMA) is a protocol that senses or listens to the medium before any transmission of data in the medium.
- CSMA is used in **Ethernet networks** where two or more network devices are connected.

► CSMA Approach/ Types/ Modes:



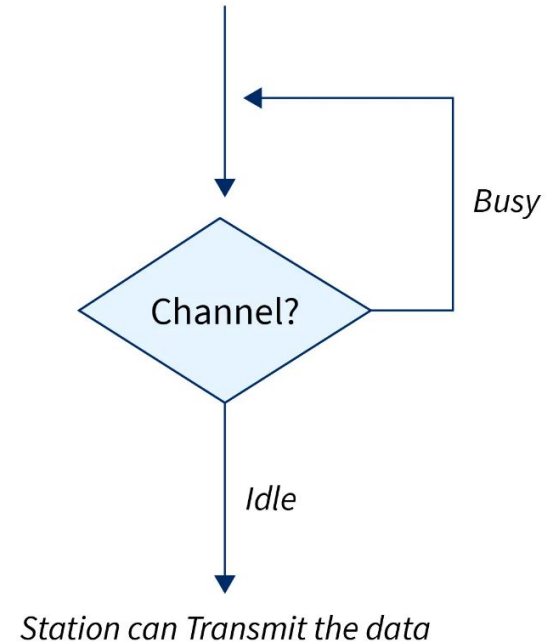
Types of CSMA

► Approaches of CSMA:

1-Persistent:

This method is considered the **straightforward** and simplest method of CSMA. In this method, if the station finds the **medium idle** then the station will **immediately** send the data frame with **1 (one) probability**.

- In this, if the station wants to transmit the data. Then the station **first senses** the medium.
- If the medium is **busy** then the station **waits** until the channel becomes idle. And the station continuously senses the channel until the medium becomes idle.
- If the station detects the channel as **idle** then the station will **immediately** send the data frame with **1 (one) probability** that's why the name of this method is 1-persistent.



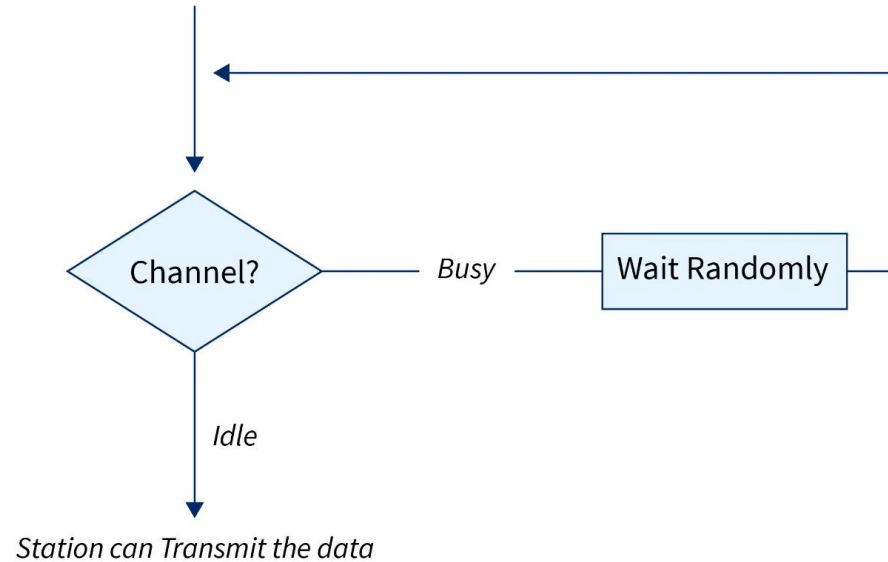
► Approaches of CSMA:

Non-Persistent:

In this method of CSMA, if the station finds the **channel busy** then it will wait for a **random amount of time before sensing** the channel **again**.

- If the station wants to transmit the data then **first of all it will sense** the medium.
- If the medium is idle then the station will **immediately** send the data.
- Otherwise, if the medium is **busy** then the station waits for a **random** amount of time and then again **senses the channel** after waiting for a random amount of time.
- In Non-persistent there is **less** chance of collision in **comparison** to the 1-persistent method as this station will not continuously sense the channel but sense the channel after waiting for a random amount of time.

► Non-Persistent Approach:

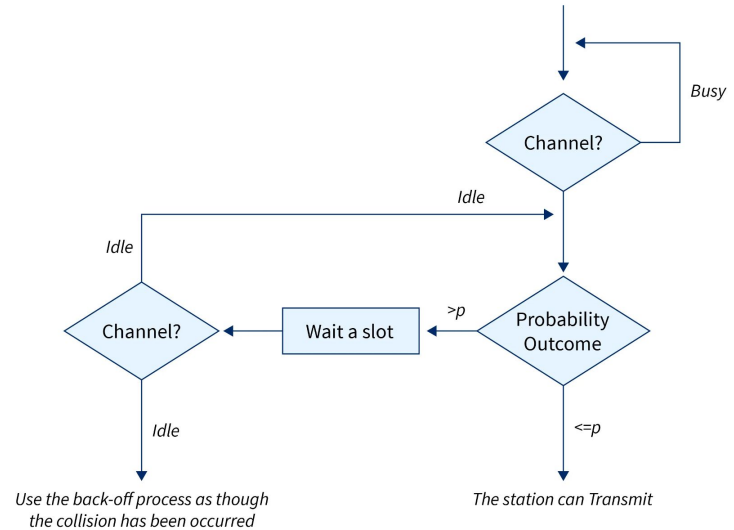


► Approaches of CSMA:

P-Persistent:

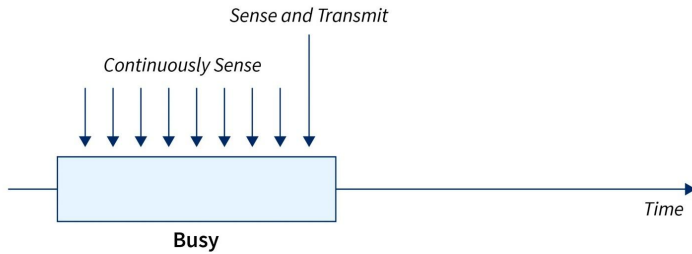
If the channel is idle then the station does the following steps.

1. The station transmits its data frame in the network by p probability.
2. If data is not transmitted ($\{q = 1 - p\}$ probability). The station waits for the start of the next time slot and after waiting again senses the channel.
3. If the channel is again idle, then it again performs step 1. If the channel is busy, then it thinks that there is a collision in the network and now this station will follow the back-off procedure.

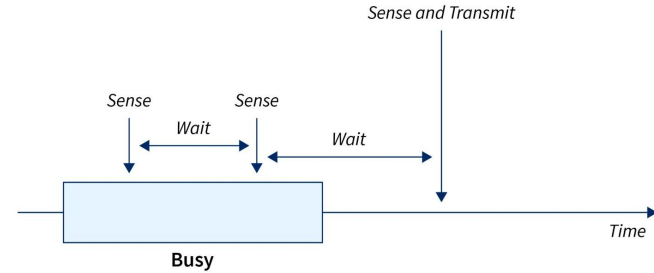


► CSMA Approach:

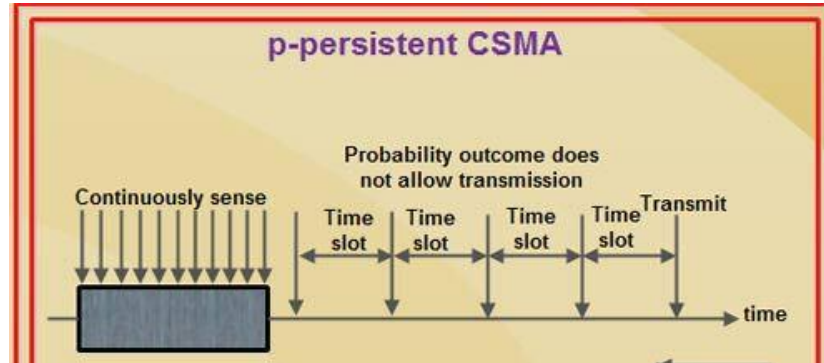
1-persistent



Non-Persistent Approach



p-persistent CSMA

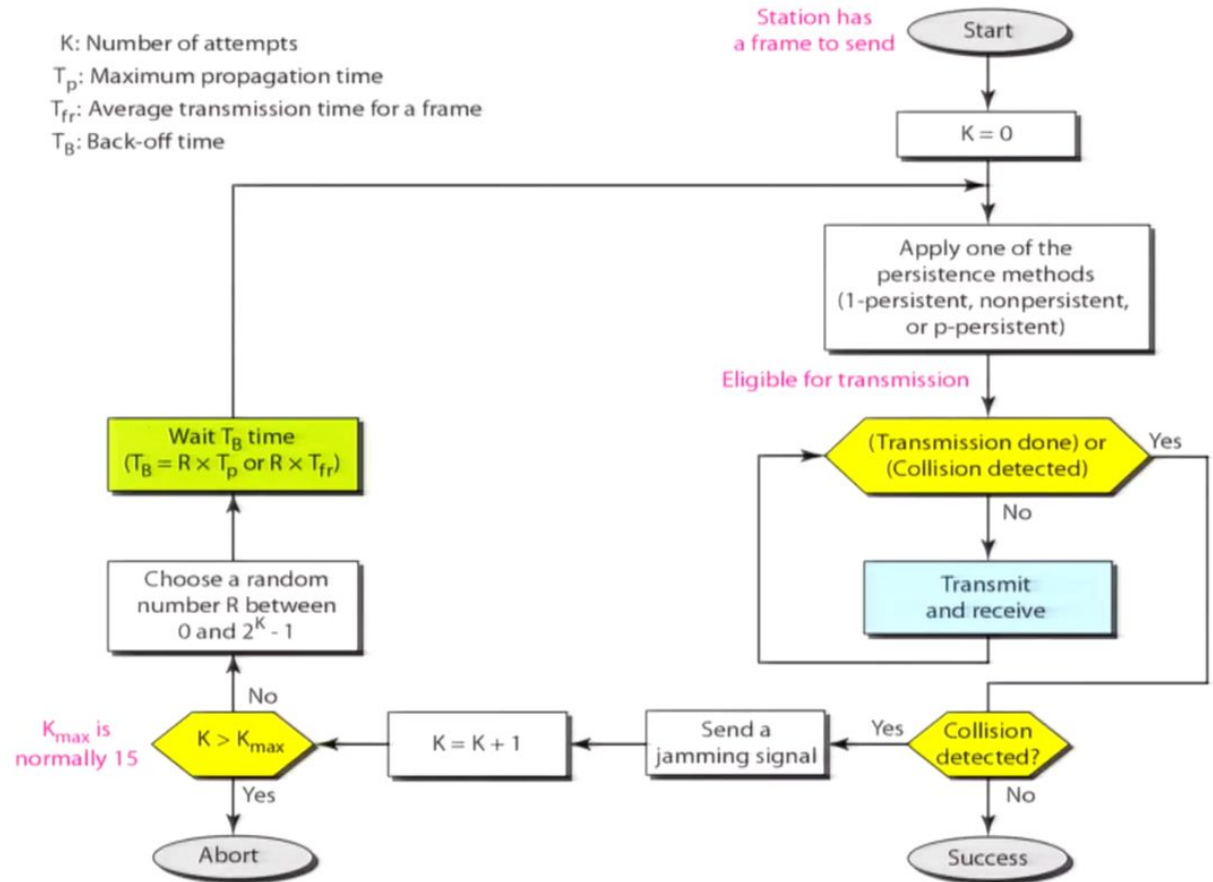


► Variation of CSMA ► CSMA/CD:

- CSMA, although **more efficient** than ALOHA or slotted ALOHA, still has one glaring inefficiency: When two frames collide, the medium **remains unusable** for the duration of transmission of both damaged frames. For long frames, compared to propagation time, the amount of **wasted capacity** can be considerable. This waste can be reduced if a station continues to listen to the medium while transmitting. This leads to the following rules for CSMA/CD:
 - a. If the medium is **idle**, transmit; otherwise, go to step b.
 - b. If the medium is **busy**, continue to listen until the channel is **idle**, then transmit **immediately**.
 - c. If a collision is **detected** during transmission, **transmit** a brief jamming signal to assure that all stations know that there has been a **collision** and then **cease** transmission.
 - d. After transmitting the **jamming signal**, wait a random amount of time, then **attempt** to transmit again.

► CSMA/CD:

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



► CSMA/CD:

Advantages:

- For collision detection, CSMA/CD is better than CSMA.
- It is used for quick collision detection on a shared channel.
- Using CSMA/CD helps prevent the waste of transmission in any way.
- Whenever mandatory, it consumes or shares all of the bandwidth.

Disadvantages:

- After reaching the maximum distance, collision detection is impossible.
- Not ideal for larger networks as collision detection is difficult to detect.
- The performance gets considerably disrupted when more devices are added.

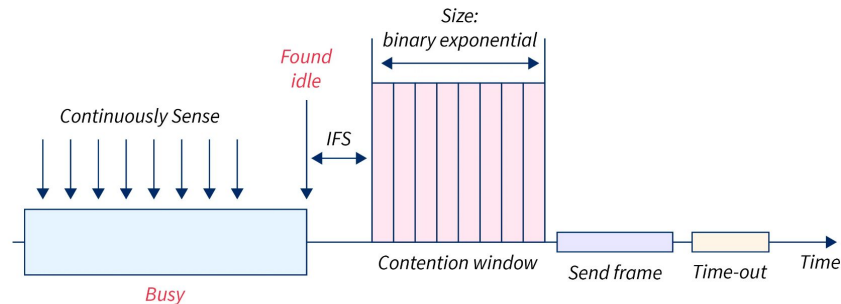
► Variation of CSMA ► CSMA/CA:

- It was invented for wireless networks.
- The process of collision detection involves sender receiving acknowledgements signals.
- The working of CSMA/CA are as follows:
 - When a **frame is ready**, the transmitting station **checks whether** the channel is idle or busy.
 - If the channel is **busy**, the station **waits** until the channel becomes idle.
 - If the channel is **idle**, the station waits for an **Inter-frame gap (IFG)** amount of time and **then** sends the frame.
 - After sending the frame, it sets a **timer**.
 - The station then **waits** for acknowledgement from the receiver. If it **receives** the acknowledgement **before expiry** of timer, it marks a **successful** transmission.
 - Otherwise, it waits for a **back-off time period** and **restarts** the algorithm.

► Variation of CSMA ► CSMA/CA:

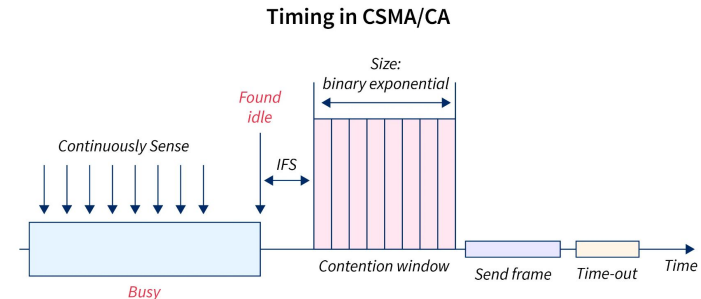
- CSMA/CA avoids the collisions using three basic techniques.
 - **Inter-frame space:** A **waiting period** before transmitting data after the channel becomes idle, determining the priority of the frame or station.
 - **Contention window:** A period divided into **time slots** where the station waits a **random** number of slots after **IFS** before attempting transmission again **if** the channel is **busy**.
 - **Acknowledgements :** Ensures **successful** data reception using **positive** acknowledgment and **time-out** mechanisms to **handle** collisions or data corruption.

Timing in CSMA/CA



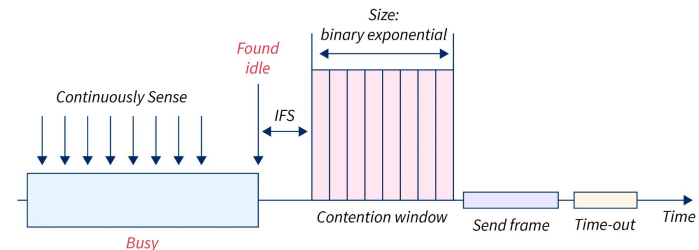
► Inter-Frame Space (IFS):

- Whenever the channel is found idle, the station **does not** transmit immediately. It **waits** for a period of time called **inter-frame space** (IFS).
- When channel is **sensed** to be idle, it may be **possible** that same distant station **may have** already started transmitting and the signal of that distant station **has not yet reached** other stations.
- Therefore the **purpose** of IFS time is to allow this transmitted signal to reach **other stations**.
- If after this IFS time, the channel is **still idle**, the station can send, but it still needs to wait a time equal to **contention time**.
- IFS variable can also be used to define the priority of a station or a frame.
- Higher the IFS lower is the priority.



► Contention Window:

- Contention window is an amount of time divided into slots.
- A station that is ready to send chooses a random number of slots as its wait time.
- The number of slots in the window changes according to the binary exponential backoff strategy. It means that it is set of one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time.
- This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station.
- In contention window the station needs to sense the channel after each time slot.
- If the station finds the channel busy, it does not restart the process. It just stops the timer & restarts it when the channel is sensed as idle.

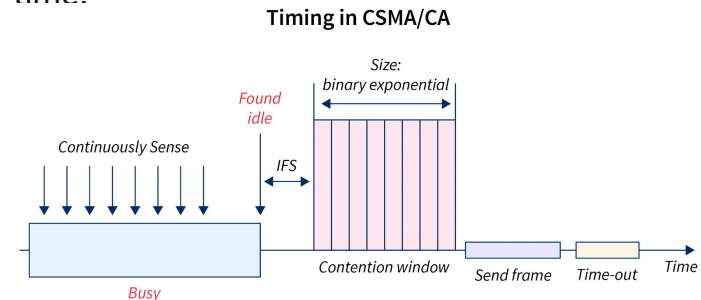


► Acknowledgement:

- Despite all the precautions, collisions may occur and destroy the data.
- The **positive** acknowledgment and the **time-out timer** can help guarantee that receiver has received the frame.

Back Off Time:

- After the occurrence of collision, station waits for some **random** back off time and then retransmits.
- This **waiting time** for which the station **waits before** retransmitting the data is called as back off time.
- Back Off Algorithm is used for calculating the back off time.



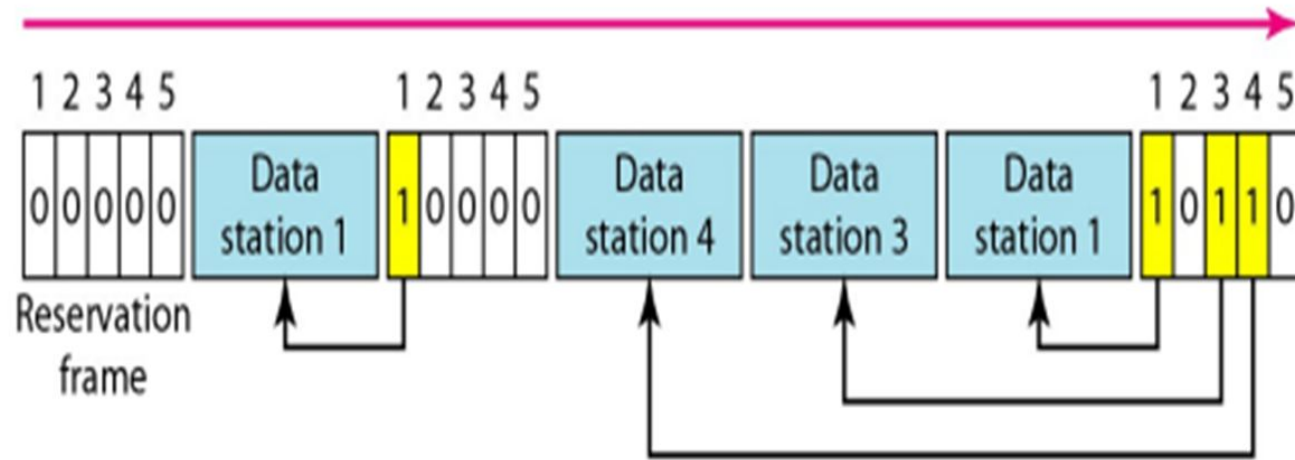
► Control Access Protocols:

- In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.
- Types of Controlled Access Techniques: The three popular controlled-access methods are:
 - Reservation
 - Polling
 - Token Passing

► Reservation:

- In the reservation method, a station needs to make a reservation before sending data.
- Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation mini slots in the reservation frame. Each mini slot belongs to a station. When a station needs to send a data frame, it makes a reservation in its own mini slot. The stations that have made reservations can send their data frames after the reservation frame.
- Figure shows a situation with five stations and a five-mini slot reservation frame. In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.

► Reservation:



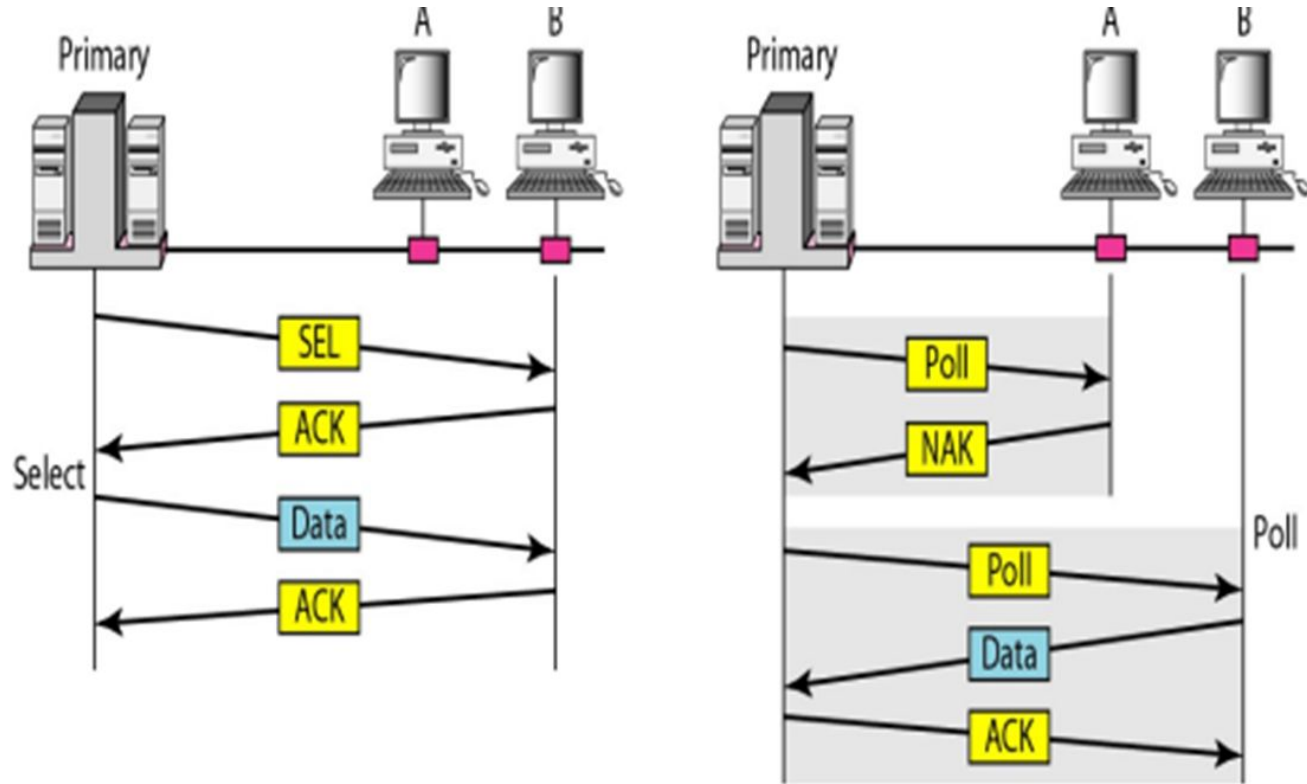
► Polling:

- Polling works with topologies in which one device is served as a primary station and the other devices are secondary stations.
- All data exchanges must be made through the primary device even when the destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions. It is up to the primary device to determine which device is allowed to use the channel at a given time. The primary device, therefore, is always the initiator of a session.
- The primary device sends and receives data by using Select and Poll function.

► Polling:

- **Select:**
 - The select function is used whenever the primary device has something to send.
 - Before sending data, the primary creates and transmits a one field which is select (SEL) frame that includes the address of the intended secondary that alerts the secondary to the upcoming transmission and wait for an acknowledgment of the secondary's ready status.
- **Poll:**
 - The poll function is used whenever the primary device has something to receive.
 - When the primary is ready to receive data, it must ask (poll) each device if it has anything to send. When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data frame if it does. When the response is a data frame, the primary reads the frame and returns an acknowledgment (ACK frame), verifying its receipt.

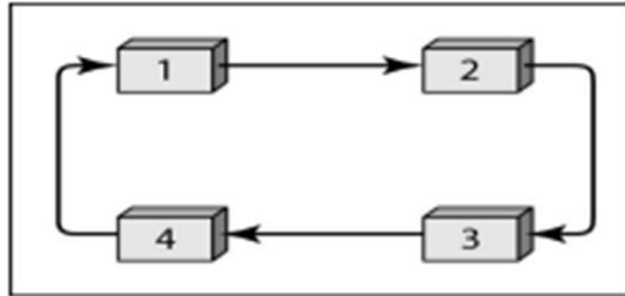
► Polling:



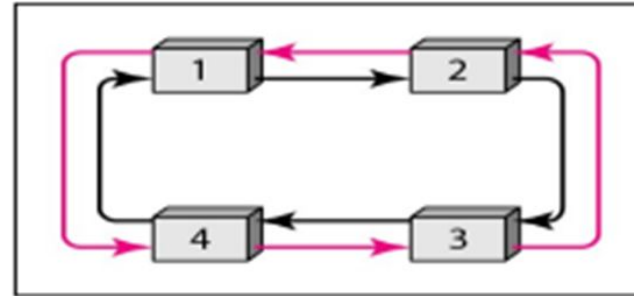
► Token Passing:

- In the token-passing method, the stations in a network are organized in a logical ring. In other words, for each station, there is a predecessor and a successor.
- In this method, a special packet called a token circulates through the ring. The possession of the token gives the station the right to access the channel and send its data.
- When a station has some data to send, it waits until it receives the token from its predecessor. It then holds the token and sends its data.
- When the station has no more data to send, it releases the token, passing it to the next logical station in the ring. The station cannot send data until it receives the token again in the next round. In this process, when a station receives the token and has no data to send, it just passes the data to the next station (successor).
- The high-speed Token Ring networks called FDDI (Fiber Distributed Data Interface) and CDDI (Copper Distributed Data Interface) use this topology. The Token Bus (also called bus ring topology) LAN, standardized by IEEE and Token Ring LAN designed by IBM, also uses this topology.

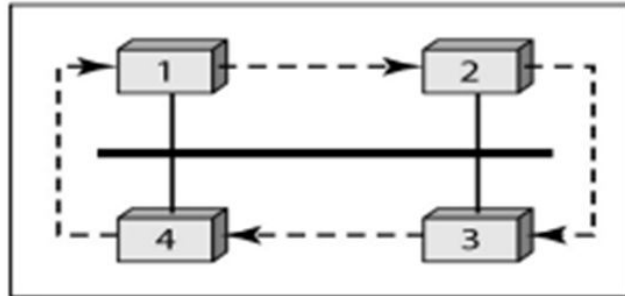
► Token Passing:



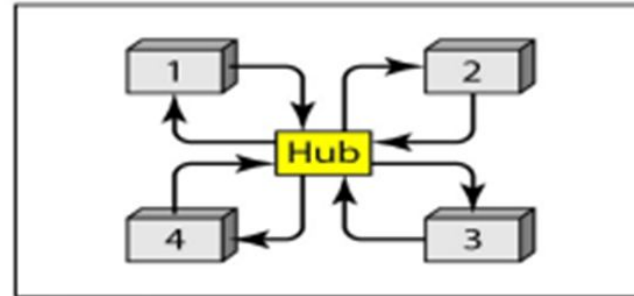
a. Physical ring



b. Dual ring



c. Bus ring



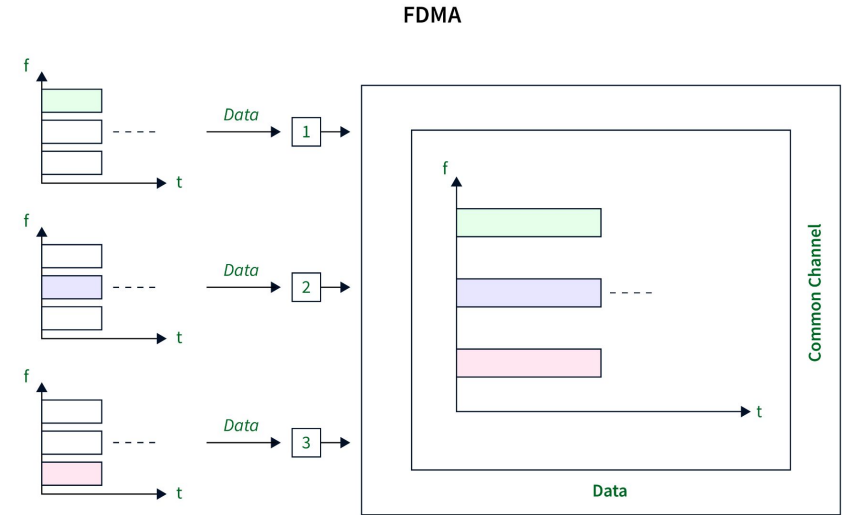
d. Star ring

► Channelization Protocols:

- It is a channelization protocol that allows the total usable bandwidth in a shared channel to be shared across multiple stations based on their time, frequency and codes. It can access all the stations at the same time to send the data frames to the channel.
- Following are the various methods to access the channel based on their time, distance and codes:
 - FDMA (Frequency Division Multiple Access)
 - TDMA (Time Division Multiple Access)
 - CDMA (Code Division Multiple Access)

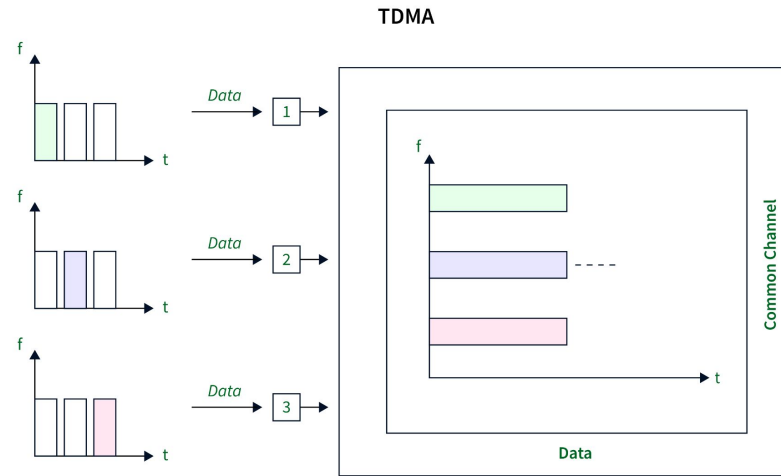
► FDMA:

- Available bandwidth is divided into frequency bands.
- Each station is allocated with a band to send its data.
- Each band is reserved for specific station all the time.
- In order to avoid station interferences, allocated bands are separated from one another by small guard bands.
- FDMA specifies a predetermined frequency band for entire period of communication.
- Stream data(a continuous flow of data that may not be packetized) can be easily used with FDMA.



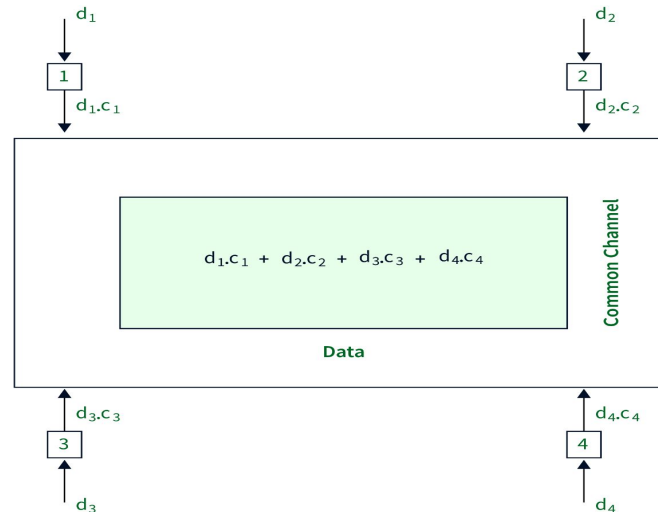
► TDMA:

- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in assigned time slot.
- The main problem with TDMA is lies in achieving synchronization between the different stations.
- Each station need to know the beginning of its slot and the location of its slot.

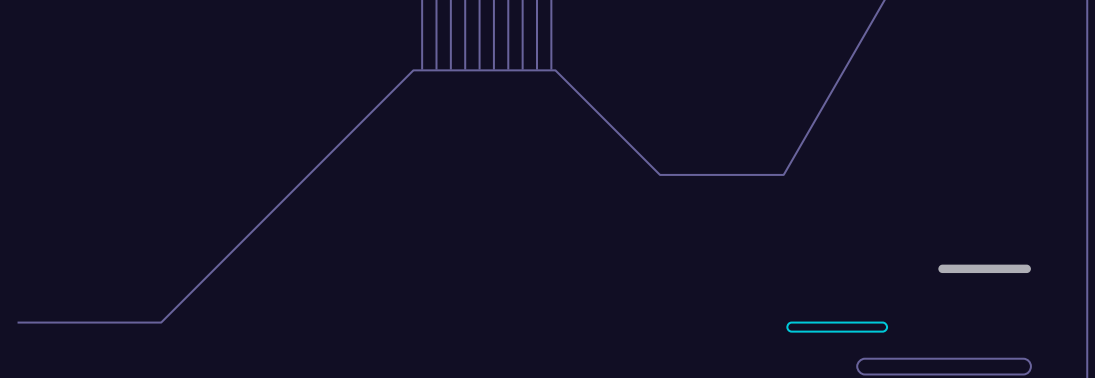


► CDMA:

- The channel resource for this protocol is code.
- In the CDMA protocol, we allow multiple users to simultaneously transmit their data signals over the entire bandwidth of the common channel by assigning unique spreading codes to each user.
- The advantages of this protocol is that we don't share the frequency as we do in FDMA, and neither do we share time as we do in TDMA.



**Ethernet Standards
(802.3 CSMA/CD, 802.4
Token Bus, 802.5 Token
Ring)**



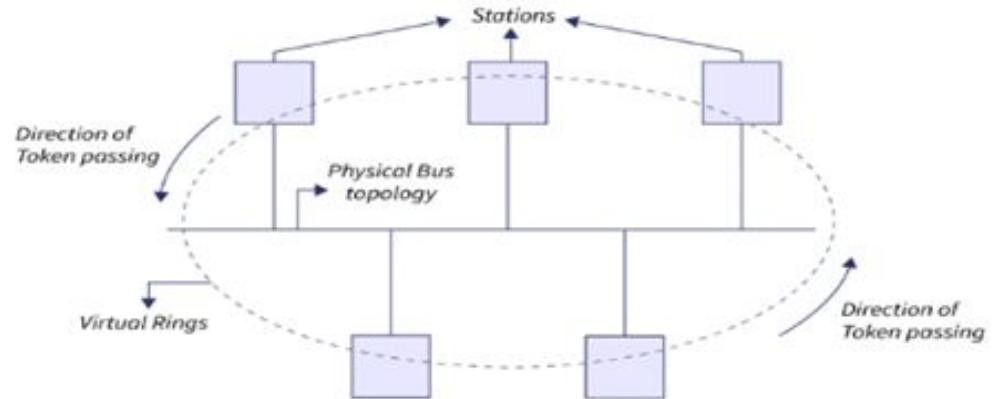
3.6

► Token Bus: (IEEE 802.4)

- Token Bus (IEEE 802.4) is a standard for **implementing** token ring over a virtual ring in LANs.
- The physical media has a **bus or a tree topology** and uses coaxial cables. A virtual ring is created with the nodes/stations, and the **token is passed** from one node to the next in a sequence along this **virtual ring**.
- Each node knows the address of its **preceding station** and its **succeeding station**. A station **can only transmit** data when it has the **token**. The working principle of Token Bus is similar to Token Ring.

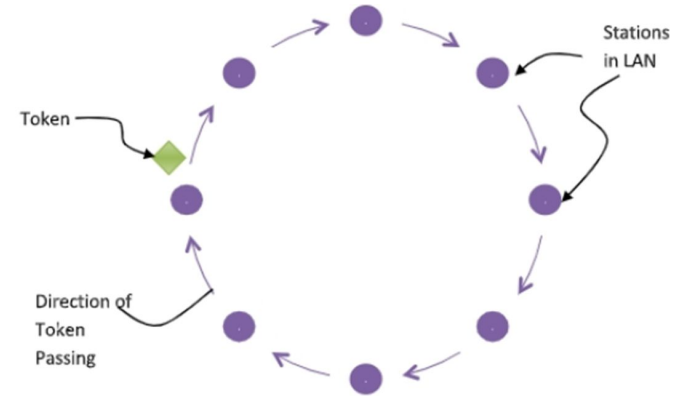
► Token Bus: (IEEE 802.4)

- A **token** is a small message that circulates among the **stations** of a computer network, providing **permission** to the stations for transmission.
- If a station has data to transmit when **it receives a token**, it sends the data and then passes the token to the **next station**; otherwise, it **simply passes** the token to the next station.
- This is depicted in the following diagram:



► Token Ring (802.5)

- A Station is authorized to send data when it receives a special frame called a **Token**.
- Stations are **arranged** around a ring (physically or logically) -A Token circulates around a ring
- If a station needs to send data ,it **waits** for the token.
- The Station **captures** the token and sends one or more frames as long as the **allocated time** has not expired
- It **releases** the token to be used by the successor station.

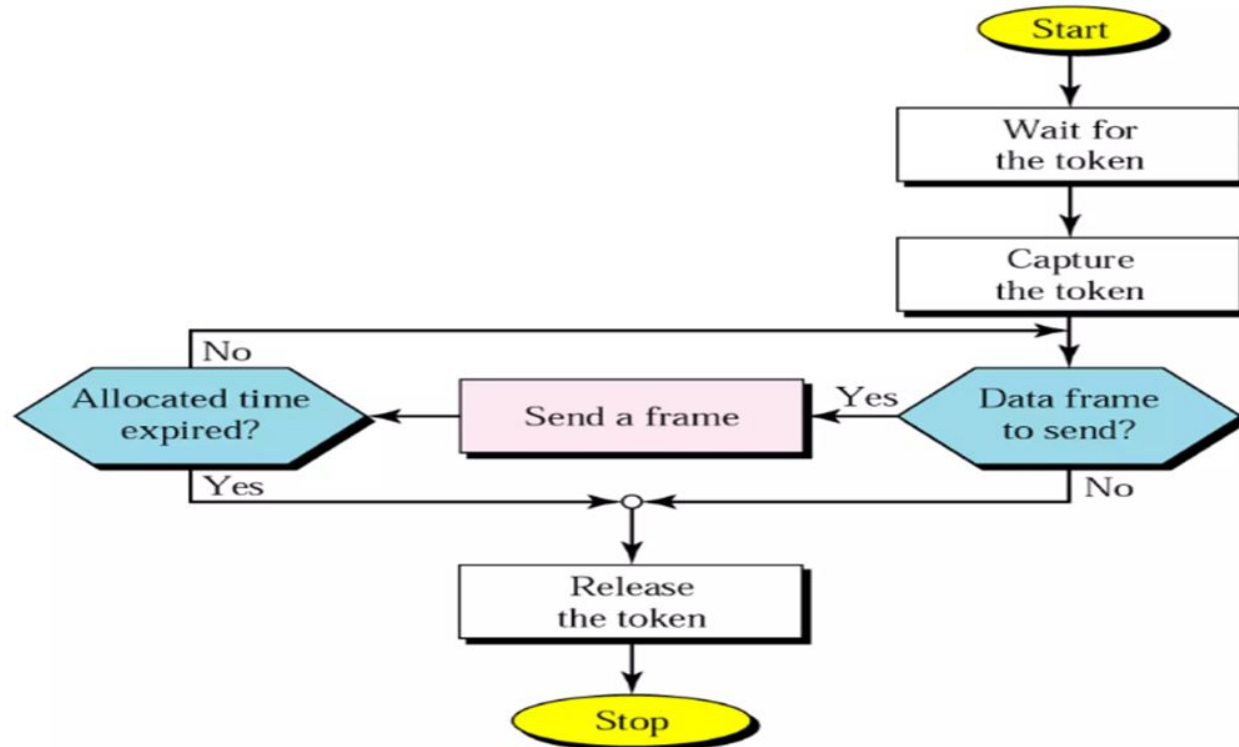


► Token Ring (802.5)

- Token Passing mechanism in token ring:
- If a station has a frame to transmit when it **receives** a token, it sends the frame and then **passes** the token to the next station; otherwise, it simply passes the token to the next station.
- Passing the token means **receiving** the token from the **preceding** station and transmitting to the **successor** station.
- The data flow is **unidirectional** in the direction of the token passing. In order that tokens are **not** circulated infinitely, they are **removed** from the network once their **purpose** is completed.

► Token Ring (802.5)

TOKEN PASSING FLOW CHART :



▶ **Wireless LAN: Spread
Spectrum, Bluetooth,
Wi-Fi**

3.7

► Wireless LAN:

- IEEE **802.11x**, commonly known as Wi-Fi, is a set of standards that define communication over wireless local area networks (**WLANs**).
- It specifies the **over-the-air** interface between a wireless client and a base station, or between two wireless clients.

Key characteristics include:

- **Frequency bands:**
 - Operates in the **2.4 GHz, 5 GHz, and 6 GHz** bands (depending on the specific standard).
 - The 5 GHz and 6 GHz bands generally offer faster speeds and less interference than the 2.4 GHz band.
- **Speed:**
 - Capable of achieving high data transfer rates.
 - The latest version, Wi-Fi 6 (802.11ax), can reach **theoretical speeds of up to** 9.6 Gbps.

► Wireless LAN:

- **Range:**

- Typical range varies depending on factors like obstacles and interference.
- Wi-Fi 6 can have a range of up to **300 feet indoors** and even farther outdoors.

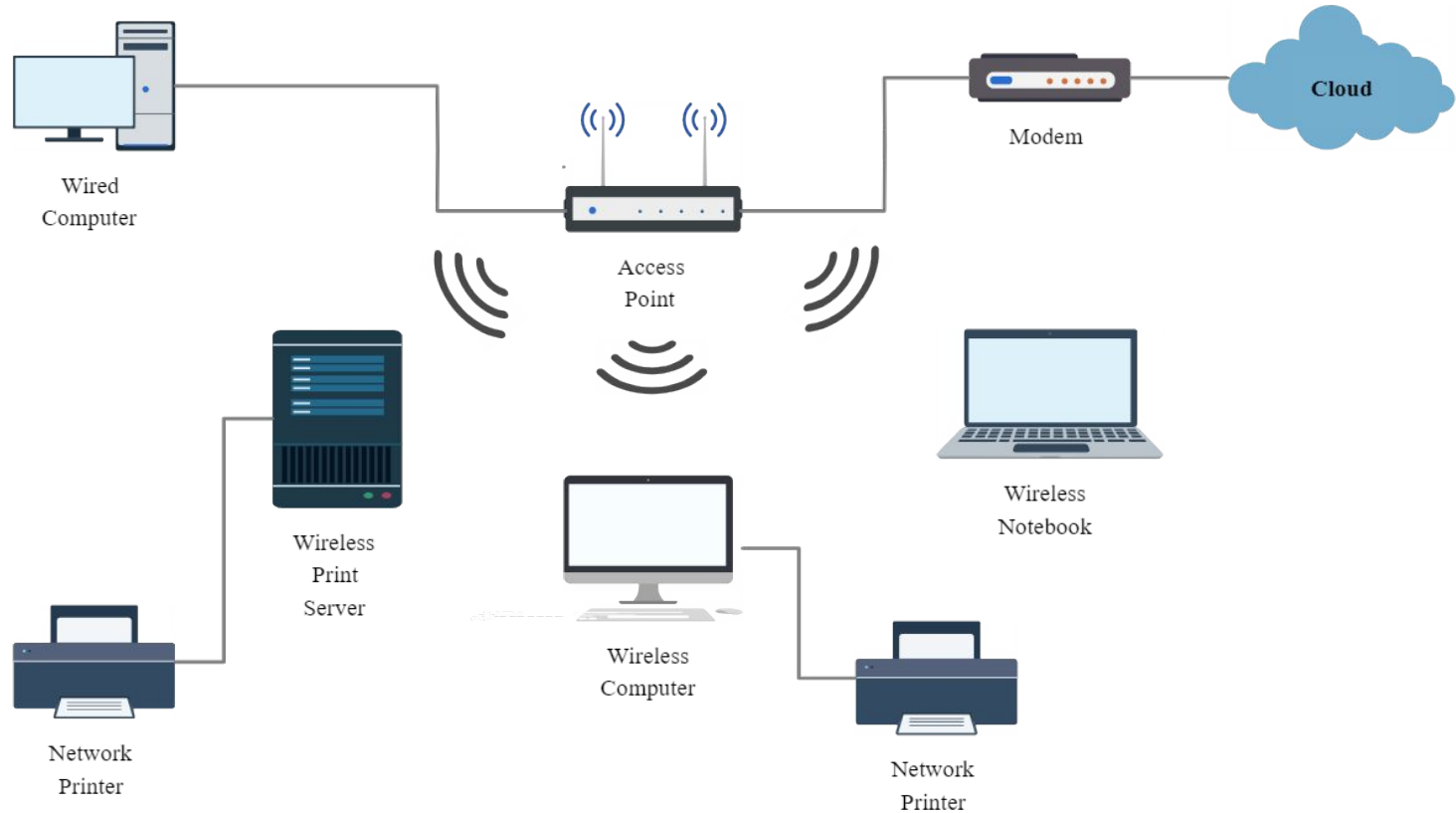
- **Security:**

- Employs various security protocols to protect data transmission, such as WPA2 and WPA3.

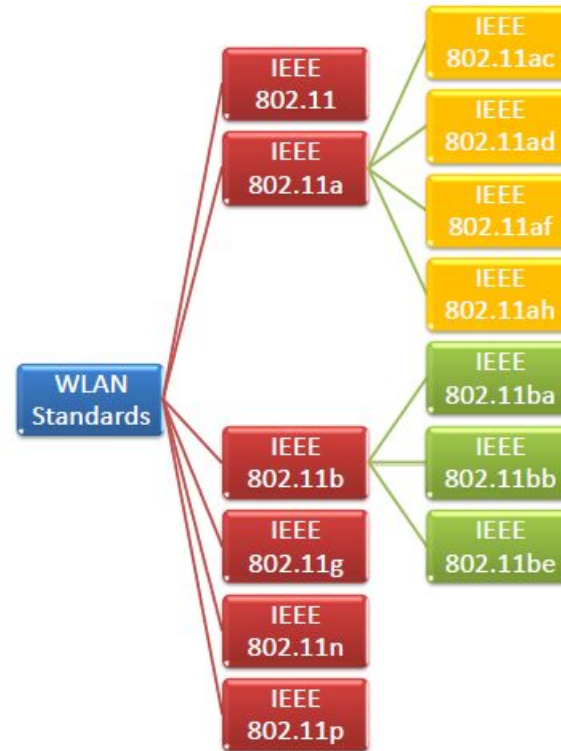
- **Applications:**

- Widely used for internet connectivity, streaming video, gaming, file sharing, and connecting various devices wirelessly.

► Wireless LAN:



► Common standards within the 802.11x family:

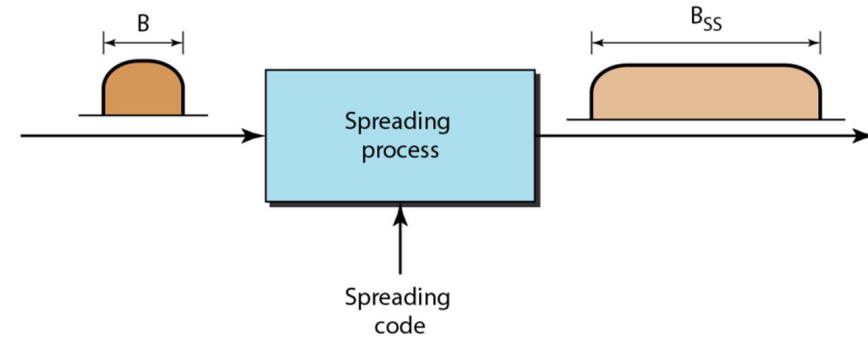


► Common standards within the 802.11x family:

Technologies	Indoor/ Outdoor	Bitrate	Freq. bands	License	Bandwidth	Modulation
IEEE 802.11	20m /100m	2 Mbps	2.4GHz	Unlicensed	20 MHz	FHSS and DSSS
IEEE 802.11b	35m/ 140m	11 Mbps	2.4GHz	Unlicensed	20 MHz	HR-DSSS
IEEE 802.11a	35m/ 119m	54 Mbps	5GHz	Unlicensed	20 MHz	OFDM
IEEE 802.11g	45m/ 90m	54 Mbps	2.4 GHz	Unlicensed	22 MHz	OFDM/ DSSS/ CCK
IEEE 802.11n	70m/ 250m	600 Mbps	2.4 GHz/ 5 GHz	Unlicensed	20 MHz/ 40 MHz	OFDM
IEEE 802.11ac wave	70m/ 250m	7000 Mbps	5 GHz	Unlicensed	80 MHz	64-QAM
IEEE 802.11ad	10m/ n/a	7000 Mbps	60 GHz	Unlicensed	2.16 GHz	Single Carrier/ OFDM
IEEE 802.11ac wave 2	70m/ 250m	7000 Mbps	5 GHz	Unlicensed	80 MHz/ 160 MHz	256-QAM

► Spreading/ Spread Spectrum (SS):

- “Spreading the bandwidth needed to transmit the data” OR Frequency of transmitted signal is varied deliberately (Ourselves).
- **Spread spectrum** is designed to be used in wireless applications in which stations must be able to **share the medium** without interception by an eavesdropper and without being subject to jamming from a malicious intruder.
- Important **encoding method** for wireless communications.
- Spread data over wide **bandwidth**
- Makes **jamming and interception** harder
- Two Types:
 - Direct Sequence Spread Spectrum Synchronous (DSSS)
 - Frequency Hopping Spread Spectrum (FHSS)

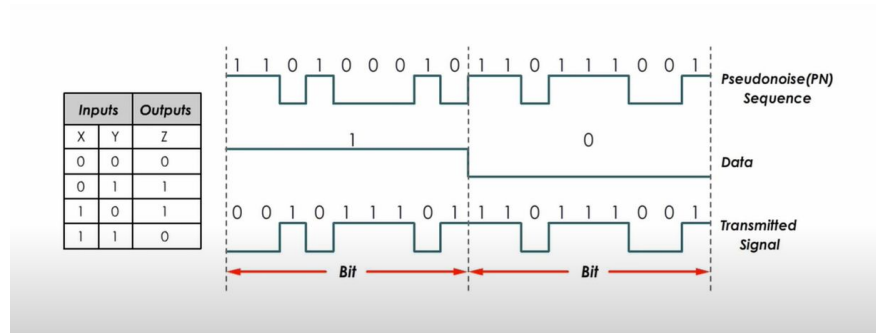
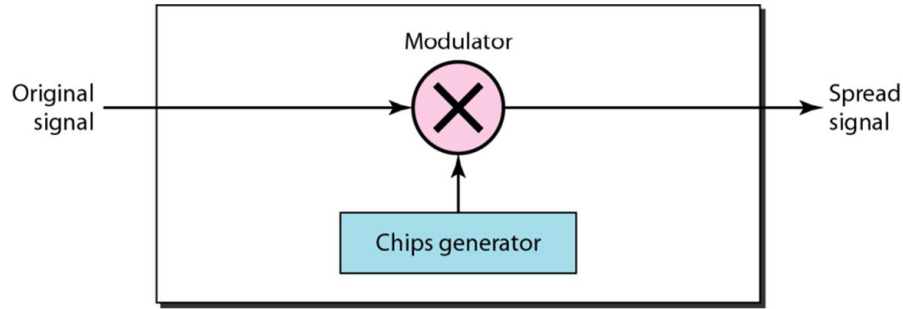


► Direct Sequence Spread Spectrum Synchronous (DSSS)

The **direct sequence spread spectrum** (DSSS) technique expands the bandwidth of a signal by replacing **each data bit with n bits using a spreading code**. In other words, each bit is assigned a code of n bits, called **chips**.

- Each bit is **multiplied** by secret code (Chipping Code).
- **Chipping Code** is used to access message.
- Used to reduce overall signal interference.
- Stream of information transmitted is divided into **small pieces**, each of which is allocated to frequency channel.
- **Chipping code** is used to divide data according to spreading ratio.

► Direct Sequence Spread Spectrum Synchronous (DSSS)

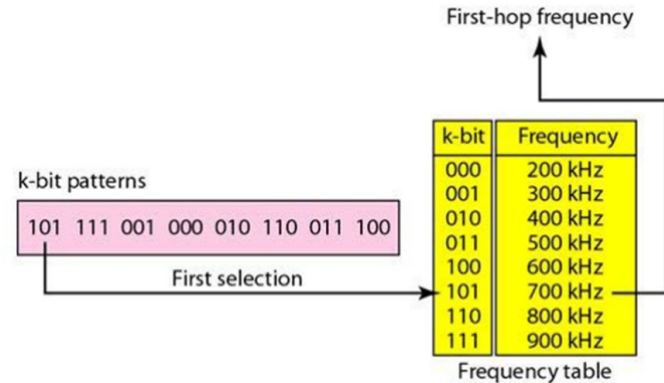
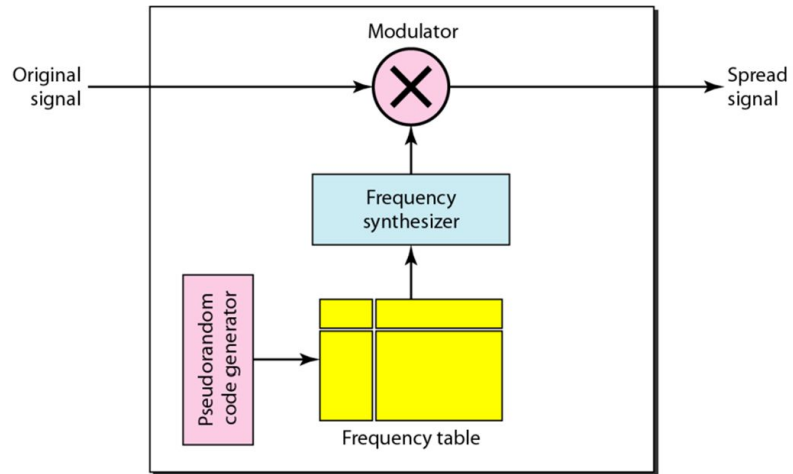


Waveform Representation of DSSS

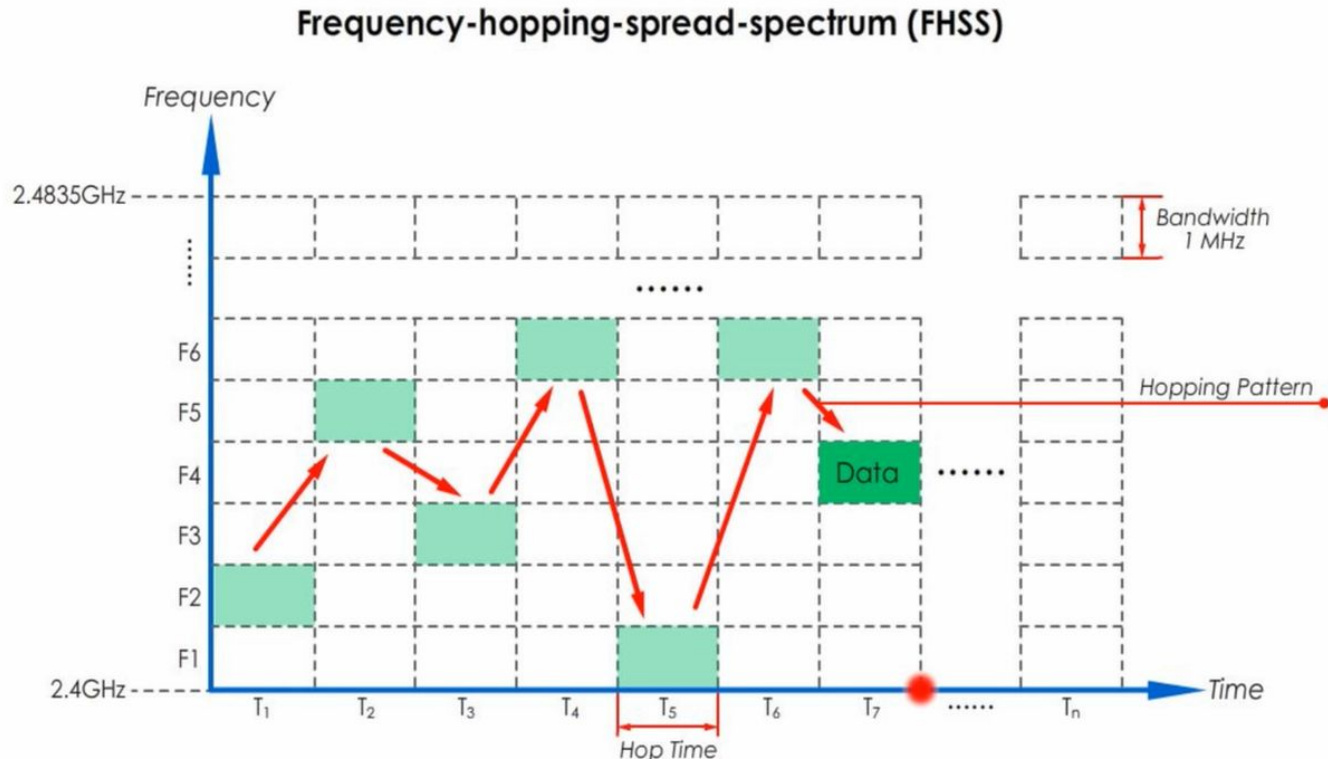
► Frequency hopping spread spectrum (FHSS):

- The **frequency hopping spread spectrum** (FHSS) technique uses M different carrier frequencies that are modulated by the source signal. **At one moment**, the signal modulates one carrier frequency; at the next moment, the signal modulates another carrier frequency.
- It is **repeated switching of frequencies** during radio transmission to reduce interference.
- Bandwidth is split into many channel of **smaller bandwidth and guard spaces**.
- Transmitter and receiver stay on these channels for certain time.
- **Hopping sequence**: Channel Usage Pattern
- **Dwell Time** : Time spent in each Hop.

► Frequency hopping spread spectrum (FHSS):



► Frequency hopping spread spectrum (FHSS):



► Bluetooth Standards (IEEE 802.15.1):

Bluetooth standards encompass a range of specifications defining **wireless data communications** over **short distances** between devices. Here's a breakdown of the key aspects:

- **Frequency:**
 - Primarily operates in the **2.4 GHz band**, unlicensed globally, minimizing interference with other devices like phones or microwaves.
- **Speed:**
 - Varies depending on the version, with:
 - Bluetooth 1.0 and 2.0: **Up to 721 kbps**
 - Bluetooth 3.0 and 4.0: **Up to 3 Mbps (24 Mbps over 802.11 link)**
 - Bluetooth 5.0 and later: **Up to 3 Mbps (2 Mbps-LE)** with significantly improved range and power efficiency.

► Bluetooth Standards:

- **Range:**
 - Typically limited to around **10 meters** (33 feet) for basic connections, though extended-range versions can reach up to **100 meters** (330 feet).
- **Power Consumption:**
 - Generally lower than Wifi due to shorter range and lower data rates, making it ideal for battery-powered devices.
- **Security:**
 - Employs various encryption and authentication protocols to protect data transmissions, with newer versions offering stronger protection.

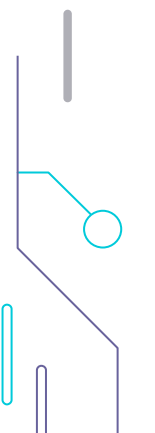
► Bluetooth Standards:

	Data Transfer Speed	Typical Max Range	Key New Features
Bluetooth 1.0 (1999)	0.7 Mbps	~10m (33 ft)	
Bluetooth 2.0 (2004)	1 Mbps core 3 Mbps with EDR	~30m (100 ft)	Enhanced Data Rate (EDR) Secure Simple Pairing (BT 2.1)
Bluetooth 3.0 (2009)	3 Mbps with EDR (24 Mbps over HS 802.11 link)	~30m (100 ft)	High Speed (HS) L2CAP Enhanced Modes Enhanced Power Control
Bluetooth 4.0 (2013)	3 Mbps with EDR 1 Mbps Low Energy	~60m (200 ft)	Low Energy (BLE) Core Specification Addenda 1-4 (BT 4.1) IoT features (BT 4.2)
Bluetooth 5 (2017)	3 Mbps with EDR 2 Mbps Low Energy	~240m (800 ft)	Slot Available Masking (SAM) LE Long Range Core Specification Addenda 6 (BT 5.1) LE Audio (BT 5.2)

Source: Android Authority

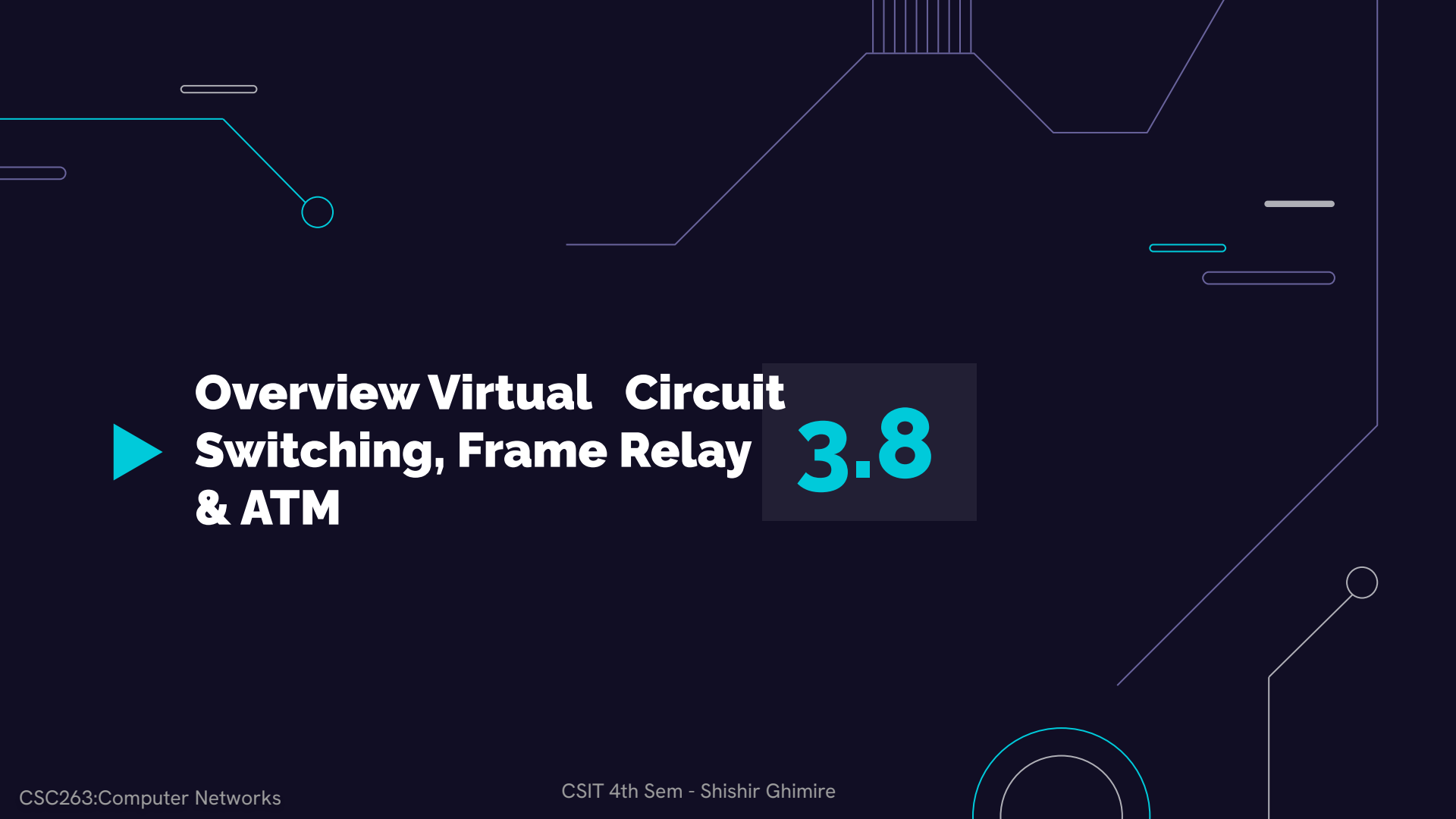


► Applications:

- Connecting wireless headphones, speakers, and headsets to phones, laptops, and tablets.
 - Enabling hands-free calling through car audio systems.
 - Transferring files between devices.
 - Building wireless sensor networks for industrial and home automation.
- 

► Different Bluetooth Versions:

1. **Bluetooth Classic:** The original version for basic data exchange and audio streaming.
2. **Bluetooth Low Energy (BLE):** Designed for low-power applications like fitness trackers and beacons.
3. **Bluetooth Audio:** Optimizes audio quality for music streaming and voice calls.
4. **Bluetooth Mesh:** Enables multi-device communication for mesh networks in smart homes and buildings.



► Overview Virtual Circuit Switching, Frame Relay & ATM

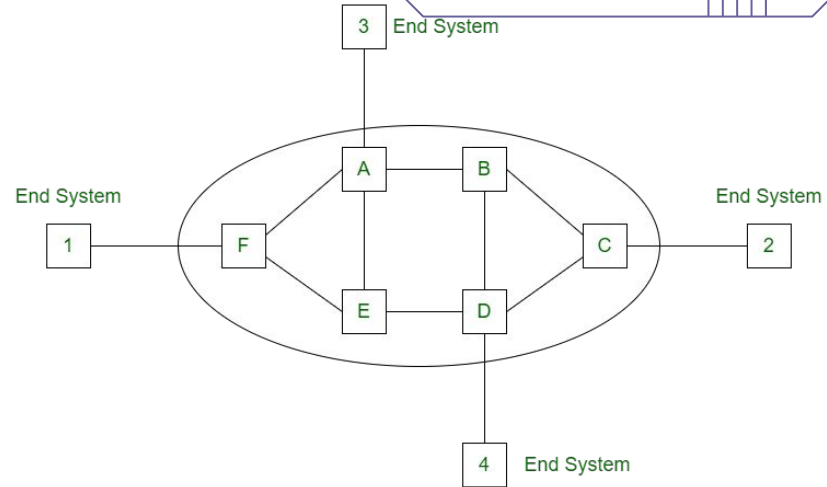
3.8

► Virtual Circuit Switching:

Virtual circuit switching is a **packet switching methodology** whereby a path is established between the source and the final destination through which all the packets will be routed during a call.

- This **path** is called a **virtual circuit** because to the user, the connection appears to be a **dedicated** physical circuit.
- However, other communications may also be sharing the parts of the same path.
- Before the data transfer begins, the source and destination **identify a suitable path** for the virtual circuit.
- All intermediate nodes between the two points **put an entry of the routing in their routing table** for the call.
- Additional parameters, such as the maximum packet size, are also exchanged between the source and the destination during call setup.
- The virtual circuit is **cleared** after the data transfer is completed.

► Virtual Circuit Switching:



Examples of virtual circuit switching are **X.25** and **Frame Relay**.

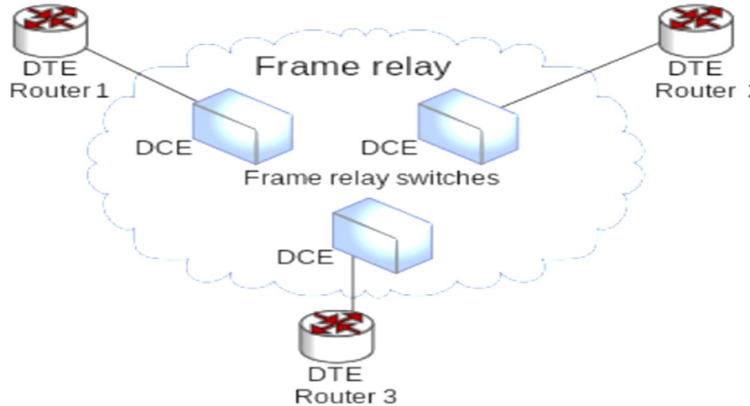
Advantages of virtual circuit switching are:

- Packets are delivered in **order**, since they all take the same route;
- The overhead in the packets is **smaller**,
- The connection is **more reliable**
- **Billing** is easier

► Frame Relay:

- Frame relay is a packet-switching telecommunications service designed for **cost-efficient** data transmission for intermittent traffic between **local area networks (LANs)** and between endpoints in wide area networks.
- In the 1980s, the X.25 networks were largely **replaced** by a new kind of network called frame relay.
- Frame Relay (FR) is a **high-performance WAN protocol** that operates at the physical and data link layers of the OSI reference Model.
- FR originally was designed for use across Integrated Service Digital Network (**ISDN**) interfaces.
- Today, it is used over a variety of other network interfaces as well.
- FR is an example of a packet-switched technology.
- Packet-switched networks enable end stations to dynamically share the network medium and the available bandwidth.

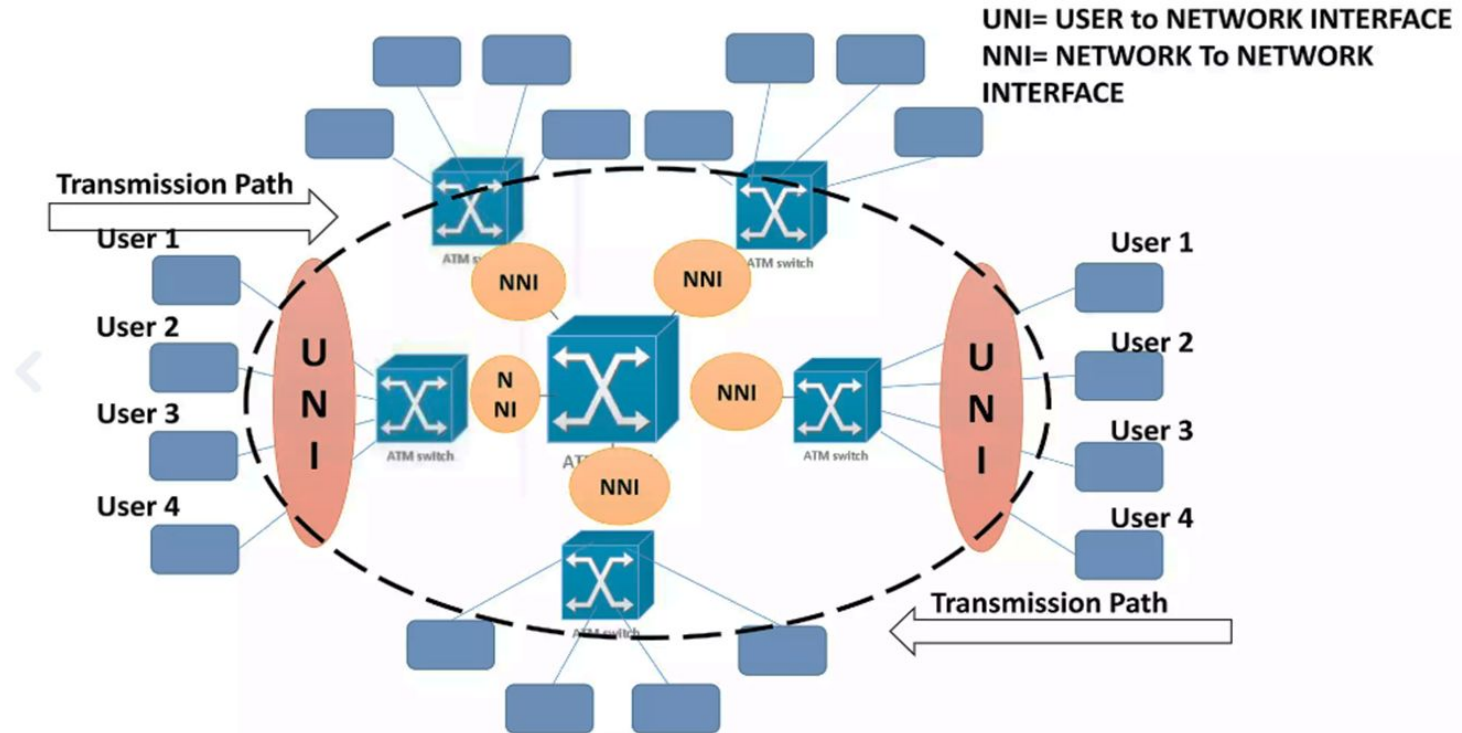
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► Asynchronous Transfer Mode (ATM):

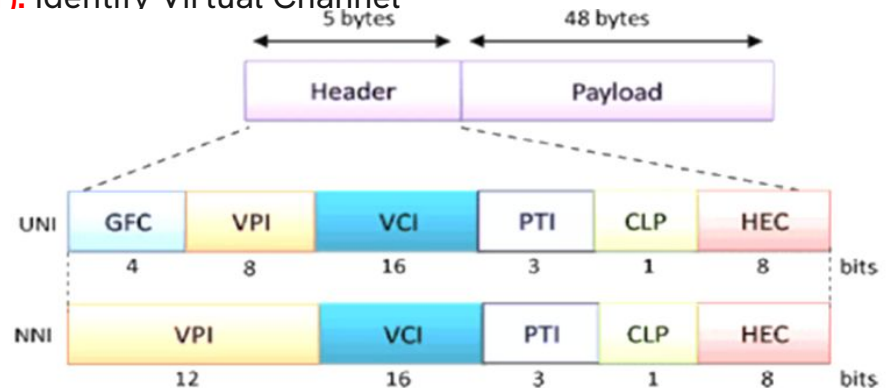
- ATM stands for Asynchronous Transfer Mode and ATM is a **switching technique** that uses **time division multiplexing** for the communication of the data.
- It's a telecommunications technology (ITU-T) used for transmitting **various types of data**, including voice, video, and data traffic, over high-speed networks.
- Cells are transmitted **asynchronously** and the network is **connection oriented**.
- Data is transmitted into **fixed size** packet called **cell** . i.e. Data unit is called cells. (53 Bytes cell format 5 byte allocated to the header and 48 bit carrying payload)
- Data is encoded in the form of **small cells** of a **fixed** size so that they become **acceptable** to TDM and can be transferred on a physical network.
- Generally,ATM uses **two types** of connection:
 - PVC(Permanent Virtual Connection)
 - SVC(Switched Virtual Connection).

► Asynchronous Transfer Mode (ATM):

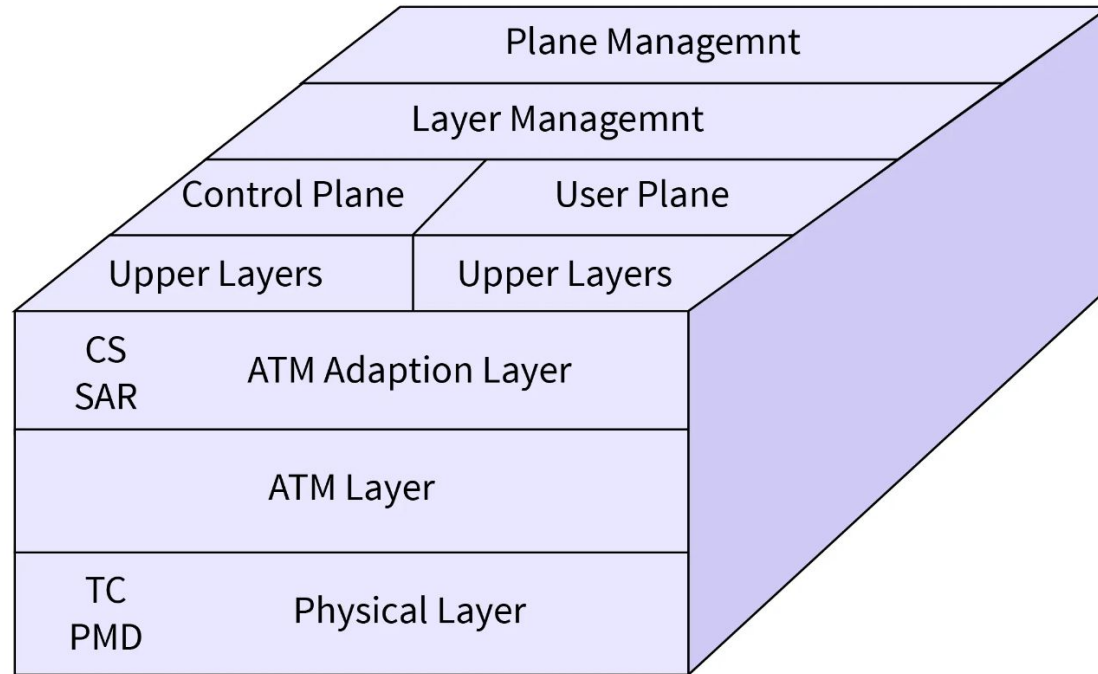


► ATM Cell Format:

- As information is transmitted in ATM in the form of **fixed size units called cells**. As known already each cell is 53 bytes long which consists of **5 bytes** header and **48 bytes** payload.
- Different fields of ATM Header are:
 - **GFC (General Flow Control)** : Field provides local function such as **identifying multiple station** that share a single ATM interface.
 - **VPI (Virtual Path Identifier)**: Used to identify **virtual path**.
 - **VCI (Virtual Channel Identifier)**: Identify Virtual Channel
 - Payload Type identifier
 - Cell Loss Priority
 - Header Error Control



► ATM Layers:



► ATM Layers:

- **ATM Adaption Layer (AAL)** : It is meant for **isolating** higher layer protocols from details of ATM processes and prepares for **conversion** of user data into cells and segments it into **48-byte cell** payloads.
 - AAL protocol **accepts** transmission from upper layer services and help them in mapping applications, e.g., voice, data to ATM cells.
- **Physical Layer** : It manages the **medium-dependent** transmission.
 - Main functions are as follows:
 - It converts **cells into a bit stream**.
 - It controls the **transmission and receipt of bits** in the physical medium.
 - Looks for the packaging of cells into appropriate type of frames.
- **ATM Layer**: It handles transmission, switching, congestion control, cell header processing, sequential delivery, etc.
-

► Asynchronous Transfer Mode (ATM):

Advantages:

- ATM is beneficial as it provides the **service of dynamic bandwidth** that is especially suited for burst traffic.
- Data transmission can be done **simply, predictably, and uniformly** as all the data are encoded into identical cells.
- Mixed traffic is efficiently handled by the **uniform packet size**.
- Packet overload is decreased by **small-sized headers** and provides effective usage of bandwidth.

Disadvantages:

- There is an **overhead of the cell header** (5 bytes/cell).
- Compared to LAN hardware, ATM switches are **very expensive**.

The background features a dark blue gradient with white and light blue geometric lines resembling a circuit board. These lines include straight paths, right-angle turns, and small circles at various points, creating a technical and modern aesthetic.

► DLL Protocol

3.9

► HDLC (High Level Data Link Control)

- High Level Data Link Control is a protocol, which **operates** at the data link layer.
- HDLC is an **ISO Standard** developed from the Synchronous Data Link Control (**SDLC**) standard proposed by IBM.
- High-level data link control (HDLC) is a **bit-oriented synchronous** data link layer protocol used to ensure the **error-free** transmission of data to respective destinations and control the **speed** at which data is transmitted. Used on both point-to-point and multipoint (multi-drop) data links.
- Data in an HDLC is organized **into units** called **frames** and is sent across networks to specified destinations. This is done using a **frame delimiter** or flag, which contains unique sequence of bits.
- Role of HDLC is to ensure that the data has been received **without any loss or errors** and in the correct order.
- Provides both connection-oriented and connectionless service.

► HDLC Station:

- Stations:
 - **Primary:** sends data, controls the link with commands.
 - **Secondary:** receives data, responds to control messages.
 - **Combined:** can issue both commands and responses.
- Link configuration:
 - **Unbalanced:** one primary station, one or more secondary stations.
 - **Balanced:** two combined station.

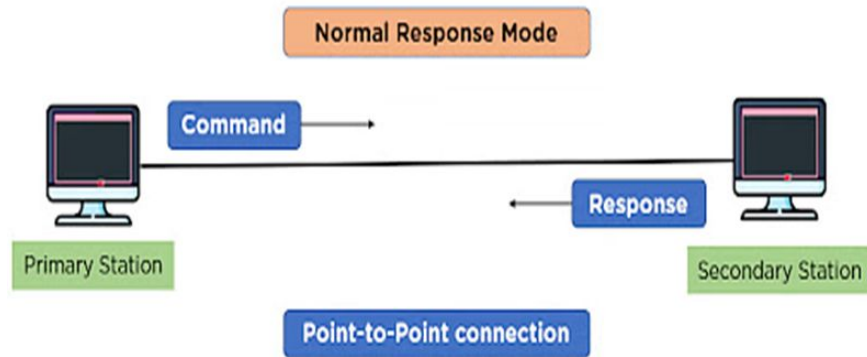
► HDLC Operation Mode:

- HDLC Operation Mode:
 - **Normal Response Mode (NRM)** where the **secondary station needs permission from the primary station** before it can transmit data. Mainly used on multi-point lines.
 - **Asynchronous Response Mode (ARM)** where the secondary station can send data **without receiving permission** from the primary station. This is hardly ever used.
 - **Asynchronous Balanced Mode (ABM)** where **either station can initiate** transmission without permission from the other. This is the most common mode used on point-to-point links..

► HDLC Operation Mode:

HDLC has two operation modes:

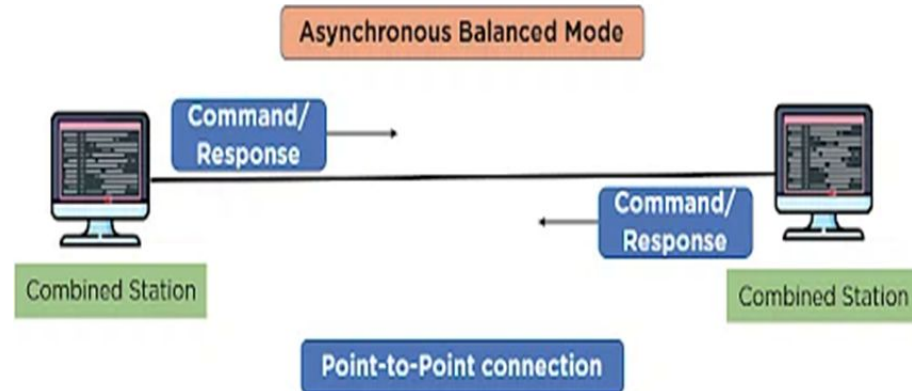
- Normal Response Mode (NRM):
 - Used with **unbalanced configuration**.
 - Primary **initiates** data transfer; secondary can **only reply**.



► HDLC Operation Mode:

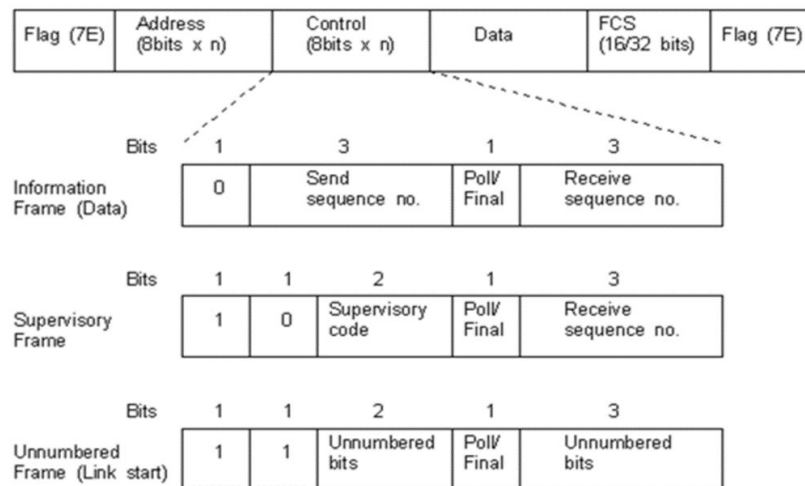
HDLC has two operation modes:

- Asynchronous Balanced Mode (ARM):
 - In this transfer model, **combined stations** are installed in a point-to-point configuration for exchange commands and responses in a balanced format.
 - Used with Balanced configuration.



► HDLC Frame Structure:

- There are three types of HDLC frames:
 - Information Frames (I-frames):** Used for carrying user data and control information.
 - Supervisory Frames (S-frames):** Provide flow and error control.
 - Unnumbered Frames (U-frames):** Used for management and control purposes without the need for acknowledgment.



► HDLC Frame Fields:

- Common Fields within HDLC frames are:
 - **Flag:** Identifies the beginning and end of a frame : 01111110 (7E Hex) .
 - **Address:**
 - It encapsulates the **receiver's address** in the field. For example, if the frame is sent from the primary station, it contains the secondary station's address and vice versa.
 - **Control :**
 - This field contains the **flow and error control information** in byte format.
 - **Data :**
 - It carries information from the network layer, and the data size may vary depending on the network.
 - **FCS :**
 - This field stands for **Frame Check Sequence** and acts as an error detection field in the HDLC protocol, which includes a 16-bit CRC check bit.

► Information Frames (I-Frame):

- Information Frames are used for the **data transfer between station**. They also include **flow and error control information** that is piggybacked on user data.
- If the **first bit** of the **control field is 0** then it is identified as I-frame.
- The send sequence, or next send N(S), and the receive sequence, or next receive N(R), hold the frame sequence numbers.
- The Poll/Final bit is called **Poll** when used by the primary station to obtain a response from a secondary station, and **Final** when used by the secondary station to indicate a response or the end of transmission.

	1	2	3	4	5	6	7	8
I: Information	0	N (S)			P/F	N (R)		
S: Supervisory	1	0	S		P/F	N (R)		
U: Unnumbered	1	1	M		P/F	M		

N (S): Send Sequence Number
N (R): Receive Sequence Number
S: Supervisory Function Bits
M: Unnumbered Function Bits
P/F: Poll/Final Bit

► Supervisory Frame (S-Frame):

- Supervisory Frames are used to **acknowledge frames**, **request for retransmissions** or to ask for **suspension** of transmission.
- The Supervisory code denotes the **type of supervisory frame** being sent.
- S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.

Supervisory Frames (S-Frames)

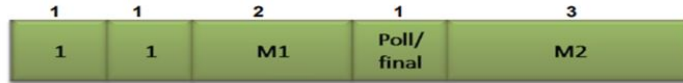


- S = 00 RR - Receiver Ready to accept more I-frames (data)
- S = 10 RNR - Receiver Not Ready to accept more I-frames
- S = 01 REJ - Go-Back-N retransmission request for an I-frame
- S = 11 SREJ - Selective retransmission request for an I-frame

► Unnumbered Frames (U-Frame):

- Unnumbered Frames are used for **link initialization or link disconnection**. The Unnumbered bits indicate the type of Unnumbered frame being used.
- U-frames or Un-numbered frames are used for miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.

Unnumbered Frames (U-Frames)



- SNRM: set normal response mode (M1 = 00, M2 = 001)
- SABM: set asynchronous balanced mode (M1 = 11, M2 = 100)
- SABME: set asynchronous balanced mode, extended (M1 = 11, M2 = 110)
- DISC: disconnect (M1=00, M2=010)
- UA: un-numbered acknowledgement (M1 = 00, M2 = 110)
- RSET: resets send and receive sequence numbers (M1 = 11, M2=001)
- FRMR: frame reject (M1 = 10, M2=001)

► PPP - Point to Point Protocol

- Point -to -Point Protocol (PPP) is a **communication protocol** of the data link layer that is used to transmit multiprotocol **data between two directly connected** (point-to-point) computers.
- It is a **byte-oriented protocol** that is widely used in broadband communications having heavy loads and high speeds.
- **Services Provided by PPP:**
 - The main services provided by Point -to -Point Protocol are :
 - **Defining the frame format** of the data to be transmitted.
 - Defining the **procedure of establishing link** between two points and exchange of data.
 - Stating the **method of encapsulation** of network layer data **in the frame**.
 - Stating **authentication rules** of the communicating devices.

► Components of PPP (PPP Stack):

- **Link Control Protocol:**

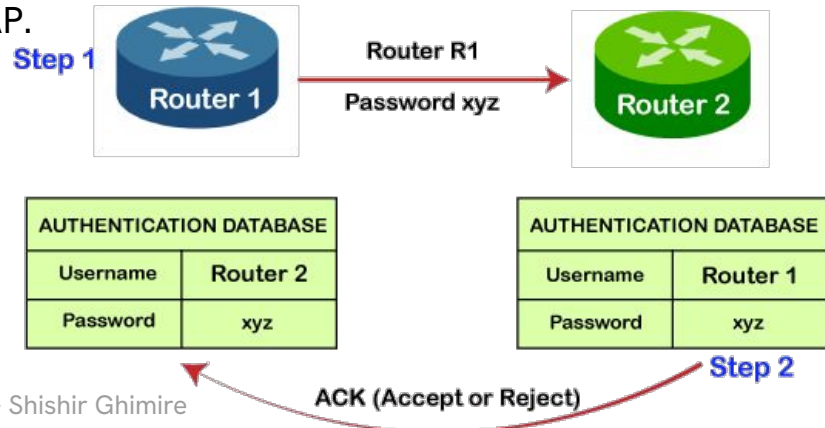
- Establishing the link between two devices.
- Maintaining this established link.
- Configuring this link.
- Terminating this link after the transfer

- **Authentication Protocol:**

- Authentication protocols help to **validate the identity of a user** who needs to access the resources.
- There are **two** authentication protocols:
 - Password Authentication Protocols (PAP)
 - Challenge Handshake Authentication Protocol (CHAP)

► PAP(Password Authentication Protocol):

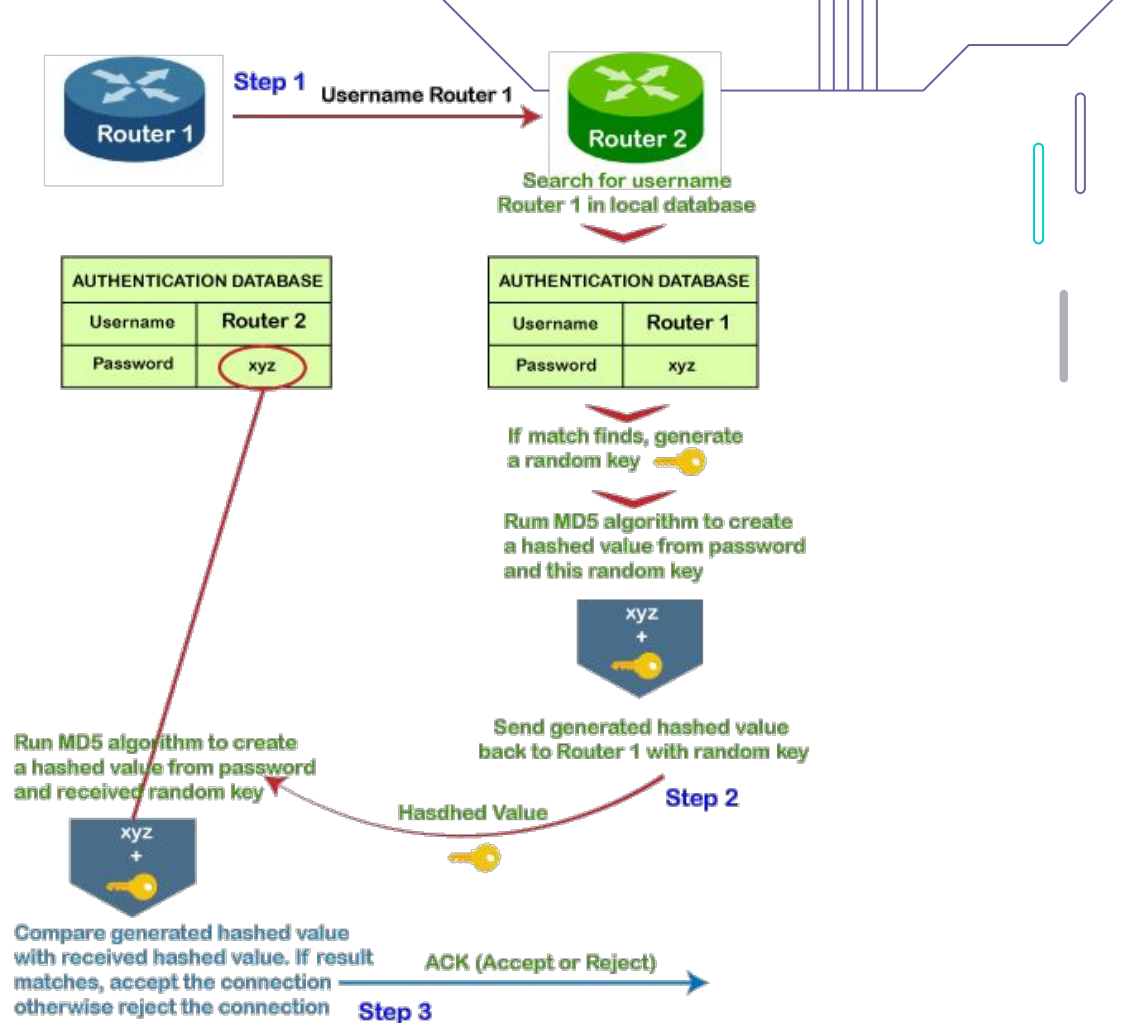
- The Password Authentication Protocol (PAP) is a simple authentication procedure with a two step process:
 - The user who wants to access a system sends an **authentication identification** (usually the user name) and a password.
 - The system checks the **validity of the identification and password** and either accepts or denies connection.
- PAP is **less secure** as compared to CHAP.



► Challenge Handshake Authentication Protocol (CHAP):

- The Challenge Handshake Authentication Protocol (CHAP) is a **three-way handshaking** authentication protocol that provides **greater security than PAP**. In this method, the password is **kept secret; it is never sent online**. The three steps in CHAP are:
 - The system sends the user a **challenge packet** containing a challenge value, usually a few bytes.
 - The user applies a **predefined function** that takes the challenge value and the user's own password and **creates a result**. The user **sends** the result in the response packet to the system.
 - The system does the same. It applies the same function to the password of the user (known to the system) and the challenge value to create a result. If the **result created** is the same as the **result sent** in the response packet, **access is granted**; otherwise, it is denied. CHAP is more secure than PAP.

► Challenge Handshake Authentication Protocol (CHAP):

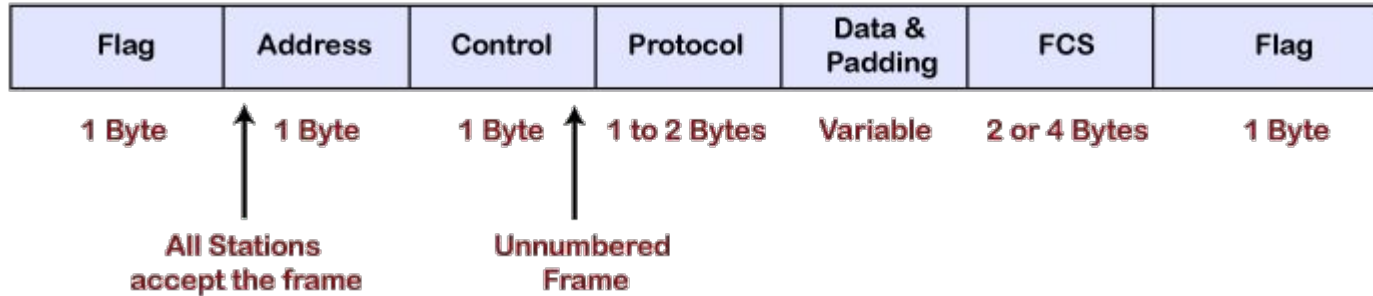


► Network Control Protocol (NCP):

- **After establishing** the link and authenticating the user, PPP **connects** to the network layer. This connection is **established by NCP**.
- Therefore NCP is a **set of control protocols** that allow the encapsulation of the data coming from network layer.
- After the network layer **configuration is done** by one of the NCP protocols, the users can exchange data from the network layer.
- PPP can carry a network layer **data packet** from protocols defined by the Internet, DECNET, Apple Talk, Novell, OSI, Xerox and so on.
- **None of the NCP packets** carry networks layer data. They just configure the link at the network layer for the incoming data.

► Frame Format in PPP:

- PPP is a **byte-oriented protocol** where each field of the frame is composed of one or more bytes. The fields of a PPP frame are :



► Frame Format in PPP:

The frame format of **PPP resembles HDLC frame**. Its various fields are:

- **Flag:**
 - The flag field is used to indicate the start and end of the frame.
 - The flag field is a 1-byte field that appears at the beginning and the ending of the frame.
 - The pattern of the flag is similar to the bit pattern in HDLC, i.e., 01111110.
- **Address:**
 - It is a 1-byte field that contains the constant value which is 11111111.
 - These 8 ones represent a broadcast message.
 - All 1's in the address field indicates that all stations are to accept the frame.
- **Control:**
 - It is a 1-byte field which is set through the constant value, i.e., 11000000.
 - It is **not a required field as PPP does not support the flow control** and a very limited error control mechanism.
 - The control field is a mandatory field where protocol supports flow and error control mechanism.

► Frame Format in PPP:

- **Protocol:**

- It is a 1 or 2 bytes field that defines **what is to be carried** in the data field.
- The data can be a user data or other information.

- **Payload:**

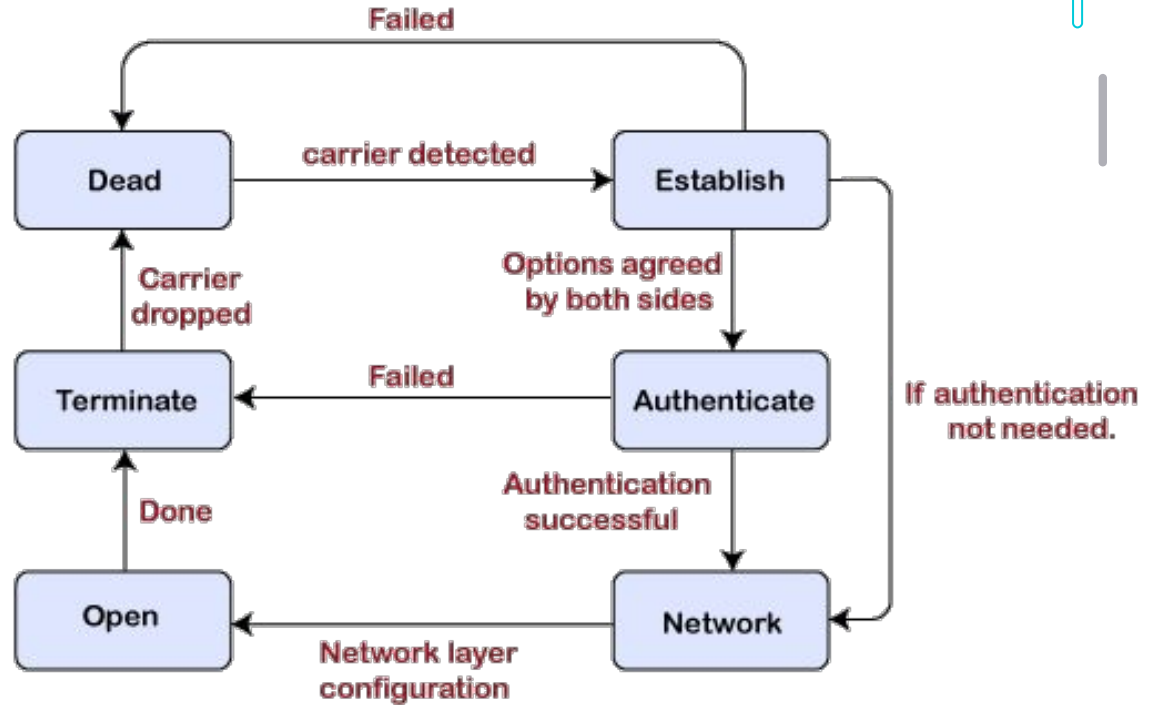
- The payload field carries either user data or other information.
- Its length is variable.
- If the length is not negotiated using LCP (Link Control Protocol) during line set up, a default length of 1500 bytes is used.

- **FCS Field:**

- FCS stands for Frame Check Sequence.
- It is either of 2 bytes or 4 bytes.
- It contains the checksum.

► Transition Phases of PPP:

- The following are the transition phases of a PPP protocol:



Transition phases

► Transition Phases of PPP:

The following are the transition phases of a PPP protocol:

- **Dead Phase:** In this initialization stage, the link undergoes **carrier detection**, signalling the physical layer readiness, and begins the **PPP process** with LCP automation.
- **Establishment Phase:** Following **peer detection**, this phase involves the exchange and negotiation of LCP frames, ultimately **establishing the link for further authentication** or network layer protocol configuration.
- **Authenticate Phase:** This optional PPP phase is activated when **peer authentication is required**, employing protocols such as PAP or CHAP for verification.

► Transition Phases of PPP:

- **Network Phase:** Once the LCP state is open, PPP enters this phase, using **NCP packets to configure network-layer protocols** like IP, facilitating proper network layer setup.
- **Open Phase:** **Data transfer** occurs in this phase, representing the active state of the connection; termination is initiated by moving to the Terminate Phase upon request.
- **Terminate Phase:** In response to a request from either endpoint, the connection enters this phase, utilizing LCP to gracefully close the link through the **exchange of termination packets**.



▶ **THANKS!**

Do you have any questions?

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